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English Version

Medical electrical equipment - Part 2-61: Particular requirements for basic safety and essential performance of pulse oximeter equipment (ISO 80601-2-61:2026)

Appareils électromédicaux - Partie 2-61: Exigences particulières pour la sécurité de base et les performances essentielles pour les oxymètres de pouls (ISO 80601-2-61:2026)

Medizinische elektrische Geräte - Teil 2-61: Besondere Festlegungen für die Sicherheit einschließlich der wesentlichen Leistungsmerkmale von Pulsoximetriegeräten (ISO 80601-2-61:2026)

This European Standard was approved by CEN on 17 March 2026.

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EUROPEAN COMMITTEE FOR STANDARDIZATION
COMITÉ EUROPÉEN DE NORMALISATION
EUROPÄISCHES KOMITEE FÜR NORMUNG

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European foreword

This document (EN ISO 80601-2-61:2026) has been prepared by Technical Committee ISO/TC 121 "Anaesthetic and respiratory equipment" in collaboration with Technical Committee CEN/TC 215 “Respiratory and anaesthetic equipment” the secretariat of which is held by BSI.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by October 2026, and conflicting national standards shall be withdrawn at the latest by October 2026.

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. CEN shall not be held responsible for identifying any or all such patent rights.

This document supersedes EN ISO 80601-2-61:2019.

Any feedback and questions on this document should be directed to the users’ national standards body/national committee. A complete listing of these bodies can be found on the CEN website.

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Endorsement notice

The text of ISO 80601-2-61:2026 has been approved by CEN as EN ISO 80601-2-61:2026 without any modification.

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International Standard

ISO 80601-2-61

Third edition
2026-04

Medical electrical equipment —

Part 2-61: Particular requirements for basic safety and essential performance of pulse oximeter equipment

Appareils électromédicaux —

Partie 2-61: Exigences particulières pour la sécurité de base et les performances essentielles pour les oxymètres de pouls

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Foreword

ISO (the International Organization for Standardization) and IEC (the International Electrotechnical Commission) form the specialized system for worldwide standardization. National bodies that are members of ISO or IEC participate in the development of International Standards through technical committees established by the respective organization to deal with particular fields of technical activity. ISO and IEC technical committees collaborate in fields of mutual interest. Other international organizations, governmental and non-governmental, in liaison with ISO and IEC, also take part in the work.

The procedures used to develop this document and those intended for its further maintenance are described in the ISO/IEC Directives, Part 1. In particular, the different approval criteria needed for the different types of document should be noted. This document was drafted in accordance with the editorial rules of the ISO/IEC Directives, Part 2 (see www.iso.org/directives or www.iec.ch/members_experts/refdocs).

ISO and IEC draw attention to the possibility that the implementation of this document may involve the use of (a) patent(s). ISO and IEC take no position concerning the evidence, validity or applicability of any claimed patent rights in respect thereof. As of the date of publication of this document, ISO and IEC had not received notice of (a) patent(s) which may be required to implement this document. However, implementers are cautioned that this may not represent the latest information, which may be obtained from the patent database available at www.iso.org/patents and https://patents.iec.ch. ISO and IEC shall not be held responsible for identifying any or all such patent rights.

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For an explanation of the voluntary nature of standards, the meaning of ISO specific terms and expressions related to conformity assessment, as well as information about ISO's adherence to the World Trade Organization (WTO) principles in the Technical Barriers to Trade (TBT) see www.iso.org/iso/foreword.html. In the IEC, see www.iec.ch/understanding-standards.

This document was prepared by Technical Committee ISO/TC 121, *Anaesthetic and respiratory equipment*, Subcommittee SC 3, *Respiratory devices and related equipment used for patient care* and Technical Committee IEC/TC 62, *Medical equipment, software, and systems*, Subcommittee SC 62D, *Particular medical equipment, software, and systems*, in collaboration with the European Committee for Standardization (CEN) Technical Committee CEN/TC 215, *Respiratory and anaesthetic equipment*, in accordance with the Agreement on technical cooperation between ISO and CEN (Vienna Agreement).

This third edition cancels and replaces the second edition (ISO 80601-2-61:2017), which has been technically revised.

The main changes are as follows:

— alignment with IEC 60601-1:2005+AMD1:2012+AMD2:2020, IEC 60601-1-8:2006+AMD1:2012 +AMD2:2020, IEC 60601-1-2:2014+AMD1:2020 and IEC 60601-1-6:2010+AMD1:2013 +AMD2:2020.

— increased disclosure requirements;

— increased the required number of participants in the clinical study and their diversity (a means to assure equal contributions across the range of skin pigmentation);

— reduced the maximum permissible A_{rms} to enhance measurement accuracy;

— required *differential bias* determination to enhance measurement accuracy;

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— clarified that *accessories* need to be included in the clinical performance *verification* and conformity to the requirements of the document

— updated the reporting requirements for the clinical performance *verification*;

— added an Annex describing the use of *transfer standard* for product development purposes;

— added an Annex mapping the requirements of this document to the IMDRF *essential principles*[25] and labelling[26] guidances; and

— harmonization with ISO 20417, where appropriate.

A list of all parts in the ISO 80601 series and the IEC 80601 series can be found on the ISO and IEC websites.

Any feedback or questions on this document should be directed to the user's national standards body. A complete listing of these bodies can be found at www.iso.org/members.html and www.iec.ch/national-committees.

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Introduction

The approximation of arterial haemoglobin saturation and pulse rate using pulse oximetry is common practice in many areas of medicine. This document covers *basic safety* and *essential performance* requirements achievable within the limits of existing technology.

The committees recognized the need to revise the first edition of this document because of the publication of IEC 60601-1:2005+AMD1:2012+AMD2:2020, IEC 60601-1-12:2014+AMD1:2020, as well as IEC 60601-1-2:2014+AMD1:2020, IEC 60601-1-11:2015+AMD1:2020, IEC 60601-1-6:2010+AMD1:2013+AMD2:2020 and IEC 60601-1-8:2006+AMD1:2012+AMD2:2020.

Annex C contains a guide to the *marking* and labelling requirements in this document.

Annex D contains a summary of the *symbols* referenced in this document.

Annex AA contains a rationale for some of the requirements. It is included to provide additional insight into the reasoning of the committees that led to a requirement and identifying the *hazards* that the requirement addresses.

Annex BB is a literature survey relevant to the determination of the maximum safe temperature of the interface between a *pulse oximeter probe* and a *patient's* tissue.

Annex CC discusses the formulae used to evaluate the *SpO_2 accuracy* of *pulse oximeter equipment* measurements, *differential bias* and the names that are assigned to those formulae.

Annex DD presents a guideline for a *controlled desaturation study* for the calibration of *pulse oximeter equipment*.

Annex EE is a tutorial introduction to several kinds of testers used in pulse oximetry.

Annex FF describes concepts of *pulse oximeter equipment* response time.

Annex GG describes data interface requirements.

Annex HH describing the clinical context of this document and its rationale;

Annex II describing the use of a *functional tester*;

Annex JJ describing the use of *transfer standard*;

Annex KK maps the requirements of this document to the IMDRF *essential principles*[25] and labelling[26] guidances

In referring to the structure of this document, the term

– "clause" means one of the six numbered divisions within the table of contents, inclusive of all subdivisions (e.g. Clause 201 includes subclauses 201.7.1, 201.7.2) and

– "subclause" means a numbered subdivision of a clause (e.g. 201.7.1, 7.2 and 201.7.2.1 are all subclauses of Clause 201.7).

References to clauses within this document are preceded by the term "Clause" followed by the clause number. References to subclauses within this document are by number only.

In this document, the conjunctive "or" is used as an "inclusive or" so a statement is true if any combination of the conditions is true.

The verbal forms used in this document conform to usage described in Annex H of the ISO/IEC Directives, Part 2. For the purposes of this document, the auxiliary verb:

– "shall" indicates a requirement;

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- “should” indicates a recommendation;
- “may” indicates a permission;
- “can” indicates a possibility or capability; and
- “must” is used to express an external constraint.

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Medical electrical equipment —

Part 2-61:
Particular requirements for basic safety and essential performance of pulse oximeter equipment

201.1 Scope, object, and related standards

IEC 60601-1:2005+AMD1:2012+AMD2:2020, Clause 1 applies, except as follows:

201.1.1 Scope

Replacement:

NOTE 1 There is guidance or rationale for this subclause contained in AA.2.1.

This document applies to the *basic safety* and *essential performance* of *pulse oximeter equipment* intended for use on humans, hereafter referred to as *ME equipment*. This includes any part necessary for *normal use*, including the *pulse oximeter monitor*, *pulse oximeter probe*, and *probe cable extender*.

These requirements apply to *pulse oximeter equipment*, including *pulse oximeter monitors*, *pulse oximeter probes* and *probe cable extenders* regardless of their origin (i.e. including *remanufactured* products).

The intended use of *pulse oximeter equipment* includes, but is not limited to, the estimation of arterial oxygen haemoglobin saturation and pulse rate of *patients* in professional healthcare institutions as well as *patients* in the *home healthcare environment* and the *emergency medical services environment*.

If a clause or subclause is specifically intended to be applicable to *ME equipment* only, or to *ME systems* only, the title and content of that clause or subclause says so. If that is not the case, the clause or subclause applies both to *ME equipment* and to *ME systems*, as relevant.

Hazards inherent in the intended physiological function of *ME equipment* or *ME systems* within the scope of this document are not covered by specific requirements in this document except in 201.11.1.2.2, IEC 60601-1:2005+AMD1:2012+AMD2:2020, 7.2.13 and 8.4.1.

NOTE 2 See also IEC 60601-1:2005+AMD1:2012+AMD2:2020, 4.2.

This document can also be applied to *ME equipment* and their *accessories* used for compensation or alleviation of disease, injury, or disability.

This document is not applicable to *pulse oximeter equipment* intended for use in laboratory research applications nor to oximeters that require a blood sample from the *patient*.

This document is not applicable to *pulse oximeter equipment* intended solely for foetal use.

This document is not applicable to remote or slave (secondary) equipment that displays SpO_2 values that are located outside of the *patient environment*.

NOTE 3 *ME equipment* that provides selection between diagnostic and monitoring functions is expected to meet the appropriate requirements of this document when configured for that function.

This document is applicable to *pulse oximeter equipment* intended for use under extreme or uncontrolled environmental conditions outside the hospital environment or physician's office, such as in ambulances and air transport. Additional standards can apply to *pulse oximeter equipment* for those environments of use.

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This document is a particular standard in the IEC 60601-1 and ISO and IEC 80601 series of standards.

201.1.2 Object

Replacement:

The object of this document is to establish particular *basic safety* and *essential performance* requirements for *pulse oximeter equipment* [as defined in 201.3.253] and its *accessories*.

NOTE 1 *Accessories* are included because the combination of the *pulse oximeter monitor* and the *accessories* needs to be adequately safe. *Accessories* can have a significant impact on the *basic safety* or *essential performance* of *pulse oximeter equipment.*

NOTE 2 This document has been prepared to address the relevant *essential principles*[25] and labelling[26] guidances of the International Medical Devices Regulators Forum (IMDRF) as indicated in Annex KK.

NOTE 3 This document has been prepared to address the relevant general safety and performance requirements of European regulation (EU) 2017/745[27].

201.1.3 Collateral standards

Amendment (add after existing text):

This document refers to those applicable collateral standards that are listed in IEC 60601-1:2005+AMD1:2012+AMD2:2020, Clause 2 and Clause 201.2 of this document.

IEC 60601-1-2:2014+AMD1:2020, IEC 60601-1-6:2010+AMD1:2013+AMD2:2020, IEC 60601-1-8:2006+AMD1:2012+AMD2:2020, IEC 60601-1-11+AMD1:2020 and IEC 60601-1-12+AMD1:2020 apply as modified in Clauses 202, 206, 208, 211 and 212, respectively. IEC 60601-1-3 and IEC 60601-1-9 do not apply. All other published collateral standards in the IEC 60601-1 series apply as published.

201.1.4 Particular standards

Replacement:

In the IEC 60601 series, particular standards may modify, replace, or delete requirements contained in the general standard, including the collateral standards, as appropriate for the particular *ME equipment* under consideration, and may add other *basic safety* or *essential performance* requirements.

NOTE The general standard is IEC 60601-1:2005+AMD1:2012+AMD2:2020.

A requirement of a particular standard takes priority over the general standard or the collateral standards.

For brevity, IEC 60601-1:2005+AMD1:2012+AMD2:2020 is referred to in this document as the general standard. Collateral standards are referred to by their document number.

The numbering of clauses and subclauses of this document corresponds to those of the general standard with the prefix “201” (e.g. 201.1 in this document addresses the content of Clause 1 of the general standard) or applicable collateral standard with the prefix “2xx” where xx is the final digits of the collateral standard document number (e.g. 202.4 in this document addresses the content of Clause 4 of the IEC 60601-1-2 collateral standard, 208.4 in this document addresses the content of Clause 4 of the IEC 60601-1-8 collateral standard, etc.). The changes to the text of the general standard are specified by the use of the following abbreviated words:

"Replacement" means that the clause or subclause of the general standard or applicable collateral standard is replaced completely by the text of this document.

"Addition" means that the text of this document is additional to the requirements of the general standard or applicable collateral standard.

"Amendment" means that the clause or subclause of the general standard or applicable collateral standard is amended as indicated by the text of this document.

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Subclauses or figures that are additional to those of the general standard are numbered starting from 201.101, additional annexes are lettered AA, BB, etc., and additional items aa), bb), etc.

Subclauses or figures that are additional to those of a collateral standard are numbered starting from 2xx, where “x” is the number of the collateral standard (e.g. 202 for IEC 60601-1-2, 203 for IEC 60601-1-3, etc).

The term "this document" is used to make reference to the general standard, any applicable collateral standards and this particular document taken together.

Where there is no corresponding clause or subclause in this document, the section, clause or subclause of the general standard or applicable collateral standard, although possibly not relevant, applies without modification; where it is intended that any part of the general standard or applicable collateral standard, although possibly relevant, is not to be applied, a statement to that effect is given in this particular document.

201.2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

NOTE 1 The way in which these referenced documents are cited in normative requirements determines the extent (in whole or in part) to which they apply.

NOTE 2 Informative references are listed in the Bibliography.

IEC 60601-1:2005+AMD1:2012+AMD2:2020, Clause 2 applies, except as follows:

Addition:

ISO 14155:2026, *Clinical investigation of medical devices for human subjects — Good clinical practice*

ISO 15223-1:2021+AMD1:2025, *Medical devices — Symbols to be used with medical device labels, labelling and information to be supplied — Part 1: General requirements*

ISO 20417:2026, *Medical devices — Information to be supplied by the manufacturer*

IEC 60068-2-27:2008, *Environmental testing — Part 2-27: Tests — Test Ea and guidance: Shock*

IEC 60068-2-31:2008, *Environmental testing — Part 2-31: Tests — Test Ec: Rough handling shocks, primarily for equipment-type specimens*

IEC 60068-2-64:2008+AMD1:2019, *Environmental testing — Part 2-64: Tests — Test Fh: Vibration, broadband random and guidance*

IEC 60601-1-11:2015+AMD1:2020, *Medical electrical equipment — Part 1-11: General requirements for basic safety and essential performance — Collateral Standard: Requirements for medical electrical equipment and medical electrical systems used in the home healthcare environment*

IEC 60601-1-12:2014+AMD1:2020, *Medical electrical equipment — Part 1-12: General requirements for basic safety and essential performance — Collateral Standard: Requirements for medical electrical equipment and medical electrical systems used in the emergency medical services environment*

IEC 60825-1:2014, *Safety of laser products - Part 1: Equipment classification and requirements*

IEC 60825-2:2021, *Safety of laser products — Part 2: Safety of optical fibre communication systems (OFCS)*

201.3 Terms and definitions

For the purposes of this document, the terms and definitions given in IEC 60601-1:2005+AMD 1:2012+AMD1:2020, and the following apply.

ISO and IEC maintain terminology databases for use in standardization at the following addresses:

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— ISO Online browsing platform: available at https://www.iso.org/obp

— IEC Electropedia: available at https://www.electropedia.org/

NOTE An alphabetized index of defined terms is found following the Bibliography.

201.3.201
accompanying information
information supplied by the manufacturer with or *marked* on a medical device or *accessory* for the user or *responsible organization*, particularly regarding safe use

Note 1 to entry: The *accompanying information* is be regarded as part of the medical device or *accessory*.

Note 2 to entry: The *accompanying information* can consist of the *label*, *marking*, *instructions for use*, *technical description*, information shown on the packaging or graphical user interface (GUI), installation manual, quick reference guide, etc. and can address the installation, use, *processing*, maintenance and disposal of the medical device or *accessory.*

Note 3 to entry: *Accompanying information* is not necessarily a written or printed document but can involve auditory, visual, or tactile materials and multiple media types (e.g. CD-ROM, DVD-ROM, USB drive, website).

Note 4 to entry: Medical devices and *accessories* that can be used safely without *accompanying information* are exempted from having *accompanying information* by some *authorities having jurisdiction.*

[SOURCE: ISO 20417:2026, 3.2]

201.3.202
accuracy
closeness of agreement between a test result and an accepted reference value

Note 1 to entry: Subclause 201.12.1.101.4 provides the method of calculating the *SpO_2 accuracy* of *pulse oximeter equipment*.

Note 2 to entry: Additional information is found in Annex CC.

[SOURCE: ISO 3534-2:2006, 3.3.1, modified — Replaced note 1 to note 3 with new notes and ‘or measurement result and the true’ with ‘and an accepted reference ’.]

201.3.203
alarm condition delay
time from the occurrence of a triggering event either in the *patient*, for *physiological alarm conditions*, or in the equipment, for *technical alarm conditions*, to when the *alarm system* determines that an *alarm condition* exists

[SOURCE: IEC 60601-1-8:2006+AMD1:2012+AMD2:2020, 3.2]

201.3.204
alarm limit
threshold used by an *alarm system* to determine an *alarm condition*

[SOURCE: IEC 60601-1-8:2006+AMD1:2012+AMD2:2020, 3.3]

201.3.205
alarm paused
state of limited duration in which the *alarm system* or part of the *alarm system* does not generate *alarm signals*

[SOURCE: IEC 60601-1-8:2006+AMD1:2012+AMD2:2020, 3.5]

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201.3.206
alarm preset
set of stored configuration parameters, including selection of algorithms and initial values for use by algorithms, which affect or modify the performance of the *alarm system*

[SOURCE: IEC 60601-1-8:2006+AMD1:2012+AMD2:2020, 3.6]

201.3.207
alarm signal generation delay
time from the onset of an *alarm condition* to the generation of its *alarm signal(s*)

[SOURCE: IEC 60601-1-8:2006+AMD1:2012+AMD2:2020, 3.10]

201.3.208
audio paused
state of limited duration in which the *alarm system* or part of the *alarm system* does not generate an auditory *alarm signal*

[SOURCE: IEC 60601-1-8:2006+AMD1:2012+AMD2:2020, 3.13]

201.3.209
colour measurement site
location on the body at which the skin pigmentation is assessed by *pigmentation measurement methods (PMMs)*

201.3.210
controlled desaturation study
hypoxaemia induced in a cohort of healthy adult human participants performed under well controlled, optimal or non-optimal laboratory conditions.

201.3.211
CO-oximeter
multiwavelength optical blood analyser that measures *total haemoglobin concentration* and the concentrations of various haemoglobin derivatives

Note 1 to entry: The relevant CO-oximetry value is functional saturation of arterial blood, SaO_2, which *pulse oximeter equipment* estimates and reports as SpO_2.

201.3.212
data update period
interval in which the *pulse oximeter equipment* algorithm provides new valid data to the display or the *functional connection*

Note 1 to entry: The *data update period* does not refer to the regular refresh period of the display, which is typically on the order of 1 s, but rather to the (typically longer) interval defined above.

201.3.213
declared range
portion of the *displayed range* of SpO_2 and pulse rate values over which there is specified *accuracy*

201.3.214
default alarm preset
alarm preset that can be activated by the *alarm system* without *operator* action

Note 1 to entry: *Manufacturer-* or *responsible organization-configured alarm presets* are possible types of *default alarm presets*.

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[SOURCE: IEC 60601-1-8:2006+AMD1:2012+AMD2:2020, 3.16]

201.3.215
differential bias
measure of the overall dependence of participant-specific *mean bias* to a factor

201.3.216
displayed range
range of SpO_2 or pulse rate values that can be displayed by the *pulse oximeter equipment*

Note 1 to entry: The *displayed range* can extend beyond the *declared range*.

201.3.217
distributed alarm system
DAS
alarm system that involves more than one item of equipment of a *ME system* intended for delivery of *alarm conditions* with technical confirmation

Note 1 to entry: The parts of a *distributed alarm system* can be widely separated in distance.

Note 2 to entry: A *distributed alarm system* is intended to notify *operators* of the existence of an *alarm condition*.

Note 3 to entry: For the purposes of this document, technical confirmation means that each element of a *distributed alarm system* confirms or guarantees the successful delivery of the *alarm condition* to the next element or appropriate *technical alarm conditions* are created as described in IEC 60601-1-8:2006+AMD1:2012+AMD2:2020, 6.11.2.2.1.

[SOURCE: IEC 60601-1-8:2006+AMD1:2012+AMD2:2020, 3.17]

201.3.218
EMS environment
emergency medical services environment
actual conditions and settings, in which *operators* interact with the *ME equipment* or *ME system*, in and around the scene of an emergency outside of a professional healthcare facility where a *patient* can be given medical care, basic or advanced life support as well as during professional transport to a professional healthcare facility or between professional healthcare facilities

EXAMPLE 1 Responding to and providing life support at the scene of an emergency to a *patient* reported as experiencing injury or illness in a pre-hospital setting, and transporting the *patient*, while continuing such life support care, to an appropriate professional healthcare facility for further care.

EXAMPLE 2 Providing monitoring, treatment or diagnosis during transport between professional healthcare facilities.

Note 1 to entry: For the purposes of this document, use of equipment intended for the *EMS environment* and temporarily used in the *home healthcare environment* by emergency medical personnel is considered use in the *EMS environment*.

Note 2 to entry: For the purposes of this document, the *operators* of equipment intended for the *EMS environment* are presumed to be professional medical personnel or personnel with relevant specialized training.

Note 3 to entry: Professional healthcare facilities include hospitals, physician offices, freestanding surgical centres, dental offices, freestanding birthing centres, limited care facilities, first aid rooms or rescue rooms and multiple treatment facilities.

Note 4 to entry: Emergency medical services are known by various names in different countries and regions.

Note 5 to entry: For the purposes of this document, transport includes road, rotary and fixed-wing ambulances.

[SOURCE: IEC 60601-1-12:2014+AMD1:2020, 3.1]

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201.3.219
essential function
function or capability that is required to maintain *basic safety*, *essential performance*, a minimum of clinical functionality as specified by the *manufacturer*, and operational availability for the medical device

Note 1 to entry: *Essential functions* include, but are not limited to, the safety instrumented function (*basic safety* and *essential performance*), the control function and the availability of urgently needed functions and such allowing the *operator* to view and manipulate the medical device safely with the most urgently needed performance (operational availability). The loss of *essential function* is commonly termed loss of protection, loss of control and loss of view respectively.

Note 2 to entry: The term is derived from IEC 62443-4-2:2019, 3.1.20, and has been refined for the purpose and scope of this document.

[SOURCE: IEC/TR 60601-4-5:2021, 3.10]

201.3.220
essential principles
essential principles of safety and performance
fundamental high-level requirements that, when conformed with, ensure a medical device or *accessory* is safe and performs as intended

[SOURCE: ISO 20417:2026, 3.10]

201.3.221
false positive alarm condition
presence of an *alarm condition* when no valid triggering event has occurred in the *patient*, the equipment or the *alarm system*

Note 1 to entry: A *false positive alarm condition* can be caused by spurious information produced by the *patient*, the *patient*-equipment interface, other equipment or the *alarm system* itself.

[SOURCE: IEC 60601-1-8:2006+AMD1:2012+AMD2:2020, 3.21]

201.3.222
firecall
method established to provide emergency access to a secure medical device

Note 1 to entry: In an emergency situation, unprivileged users can gain access to key systems to correct the problem. When a *firecall* is used, there is usually a review *process* to ensure that the access was used properly to correct a problem. These methods generally either provide a one-time use user identifier (ID) or one-time password or other suitable measures.

Note 2 to entry: Also referred to as "break glass" feature.

[SOURCE: IEC/TR 60601-4-5:2021, 3.11]

201.3.223
fractional oxyhaemoglobin
DEPRECATED :FO_2Hb fractional saturation
oxyhaemoglobin concentration cO_2Hb divided by the *total haemoglobin concentration*, *ctHb*, in the blood

$$FO_2\mathrm{Hb} = \frac{cO_2\mathrm{Hb}}{c\mathrm{tHb}}$$

Note 1 to entry: cO_2Hb is the concentration of oxyhaemoglobin; *ctHb* is the *total haemoglobin concentration*.

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Note 2 to entry: This is sometimes reported as a percentage (multiplying the fraction by 100).

Note 3 to entry: *Fractional oxyhaemoglobin* is the term used by the Clinical and Laboratory Standards Institute (CLSI), (formerly NCCLS or National Committee for Clinical Laboratory Sciences) for this ratio.

Note 4 to entry: CLSI denotes "concentration" by a prefixed letter *c*, while in the past the convention of square brackets (e.g. [O_2Hb]) was used.

Note 5 to entry: CLSI[3] uses the following notations:

- oxyhaemoglobin (O_2Hb);
- deoxyhaemoglobin (HHb);
- carboxyhaemoglobin (COHb);
- methaemoglobin (MetHb);
- sulfhaemoglobin (SulfHb); and
- *total haemoglobin concentration* (*ctHb*), which is derived by the cyanmethaemoglobin method of CLSI H15[23].

201.3.224
functional oxygen saturation
percentage saturation given by the oxyhaemoglobin concentration (cO$_2$Hb) divided by the sum of the oxyhaemoglobin concentration and the deoxyhaemoglobin concentration (cHHb)

$$\frac{100 \times c\mathrm{O}_2\mathrm{Hb}}{c\mathrm{O}_2\mathrm{Hb} + c\mathrm{HHb}}$$

Note 1 to entry: The CLSI[3] term for this ratio is haemoglobin oxygen saturation, and its notation is $S\mathrm{O}_2$.

Note 2 to entry: As related to SpO_2, this percent saturation is for arterial blood.

201.3.225
functional tester
test equipment which presents *pulse oximeter equipment* with a signal having a predictable value of *modulation ratio* so that the *operator* can observe the resulting displayed value of SpO_2, and compare it to the expected value derived from the calibration curve for that particular *pulse oximeter equipment*

Note 1 to entry: The *accuracy* of the SpO_2 value given by the *pulse oximeter equipment* depends in part on whether the calibration curve of the *pulse oximeter monitor* properly reflects the optical characteristics of the *pulse oximeter probe* and *pulse oximeter probe*-tissue interaction. *Functional testers* are not able to confirm the SpO_2 *accuracy* of the calibration curve or sufficiently assess the optical characteristics of *pulse oximeter probes* to determine their proper calibration. Additional information is found in EE.4.

Note 2 to entry: Not all *functional testers* and *pulse oximeter equipment* are compatible. *Functional testers* can vary in pulse simulation methods, pulse contours, and amplitude. It is possible that a *functional tester* cannot accurately reproduce the calibration of the *pulse oximeter equipment* and can yield different results between *pulse oximeter equipment*.

201.3.226
healthcare professional, adj
appropriately trained, knowledgeable, and skilled, providing systematic preventive, curative, promotional or rehabilitative healthcare services

[SOURCE: ISO 80601-2-12:2023, 201.3.247]

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201.3.227
HF surgical equipment
medical electrical equipment which generates high frequency currents intended for the performance of surgical tasks, such as the cutting or coagulation of biological tissue by means of these high frequency currents

Note 1 to entry: *HF surgical equipment* is also variously known as surgical diathermy, electrosurgical equipment, electrosurgical generator, RF generator or HF generator.

Note 2 to entry: A footswitch is an example of an associated *accessory* that is part of *HF surgical equipment*.

[SOURCE: IEC 60601-2-2:2017+AMD1:2023, 3.224]

201.3.228
home healthcare environment
dwelling place in which a *patient* lives or other places where *patients* are present, excluding professional healthcare facility environments where *operators* with medical training are continually available when *patients* are present

EXAMPLE In a car, bus, train, boat or plane, in a wheelchair or walking outdoors.

Note 1 to entry: Professional healthcare facilities include hospitals, physician offices, freestanding surgical centres, dental offices, freestanding birthing centres, limited care facilities, first aid rooms or rescue rooms, multiple treatment facilities and emergency medical services.

Note 2 to entry: For the purpose of this document, nursing homes are considered *home healthcare environments*.

Note 3 to entry: Other places where a *patient* is present include the outdoor environment, while working and in vehicles.

[SOURCE: IEC 60601-1-11:2015+AMD1:2020, 3.1, modified — replaced ' collateral standard' with 'document'.]

201.3.229
immunity
ability of *ME equipment* or a *ME system* to perform without degradation in the presence of an electromagnetic disturbance

[SOURCE: IEC 60601-1-2:2014+AMD1:2020, 3.8, modified — deleted 'the'.]

201.3.230
immunity test level
level of a test signal used to simulate an electromagnetic disturbance when performing an *immunity* test

[SOURCE: IEC 60601-1-2:2014+AMD1:2020, 3.9, modified — deleted 'the'.]

201.3.231
individual typology angle
ITA
objective colorimetry measure of skin pigmentation that is inversely related to the amount of pigmentation

$$ITA = \left(\frac{180}{\pi}\right) \bullet \arctan\left(\frac{L^* - 50}{b^*}\right)$$ [8][52]

where

L^* is the luminance ranging from black (0) to white (100); and

b^* is the yellow/blue component.

Note 1 to entry: *ITA* is the vector direction on the L*-b* plane where *ITA* is given in degrees.

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Note 2 to entry: *ITA* is collected with the use of equipment that provides the measure directly or derived from the parameters L* and b* according to Reference [52].

Note 3 to entry: For the purposes of this document, skin pigmentation and the quantification thereof by *ITA* only refers to melanin.

201.3.232
information supplied by the manufacturer
information related to the identification and use of a medical device or *accessory*, in whatever form provided, intended to ensure the safe and effective use of the medical device or *accessory*

Note 1 to entry: e-documentation is included in the *information supplied by the manufacturer*.

Note 2 to entry: Shipping documents (e.g. packing list and customs documents) and promotional material are excluded from *information supplied by the manufacturer.* However, some *authorities having jurisdiction* can consider such supplemental information as *information supplied by the manufacturer*.

Note 3 to entry: The primary purpose of *information supplied by the manufacturer* is to identify the medical device or *accessory* and its *manufacturer*, and provide essential information about its safety, performance, and appropriate use.

[SOURCE: ISO 20417:2026, 3.15, modified — deleted note 4.]

201.3.233
instructions for use
IFU
package insert
portion of the *accompanying information* that is
- directed to the *operator*
- essential for the safe and effective use of a medical device or *accessory*

Note 1 to entry: An *operator* can be either a *lay operator* or professional *operator* with relevant specialized training.

Note 2 to entry: Instructions for the *processing* between uses of a medical device or *accessory* can be included in the *instructions for use*.

Note 3 to entry: The *instructions for use,* or portions thereof, can be located on the display of a medical device or *accessory*.

Note 4 to entry: Medical devices or *accessories* that can be used safely and effectively without *instructions for use* are exempted from having *instructions for use* by some *authorities having jurisdiction*.

[SOURCE: ISO 20417:2026, 3.16, modified — replaced "user" with "*operator*" and deleted note 6.]

201.3.234
label
<medical device, *accessory*> written, printed, or graphic information appearing on the item itself, on the packaging of each item or on the packaging of multiple items

Note 1 to entry: The word *labelled* is used to designate the corresponding act.

Note 2 to entry: *Label* includes the *marking* on the medical device or *accessory*.

Note 3 to entry: Information indicated on a graphical *user* interface (GUI) is considered as appearing on the item.

[SOURCE: ISO 20417:2026, 3.18, modified — deleted note 4.]

201.3.235
lay, adj
term referring to non-professional or professional without relevant specialized training

EXAMPLE *Lay operator*, *lay responsible organization*.

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[SOURCE: IEC 60601-1-11:2015+AMD1:2020, 3.2]

201.3.236
local bias
b
difference between the expectation of a test result (SpO_2) and an accepted reference value (SaO_2)

Note 1 to entry: For *pulse oximeter equipment*, this is, at a given value of the reference oxygen saturation, the difference between the *y*-value of the regression line at that coordinate and the *y*-value of the line of identity, in a plot of SpO_2 versus S_R, or given by:

$$b_i = SpO_{2\,\mathrm{fit},i} - S_{\mathrm{R}i}$$

where

$SpO_{2\mathrm{fit},i}$ is the value of the curve fitted to the test data at the ith reference oxygen saturation value; and

$S_{\mathrm{R}i}$. is the ith reference oxygen saturation value.

Note 2 to entry: Additional information is found with the term *mean bias* and in the discussion in Annex CC.

[SOURCE: ISO 3534-2:2006, 3.3.2, modified — Replaced note 1 to note 3 with new notes and 'or measurement result and the true value' with '(SpO_2) and an accepted reference value (SaO_2)'.]

201.3.237
marking
information, in text or graphical format, durably affixed, printed, etched or by equivalent means to a product

Note 1 to entry: The word *marked* is used to designate the corresponding act.

Note 2 to entry: *Marking* is different from 'direct marking' as commonly described in unique device identification (UDI) standards and regulations. A UDI 'direct marking' is a type of *marking*.

[SOURCE: ISO 20417:2026, 3.22, modified — deleted notes 3 and 4.]

201.3.238
mean bias
B
mean difference between the test and reference values, preserving sign

$$B = \frac{\sum_{i=1}^{n}\left(SpO_{2i} - S_{\mathrm{R}i}\right)}{n}$$

where

n is the number of data pairs in the sample within the range of interest;

SpO_{2i} is the *i*th SpO_2 datum; and

$S_{\mathrm{R}i}$ is the ith reference oxygen saturation value.

Note 1 to entry: Additional information also is found with the term *local bias* and in the discussion in Annex CC.

Note 2 to entry: When defined in this way, *mean bias* is the average of all *local bias* values, b_i.

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201.3.239
modulation ratio
ratio of ratios
R
basic quantity derived by *pulse oximeter equipment* from time-dependent light intensity measurements

Note 1 to entry: *Pulse oximeter equipment* uses an empirical calibration curve to derive SpO_2 from *R*. Additional information is found in EE.4.

Note 2 to entry: *Pulse oximeter equipment* typically uses multiple calibration curves where each curve is derived for range of light emitter wavelengths.

201.3.240
multiple patient multiple use
<medical device, *accessory*> intended by the *manufacturer* to be reused on multiple *patients* for multiple uses

Note 1 to entry: A *multiple patient multiple use* medical device or *accessory* typically requires *processing* between *patients*.

Note 2 to entry: A *multiple patient multiple use* medical device or *accessory* can require *processing* between uses on a single *patient*.

[SOURCE: ISO 20417:2026, 3.26]

201.3.241
normalized
displayed at constant pulse amplitude independent of the actual pulse magnitude of the signal being displayed

201.3.242
objective pigmentation measurement method
PMM that is independent of human interpretation

Note 1 to entry: *Objective pigmentation measurement methods* are generally based on spectroscopic or colorimetric methods that provide *ITA* values (or data such as RGB from which *ITA* can be derived) and are dependent on the use of proper measurement technique.

Note 2 to entry: An *objective pigmentation measurement method* typically is performed by the correct use of a calibrated scientific equipment as described in Reference [4].

201.3.243
operator interface
means by which the *operator* and the *ME equipment* interact

Note 1 to entry: The *accompanying documents* are considered part of the *ME equipment* and its *operator interface*.

Note 2 to entry: *Operator interface* includes all the elements of the *ME equipment* with which the *operator* interacts including the physical aspects of the *ME equipment* as well as visual, auditory, tactile displays and is not limited to a software interface.

Note 3 to entry: For the purposes of this standard, the *manufacturer* may treat the combination of *ME equipment* and other equipment as a single *operator interface*.

Note 4 to entry: See IEC 62366-1:2015, 3.26.

[SOURCE: IEC 60601-1-6:2010+AMD1:2013+AMD2:2020, 3.1]

201.3.244
operator settings
current state of any *pulse oximeter monitor* controls, including *alarm settings*

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201.3.245
percent modulation
%mod
indicator of pulsatile signal strength resulting only from tissue arterial blood content changes at the *pulse oximeter probe* site over the cardiac cycle

Note 1 to entry: *Percent modulation* may be approximated as AC/DC of the detected light signal multiplied by 100, where "AC" refers to the modulated signal portion and "DC" refers to the baseline (non-modulated) signal portion.

Note 2 to entry: *Percent modulation* may be used in referring to different light signals independently, e.g. "infrared *%mod*" and "red *%mod*.

Note 3 to entry: When used to indicate an overall pulsatile signal strength at a *pulse oximetry probe* site, the infrared signal or a signal best representing the isosbestic point for oxyhaemoglobin (O_2Hb) and deoxyhaemoglobin (HHb) absorption should be the primary factor used in determining the value of *%mod*.

201.3.246
physiological alarm condition
alarm condition arising from a monitored *patient*-related variable

EXAMPLE 1 High exhaled anaesthetic agent concentration.

EXAMPLE 2 Low exhaled tidal volume.

EXAMPLE 3 Low oxygen saturation measured by pulse oximetry.

EXAMPLE 4 High arterial pressure.

EXAMPLE 5 High heart rate.

[SOURCE: IEC 60601-1-8:2006+AMD1:2012+AMD2:2020, 3.31]

201.3.247
pigmentation measurement method
PMM
method for quantification or classification of skin pigment

201.3.248
precision
closeness of agreement between independent test results obtained under stipulated conditions

$$s_{\mathrm{res}} = \sqrt{\frac{\sum_{i=1}^{n}\left(SpO_{2i} - SpO_{2\mathrm{fit},i}\right)^2}{\left(n-2\right)}}$$

where

n is the number of data pairs in the sample within the range of interest;

$(SpO_{2i} - SpO_{2\mathrm{fit},i})$ is the difference between the ith SpO_2 datum and the value of the fitted curve corresponding to the ith reference oxygen saturation value, $S_{\mathrm{R}i}$.

Note 1 to entry: This is equivalent to the formula for standard deviation of the residuals, and is considered the relevant metric for *pulse oximeter equipment*. Additional information is found in Annex CC.

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[SOURCE: ISO 3534-2:2006, 3.3.4, modified — Replaced note 1 to note 3 with a new note, deleted '/measurement results' and added formula.]

201.3.249
primary operating function
function that involves user interaction that is related to the safety of the medical device

Note 1 to entry: Often a *primary operating function* is interacted with by a series of tasks that can be broken down into a series of user interactions.

Note 2 to entry: The concept of safety includes loss or degradation of performance resulting in an unacceptable *risk* to the *patient*, including use error that prevents the user from effectively using the medical device to achieve its intended medical purpose. In IEC 60601-1, this is referred to as *essential performance*.

[SOURCE: IEC 62366-1:2015+AMD1:2020, 3.11]

201.3.250
probe cable extender
cable that connects a *pulse oximeter monitor* to a *pulse oximeter probe*

Note 1 to entry: Not every *pulse oximeter equipment* utilizes a *probe cable extender* (e.g. standalone finger clip style).

Note 2 to entry: A *probe cable extender* can be an *applied part*.

201.3.251
probe fault
abnormal condition of the *pulse oximeter probe* or *probe cable extender*

Note 1 to entry: *Probe faults* can cause *patient harm*, if not detected. *Patient harm* can be caused by providing incorrect values, by exposing the *patient* to high *pulse oximeter probe* temperatures or by introducing a *risk* of electric shock.

201.3.252
processing
<preparation of medical device, *accessory*> *processes* to prepare a new or used healthcare product for its *intended use*

Note 1 to entry: *Processing* includes cleaning, disinfection and sterilization, as appropriate.

[SOURCE: ISO 20417:2026, 3.31]

201.3.253
pulse oximeter equipment
ME equipment for the non-invasive estimation of *functional oxygen saturation* of arterial haemoglobin (SpO_2) from a light signal interacting with tissue, by using the time-dependent changes in tissue optical properties that occur with pulsatile blood flow

Note 1 to entry: *Pulse oximeter equipment* comprises a *pulse oximeter monitor*, a *probe cable extender*, if provided, and a *pulse oximeter probe*, which can be combined in a single assembly.

Note 2 to entry: Light is more technically referred to as electromagnetic radiation (optical radiation). This document uses the common term.

201.3.254
pulse oximeter monitor
part of the *pulse oximeter equipment* that encompasses the electronics, display and *operator interface*, excluding the *pulse oximeter probe* and *probe cable extender*

Note 1 to entry: A *pulse oximeter monitor* can consist of multiple pieces of hardware in separate locations (e.g. a telemetry system in which the *applied part* and primary display are in physically separated).

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201.3.255
pulse oximeter probe
part of the *pulse oximeter equipment* that includes the *applied part* and transducer component

Note 1 to entry: The *pulse oximeter probe* typically consists of a cable and a rigid or flexible assembly containing two photo emitters and a photo detector.

201.3.256
remanufacturing
DEPRECATED: reprocessing
any activity, which is not specified in the *accompanying document*, which renders a used product suitable for use

Note 1 to entry: Such activities are often referred to as refinishing, restoring, recycling, refurbishing, repairing or *remanufacturing*.

Note 2 to entry: Such activities can occur in healthcare facilities.

Note 3 to entry: The word "*remanufactured*" is used to designate the corresponding status.

Note 4 to entry: There is guidance or rationale for this definition contained in AA.2.2.

201.3.257
SaO_2
fraction of functional haemoglobin in arterial blood that is saturated with oxygen

Note 1 to entry: Subclause 201.12.1.101.3 provides the acceptable method for determining SaO_2.

Note 2 to entry: Additional information about SaO_2 is found in 201.3.224.

Note 3 to entry: SaO_2 is normally expressed as a percentage (by multiplying the fraction by 100).

201.3.258
security level
level corresponding to the required set of countermeasures and inherent *cybersecurity* properties of devices and systems for a zone or conduit based on assessment of *risk* for the zone or conduit

[SOURCE: IEC/TR 60601-4-5:2021, 3.23, modified — replaced 'security' with 'cybersecurity'.]

201.3.259
single patient multiple use
<*medical device*, *accessory*> intended by the *manufacturer* to be reused on an individual *patient* for multiple uses

Note 1 to entry: A *single patient multiple use medical device* or *accessory* can require *processing* between uses.

Note 2 to entry: For an implantable *medical device*, the duration from implanting to explanting the *medical device* is considered *single use* and not *single patient multiple use.*

[SOURCE: ISO 20417:2026, 3.38]

201.3.260
single use
do not reuse
use only once
<medical device, *accessory*> intended by the *manufacturer* to be used on an individual *patient* or an individual specimen during a single *procedure* and then disposed of

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Note 1 to entry: A *single use* medical device or *accessory* is not intended by its *manufacturer* to be further *processed* and used again.

Note 2 to entry: For an implantable medical device, the duration from implanting to explanting the medical device is considered *single use* and not *single patient multiple use.*

[SOURCE: ISO 20417:2026, 3.39]

201.3.261
SpO_2
estimate of SaO_2 made by *pulse oximeter equipment*

Note 1 to entry: Two-wavelength *pulse oximeter equipment* cannot compensate for the interference caused by the presence of dyshaemoglobins in their estimation of SaO_2[124] [125].

Note 2 to entry: SpO_2 is normally reported as a percentage (multiplying the fraction by 100).

201.3.262
standardized skin colour scale
scale that defines a range of colours that captures a majority of colours seen in human skin and is used as a subjective *PMM*

Note 1 to entry: A *standardized skin colour scale* has specified RGB or equivalent values that can be used by an observer using a properly calibrated printed colour swatch.

Note 2 to entry: The Monk Skin Tone scale (MST) is one such scale[104].

201.3.263
symbol
graphical representation appearing on the *label* or in *accompanying information* of a product that communicates characteristic information without the need for the supplier or receiver of the information to have knowledge of the language of a particular nation or people

Note 1 to entry: The *symbol* can be an abstract representation or a graphical representation, or one that uses familiar objects, including alphanumeric characters (with sufficient justification).

[SOURCE: ISO 20417:2026, 3.42]

201.3.264
technical alarm condition
alarm condition arising from a monitored equipment-related or *alarm system*-related variable

EXAMPLE 1 An electrical, mechanical, or other failure.

EXAMPLE 2 A failure of a probe or component (unsafe voltage, high impedance, signal impedance, artefact, noisy signal, disconnection, calibration error, tubing obstruction, etc.).

EXAMPLE 3 An algorithm that cannot classify or resolve the available data.

[SOURCE: IEC 60601-1-8:2006+AMD1:2012+AMD2:2020, 3.36]

201.3.265
technical description
portion of the *accompanying information* that is
- directed to the *responsible organization* and *service personnel*
- essential for preparation for the first use and safe use, maintenance or repair as well as *processing* transport or storage for the expected lifetime including decommissioning or disposal of a medical device or *accessory*

Note 1 to entry: The *technical description* can be included in the *instructions for use*.

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[SOURCE: ISO 20417:2026, 3.43, modified — deleted note 2.]

201.3.266
total haemoglobin concentration
ctHb
sum of concentrations of all haemoglobin species in the blood including, but not limited to, oxyhaemoglobin (cO_2Hb), methaemoglobin (*c*MetHb), deoxyhaemoglobin (*c*HHb), sulfhaemoglobin (*c*SHb) and carboxyhaemoglobin (*c*COHb)[34]

201.3.267
transfer standard
DEPRECATED: secondary standard
pulse oximeter equipment for non-invasive estimate of SaO_2

Note 1 to entry: Bias and *precision* of the *transfer standard* are directly traceable to *CO-oximeter* SaO_2 data and have been *verified* to assess *accuracy* performance using an error transfer equation. The *transfer standard* is comprised of a *pulse oximeter monitor*, a *probe cable extender*, and a *pulse oximeter probe* (each component specially kept together as a single *transfer standard*). See Annex JJ for details.

Note 2 to entry: A *transfer standard* can be used during *accuracy* performance evaluation of a *pulse oximeter equipment* under test for development purposes.

Note 3 to entry: A *transfer standard* cannot be used in a *controlled desaturation study* for performance *verification* of *accuracy* and *differential bias* [see 201.12.1.101.3 a) 1)].

201.3.268
validation
confirmation, through the provision of *objective evidence*, that the requirements for a specific intended use or application have been fulfilled

Note 1 to entry: The *objective evidence* needed for a *validation* is the result of a test or other form of determination such as performing alternative calculations or reviewing documents.

Note 2 to entry: The term "*validated*" is used to designate the corresponding status.

Note 3 to entry: The use conditions for *validation* can be real or simulated.

[SOURCE: ISO 9000:2015, 3.8.13]

201.4 General requirements

IEC 60601-1:2005+AMD1:2012+AMD2:2020, Clause 4 applies, except as follows:

Additional subclauses:

201.4.3.101 Additional requirements for *essential performance*

NOTE There is guidance or rationale for this subclause contained in AA.2.3.

Additional *essential performance* requirements are found in the subclauses listed in Table 201.101.

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Table 201.101 — Distributed *essential performance* requirements

Requirement	Subclause
For *pulse oximeter equipment* provided with an *alarm system* that includes the capability to detect a *physiological alarm condition*: *SpO_2 accuracy*, pulse rate *accuracy* and limit *alarm conditions* [a] or generation of a *technical alarm condition*	201.12.1.101 201.12.1.104 208.6.1.2.101 201.11.8.101.1 201.12.4.101 201.13.101
For *pulse oximeter equipment* not provided with an *alarm system* that includes the capability to detect a *physiological alarm condition*: *SpO_2 accuracy* and pulse rate *accuracy* [a] or indication of abnormal operation	201.12.1.101 201.12.1.104 201.12.4 201.13.101
[a] Subclause 202.8.2 indicates methods of evaluating *SpO_2 accuracy* and pulse rate *accuracy* as acceptance criteria following specific tests required by this document.	

201.4.101 Additional requirements for acceptance criteria

Many of the clauses and subclauses within this document establish acceptance criteria for performance aspects. These acceptance criteria shall always be met.

When the *manufacturer* specifies in the *accompanying document* performance levels better than those specified within this document, these *manufacturer*-specified levels become the acceptance levels.

EXAMPLE For a specified level of *SpO_2 accuracy* of 1,5 %, the *pulse oximeter equipment* is required to have 1,5 % *SpO_2 accuracy* for all requirements (e.g. during *immunity* tests).

201.4.102 Additional requirements for *pulse oximeter equipment*, parts and *accessories*

a) The *pulse oximeter equipment*, as well as all individual parts and *accessories* specified for use with a *pulse* oximeter *monitor*, shall conform with all requirements specified in this document. This includes all combinations of parts or *accessories* that are specified by a *manufacturer* for use in *pulse oximeter equipment.*

NOTE 1 This requirement ensures *basic safety* and *essential performance* of parts and *accessories* of the *pulse oximeter equipment*, in combination with their intended *pulse oximeter monitors*.

NOTE 2 This requirement applies independent of the *manufacturer* of the individual parts and *accessories* that, when combined, comprise the *pulse oximeter equipment*. *Pulse oximeter monitors* are frequently used with *pulse oximeter probes* and *probe cable extenders* from different *manufacturers*. This requirement ensures the compatibility of such combinations.

b) All specified combinations of *pulse oximeter equipment*, as well as all individual parts and *accessories* specified for use with a *pulse oximeter monitor*, shall be disclosed in the *instructions for use*.

NOTE 3 Additional information is found in 201.7.9.2.1.101 g) and 201.7.9.2.14.101 a) and b).

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201.5 General requirements for testing of *ME equipment*

IEC 60601-1:2005+AMD1:2012+AMD2:2020, Clause 5 applies.

201.6 Classification of *ME equipment* and *ME systems*

IEC 60601-1:2005+AMD1:2012+AMD2:2020, Clause 6 applies.

201.7 *ME equipment* identification, *marking* and documents

IEC 60601-1:2005+AMD1:2012+AMD2:2020, Clause 7 applies, except as follows:

Additional subclause:

201.7.1.101 *Information to be supplied by the manufacturer*

a) The *information supplied by the manufacturer* of a *pulse oximeter equipment* and its *accessories* shall be in accordance with ISO 20417:2026.

b) In applying ISO 20417:2026, the terms in this document and those in IEC 60601-1:2005+AMD1:2012+AMD2:2020 shall be used as follows.

   1) The term "*accompanying information*" assumes the same meaning as *accompanying documents*.

   2) The term "*medical device*" assumes the same meaning as *ME equipment*.

   3) The term "*user*" assumes the same meaning as *operator*.

Check conformity by application of ISO 20417:2026.

201.7.2.3 Consult *accompanying documents*

Replacement:

The *pulse oximeter equipment* shall be *marked* with the *safety sign* for the mandatory action: "follow instructions for use", ISO 7010-M002 (see IEC 60601-1:2005+AMD1:2012+AMD2:2020, Table D.2, Number 10).

Additional subclauses:

201.7.2.4.101 Additional requirements for *accessories*

a) *Accessories* supplied separately shall:

   1) fulfil the *marking* requirements of ISO 20417:2026, 6.1.1 c); and

   2) be *marked* with an indication of any limitations or adverse effects of the *accessory* on the *basic safety* or *essential performance* of the *pulse oximeter monitor*, if applicable. See also 201.7.2.101.

b) If *marking* the *accessory* is not practicable, this information may be placed in the *instructions for use*.

Check conformity by inspection, inspection of the *risk management file* for any limitations or adverse effects of the *accessory*, if applicable, and where necessary, inspection of the *instructions for use*.

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201.7.2.9.101 IP classification

a) Notwithstanding the requirements of IEC 60601-1:2005+AMD1:2012+AMD2:2020, 7.2.9, the *enclosure* of *ME equipment* shall be *marked* with the IP classification required by 201.11.6.5.101.

b) If some or all of the protection against the ingress of water or particulate matter is provided by a carrying case, then:

   1) the degree of protection provided by the *enclosure* shall be *marked* on the *enclosure*; and

   2) the degree of protection provided by the carrying case shall be *marked* on the carrying case.

   EXAMPLE If for *portable ME equipment*, the *enclosure* provides the protection against the ingress of particulate matter and the carrying case provides the protection against the ingress of water, the *enclosure* of the *ME equipment* would be *marked* IP2X and the carrying case would be *marked* IPX2.

c) An *enclosure* or a carrying case that is classified IPX0 need not be *marked* as such.

d) If an *enclosure* does not provide the minimum required degree of protection against the ingress of water, it shall be *marked*:

   1) 'keep dry'; or

   2) with ISO 15223-1:2021, *Symbol* 5.3.4 (see Table 201.D.1.101, *Symbol* 1).

Check conformity by inspection and by application of the tests and criteria of IEC 60601-1:2005+AMD1:2012+AMD2:2020, 7.1.2 and 7.1.3.

201.7.2.101 Additional requirements for *marking* on the outside of *ME equipment* parts

ME equipment, parts or *accessories* shall be *marked* as follows.

a) If a *pulse oximeter monitor* is not provided with a low *SpO_2 alarm condition*, it shall be *marked* with a statement to the effect "No SpO_2 Alarms" or *Symbol* IEC 60417-5319 (DB-2002-10) (see IEC 60601-1-8:2006+AMD1:2012+AMD2:2020, Table C.1, *Symbol* 3).

b) A *remanufactured pulse oximeter probe* shall be *marked* as such.

c) These *markings* shall be *clearly legible*.

Check conformity by inspection.

201.7.4.3 Units of measurement

Amendment (add to the bottom as a new row in Table 1):

aa) *Functional oxygen saturation* shall be expressed in units of percent *SpO_2* and shall be *marked* as % *SpO_2* or *SpO_2*.

bb) Pulse rate shall be expressed in units of reciprocal minutes (1/min).

Additional subclauses:

201.7.9.2.1.101 Additional general requirements

The *instructions for use* shall indicate the following:

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a) that the *pulse oximeter equipment* is calibrated to estimate *functional oxygen saturation*;

b) the range of the peak wavelengths and maximum optical output power of the light emitted by the *pulse oximeter probe* and a statement to the effect that information about wavelength range can be especially useful to clinicians;

EXAMPLE 1 Clinicians performing photodynamic therapy.

c) description of the effect on displayed and transmitted SpO_2 and pulse rate data values, including the effects of any selectable operating mode that affects these properties, by:

1) the data averaging and other signal processing;

2) the *data update period*;

3) the *alarm condition delay*; and

4) *alarm signal generation delay;*

NOTE 1 Annex FF provides an example of how to assess and describe response time graphically.

d) the *displayed ranges* of SpO_2.;

e) the *displayed ranges* of pulse rate;

f) for a *pulse oximeter monitor* without an *alarm system* that includes the capability to detect an SpO_2 or pulse rate *physiological alarm condition*, a statement to that effect; and

g) for a *pulse oximeter monitor*, the *pulse oximeter probes* and *probe cable extenders* with which the *pulse oximeter monitor* has been *verified* and tested for conformity with this document.

1) The list may be made available by electronic means.

NOTE 2 Additional information is found in 201.4.102.

Check conformity by inspection of the *instructions for use*.

201.7.9.2.2.101 Additional requirements for warnings and safety notices

The *instructions for use* shall include:

a) for each *pulse oximeter probe* and *probe cable extender*, a warning to the effect that probes and cables are designed for use with specific monitors;

b) a warning statement to the effect that the responsible organization or operator needs to verify the compatibility of the pulse oximeter monitor, pulse oximeter probe and probe cable extender before use to prevent patient injury or inaccurate readings;

c) a warning statement to the effect that misapplication of a pulse oximeter probe with excessive pressure for prolonged periods can induce pressure injury; and

d) for *pulse oximeter equipment* with *fingertip pulse oximeter probes*, a warning statement to the effect that artificial acrylic nails, gel-based manicures, nail jewellery, specific colours of nail polish, varied nail colour patterns, long nails and bruised nail beds interfere with the optical pathway or probe placement and can cause inaccurate readings[56][139].

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Such ornamentations can result in finger-tip *pulse oximeter probe* SpO_2 values that differ from SaO_2, which can cause a missed detection of hypoxaemia or the initiation of unnecessary therapy. Many investigators who studied the phenomena concluded that these nail types should be routinely removed or an alternative *pulse oximeter probe* site should be sought. Additionally, when excessive nail length has the *pulse oximeter probe* optics no longer over an area of pulsing arterial blood, the SpO_2 value cannot reflect SaO_2.

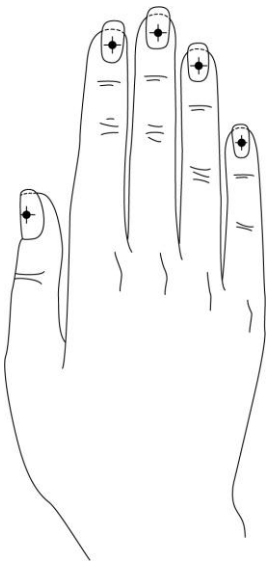
Check conformity by inspection of the *instructions for use.*

201.7.9.2.9.101 Additional requirements for operating instructions

The *instructions for use* shall indicate the following.

a) A description with graphic indicating proper orientation and placement of the *pulse oximeter probe*.

   1) Proper positioning shall be visually confirmed by the user, which for a finger site typically is the emitter over the nailbed centrum. See Figure 201.101 for a typical example finger probe graphic.



Key

✦ Fingertip *probe* locations for optimal performance.

Figure 201.101 — Example finger *pulse oximeter probe* locations and orientation

b) A statement that the signal inadequacy indicator cannot identify all sources of error that result in potentially incorrect SpO_2 or pulse rate.

c) The recommended maximum application time for each type of *pulse oximeter probe* at a single site.

d) The IP classification and meaning of the IP classification of the *pulse oximeter equipment enclosure* and,

   1) if applicable, on any carrying case provided with the *pulse oximeter equipment.*

      EXAMPLE IPX2 means this *pulse oximeter equipment* is protected against harmful effects of dripping water when tilted at 15° according to IEC 60529.

e) *Pulse oximeter equipment* is not typically capable of, nor intended as a means for, the measurement of blood flow or blood volume.

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f) If the *pulse oximeter equipment* is provided with temperature capability such that the *pulse oximeter probe* can operate at a *pulse oximeter probe* contact surface temperature greater than 41 °C, specific instructions emphasizing the importance of proper *pulse oximeter probe* application, without excessive pressure.

   1) In addition, specific instructions for any changes in recommended maximum application time when using temperatures greater than 41 °C.

Check conformity by inspection of the *instructions for use*.

201.7.9.2.14.101 Additional requirements for *accessories*, supplementary equipment, used material

The *instructions for use* shall include the following.

a) For *pulse oximeter probes*, the *pulse oximeter monitors* and *probe cable extenders* with which the *pulse oximeter probes* have been *verified* and tested for conformity with this document.

   1) The list may be made available by electronic means.

      NOTE 1 Additional information is found in 201.4.102.

b) For *probe cable extenders*, the *pulse oximeter monitors* and *pulse oximeter probes* with which the *probe cable extenders* have been *verified* and tested for conformity with this document.

   1) The list may be made available by electronic means.

      NOTE 2 Additional information is found in 201.4.102.

   2) Any reasonably foreseeable adverse effect of any recommended *accessory* on the *basic safety* or *essential performance* of the *pulse oximeter monitor.*

      NOTE 3 Additional requirements are found in 201.16.

Check conformity by inspection of the *instructions for use* and inspection of the *risk management file* for any adverse effect of any recommended *accessory*.

201.7.9.3.1.101 Additional general requirements

NOTE 1 There is guidance or rationale for this subclause contained in AA.2.4.

a) The *technical description* shall include a statement to the effect that a *functional tester* cannot be used to assess the *accuracy* of a *pulse oximeter probe* or a *pulse oximeter monitor*.

   NOTE 2 Additional information is found in Annex EE.

b) The *technical description* should provide descriptions on how the *responsible organization* can verify operation of the *pulse oximeter equipment*.

   1) If the use of a *functional tester* is specified, the *technical description* should indicate the *model or type reference* and its software unique identifier, if applicable, of at least one *functional tester* that is compatible with the basic functions of the *pulse oximeter equipment* (e.g. it can ensure that the *pulse oximeter equipment* generates a display of SpO_2 and pulse rate).

Check conformity by inspection of the *technical description*.

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201.8 Protection against electrical *hazards* from *ME equipment*

IEC 60601-1:2005+AMD1:2012+AMD2:2020, Clause 8 applies, except as follows:

Additional subclause:

201.8.3.101 Additional requirements for classification of *applied parts*

Applied parts of *pulse oximeter equipment* shall be:

a) *type BF applied parts*; or

b) *type CF applied parts*.

Check conformity by inspection.

201.9 Protection against mechanical *hazards* of *ME equipment* and *ME systems*

IEC 60601-1:2005+AMD1:2012+AMD2:2020, Clause 9 applies.

201.10 Protection against unwanted and excessive radiation *hazards*

IEC 60601-1:2005+AMD1:2012+AMD2:2020, Clause 10 applies, except as follows:

201.10.4 Lasers

Replacement:

aa) Depending on the light source used in a *pulse oximeter probe*, the relevant requirements of IEC 60825-1:2014 shall apply to a *pulse oximeter probe*.

bb) In the case of laser fibre optics, the requirements of IEC 60825-2:2021 shall apply.

Check conformity by application of the requirements of IEC 60825-1:2014 and IEC 60825-2:2021, as applicable.

201.11 Protection against excessive temperatures and other *hazards*

IEC 60601-1:2005+AMD1:2012+AMD2:2020, Clause 11 applies, except as follows:

201.11.1.2.2 *Applied parts* not intended to supply heat to a *patient*

Amendment (add at the end of subclause 11.1.2.2):

aa) The *pulse oximeter probe*-tissue interface shall be evaluated when the skin temperature is initially at 35 °C for each *pulse oximeter monitor* and *pulse oximeter probe* indicated in the *instructions for use*.

NOTE 100 Additional information is found in Annex BB.

bb) If the surface temperature of the *pulse oximeter probe* at the tissue interface is capable of exceeding 41 °C, then

1) the *pulse oximeter equipment* shall have an *operator*-adjustable control for activating any elevated temperature mode that permits the tissue interface to exceed 41 °C.

i) A deliberate sequence of *operator* actions shall be required to activate this mode.

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ii) The *instructions for use* shall describe this sequence of *operator* actions.

2) the *pulse oximeter equipment* shall provide a means to limit the duration of an elevated temperature mode that permits the tissue interface temperature to exceed 41 °C.

i) The duration of the elevated temperature mode shall not exceed 4 h at 43 °C or 8 h at 42 °C.

3) the *instructions for use* shall include a statement to the effect that the use of temperature settings greater than 41 °C requires special attention in *patients* with susceptible skin, such as neonates, geriatric *patients* or burn victims.

4) the *pulse oximeter equipment* shall indicate when it is in the elevated temperature mode.

5) the *technical description* shall describe the test method used to measure the maximum temperature at the *pulse oximeter probe*-tissue interface.

i) When performing the temperature measurements for the *pulse oximeter probe*-tissue interface, as specified in IEC 60601-1:2005+AMD1:2012+AMD2:2020, 11.1.3, the test method disclosed in the *technical description* may be utilized.

NOTE 101 Additional information is found in BB.3.

Check conformity by functional testing and inspection of the *technical description*.

Additional subclauses:

201.11.6.5.101 Additional requirements for ingress of water or particulate matter into *ME equipment* or *ME system*

NOTE 1 There is guidance or rationale for this subclause contained in AA.2.5.

a) The *enclosure* of *transportable pulse oximeter equipment* shall provide a degree of protection to the harmful ingress of water of at least an IPX2 for *pulse oximeter equipment.*

1) For *portable ME equipment* that is only intended to be used within a protective case, this requirement may be met while the *ME equipment* is inside the case.

b) Notwithstanding a), *pulse oximeter equipment* incorporated as a module into an *enclosure* of another *ME equipment* (e.g. a multifunction *patient* monitor or ventilator) or a module is incorporated in a cable and connected to another *ME equipment*, the requirements for ingress protection of that other *ME equipment* shall apply.

NOTE 2 Additional requirements for ingress of water or particulate matter into the *enclosure* of *pulse oximeter equipment* intended for use in the *home healthcare environment* are found in IEC 60601-1-11 and in the *emergency medical services environment* are found in IEC 60601-1-12.

Check conformity in accordance with the tests of IEC 60529:2013 with the *pulse oximeter equipment* placed in the least favourable position of *normal use* and by inspection. After these *procedures*, confirm that *basic safety* and *essential performance* are maintained.

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201.11.8.101 Additional requirements for interruption of the power supply/*supply mains* to *ME equipment*

201.11.8.101.1 *Technical alarm condition* for power supply failure

a) If *pulse oximeter equipment* is equipped with an *alarm system* that detects a *physiological alarm condition*, the *alarm system* shall provide at least a *medium priority technical alarm condition* to indicate when the power supply falls outside the values specified for normal operation.

NOTE After the loss of power, the *alarm system* is not expected to repeat *alarm signals* indefinitely.

b) If the function of the *pulse oximeter equipment* is maintained by the switchover to an *internal electrical power source*, the supply failure *medium priority technical alarm condition* shall not be activated.

1) Any such switchover to an *internal electrical power source* shall be indicated by an *information signal* or a *low priority technical alarm condition*.

Check conformity by functional testing.

201.11.8.101.2 Settings and data storage following short interruptions or automatic switchover

When the *supply mains* to the *pulse oximeter equipment* is interrupted for less than 30 s or automatic switchover to an *internal electrical power source* occurs, the following shall be preserved unchanged:

a) all settings; and

b) all stored *patient* data.

NOTE 1 The *pulse oximeter equipment* does not have to continue to operate during the interruption of the *supply mains*.

NOTE 2 Settings include *operator settings*, *responsible organization* settings, and the mode of operation.

Check conformity by observing the *pulse oximeter equipment* settings and stored *patient* data and then interrupting the *supply mains* for a period of between 25 s and 30 s by disconnecting the *power supply cord*. After reestablishment of power, confirm that the above settings and stored data are the same.

201.11.8.101.3 Operation following long interruptions

The *instructions for use* shall disclose the operation of the *pulse oximeter equipment* after the *supply mains* has been interrupted when the “on-off” switch remains in the "on" position and is restored after a period of time longer than 30 s.

Check conformity by inspection of the *instructions for use*.

201.12 *Accuracy* of controls and instruments and protection against hazardous outputs

IEC 60601-1:2005+AMD1:2012+AMD2:2020, Clause 12 applies, except as follows:

201.12.1 *Accuracy* of controls and instruments

Additional subclauses:

201.12.1.101 *SpO_2 accuracy* of *pulse oximeter equipment*

NOTE There is guidance or rationale for this subclause contained in AA.2.6.

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201.12.1.101.1 Specification of *SpO_2 accuracy*

NOTE There is guidance or rationale for this subclause contained in AA.2.7.

a) The *SpO_2 accuracy* of *pulse oximeter equipment* shall be a root-mean-square difference (A_{rms}) of less than or equal to 3,0% *SpO_2* with 95 % confidence over the range of 70 % to 100 % *SaO_2*.

   1) The specified *SpO_2 accuracy* over the range 70 % *SaO_2* to 100 % *SaO_2* for each combination of *pulse oximeter probe* and *pulse oximeter monitor* shall be disclosed in the *instructions for use.*

   2) The *SpO_2* shall be indicated as functional oxygen saturation.

   3) The *SpO_2* shall not be indicated as *fractional oxyhaemoglobin*.

   4) *SpO_2 accuracy* information shall be accompanied by a disclosure indicating to the effect that because pulse oximeter equipment measurements are statistically distributed, only about two-thirds of pulse oximeter equipment measurements in a larger population can be expected to fall within $\pm A_{\mathrm{rms}}$ and about 95 % falling within ±2 times A_{rms} of the value measured by a CO-oximeter.

   5) *SpO_2 accuracy* information shall be accompanied by a disclosure indicating to the effect that the A_{rms} value is determined in a controlled desaturation study using specific probes and monitor combinations that is performed on healthy participants and represents the best-case scenario with real-world clinical inaccuracy expected to be worse than the A_{rms} value disclosed from the controlled desaturation study[42].

   6) When a *pulse oximeter monitor* is suitable for use with a variety of *pulse oximeter probes*, *SpO_2 accuracy* information shall be made available for each type of *pulse oximeter probe*.

b) Additional subrange *SpO_2 accuracy* specifications over other ranges may also be provided in the *instructions for use* as long as the range is greater than or equal to 10 % *SpO_2*.

   EXAMPLE 1 A specified *SpO_2 accuracy* of ±3 % for 70 % to 90 % *SaO_2*.

c) If subrange *SpO_2 accuracy* claims below 65 % *SaO_2* are made, *SpO_2 accuracy* shall be stated in an additional range over a span of saturation not to exceed 20 % *SaO_2*.

   EXAMPLE 2 A specified *SpO_2 accuracy* range of 60 % to 80 % *SaO_2*.

Check conformity by following the requirements of 201.12.1.101.3 and by inspection of the *instructions for use*.

201.12.1.101.2 Pigmentation *differential bias*

Using the objective pigmentation value at the probe site, the absolute value of the pigmentation *differential bias* of *pulse oximeter equipment* across populations should be less than or equal to 2,5 % with 95 % confidence for the interval between 80 % and 90 % *SaO_2*, centred at 85 %.

NOTE *Pulse oximeter equipment* with smaller absolute values of *differential bias* are desirable.

201.12.1.101.3 Data collection for determination of *SpO_2 accuracy* and pigmentation *differential bias*

NOTE 1 There is guidance or rationale for this subclause contained in AA.2.8.

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a) The claims of *SpO_2 accuracy* shall be supported by *controlled desaturation study* measurements taken at least over the full range of *SaO_2* values +3 % of the lower value and -3 % of the upper value for which *SpO_2 accuracy* is claimed.

EXAMPLE 1 A *controlled desaturation study* supporting a claimed range of *SpO_2 accuracy* from 70 % *SaO_2* to 100 % *SaO_2* can be supported with *SaO_2* data collected over the range of 73 % *SaO_2* to 97 % *SaO_2*.

EXAMPLE 2 A *controlled desaturation study* supporting a claimed range of *SpO_2 accuracy* from 70 % *SaO_2* to 100 % *SaO_2* can be supported with *SaO_2* data collected over the range of 71 % *SaO_2* to 100 % *SaO_2*.

1) *SaO_2* shall be determined by invasive sampling of arterial blood and measured using a *CO-oximeter*.

b) The *controlled desaturation study* shall be performed in accordance with the requirements of:

1) Annex DD; and

2) ISO 14155:2026.

c) The *residual risk* inherent in a controlled hypoxia study on healthy adult volunteers, can be reduced to a non-significant level by following recommended additional *procedures*[39] (see also Annex AA.2.8).

d) The *accuracy* of *pulse oximeter equipment* for paediatric *patients* shall be supported via *controlled desaturation study* measurements on adult participants.

NOTE 2 Paediatric participants are a vulnerable population.

e) Data points should be collected with comparable density over the full range claimed.

NOTE 3 Additional information is found in Annex DD.

f) Any types of interference known to influence or affect the *SpO_2 accuracy*

1) need not be stated as part of the *SpO_2 accuracy* specification, but

2) shall be disclosed in the *instructions for use*.

EXAMPLE 3 Ambient light (including photodynamic therapy); physical movement (*patient's* voluntary or imposed motion); *procedures* being performed on the *patient*; low perfusion; electromagnetic interference; *HF surgical equipment*; dysfunctional haemoglobin; presence of certain dyes; inappropriate positioning of the *pulse oximeter probe*.

g) A summary of the test methods used to establish the *SpO_2 accuracy* claims shall be disclosed in the *technical description*.

h) *Functional testers* or *patient* simulators shall not be used for *verification* the *SpO_2 accuracy* of *pulse oximeter equipment*.

NOTE 4 Some *functional testers* or *patient* simulators can be used to verify aspects of design changes of a *pulse oximeter monitor* but not a *pulse oximeter probe*. Additional information is found in Annex EE.

Check conformity by following the requirements of Annex DD in accordance with ISO 14155:2026 and by inspection of the *instructions for use* and *technical description*.

201.12.1.101.4 Data analysis for determination of *SpO_2 accuracy*

NOTE 1 There is guidance or rationale for this subclause contained in AA.2.9.

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a) For each range specified, *SpO_2 accuracy* of the *pulse oximeter equipment* shall be stated in terms of the root-mean-square (rms) difference between measured values (SpO_{2i}) and reference values of SaO_2 determined *from CO-oximetry* (S_{Ri}), as given by Formula (1).

$$A_{\mathrm{rms}} = \sqrt{\frac{\sum_{i=1}^{n}\left(SpO_{2i} - S_{\mathrm{R}i}\right)^2}{n}} \qquad (1)$$

NOTE 2 The concepts of bias and precision as given in ASTM E456-13a[20] and ambiguity as given in Reference[127] also have value in representing the *accuracy* of *ME equipment*. The decision to require the form of *SpO_2 accuracy* stated above (which has been traditional in pulse oximetry, although under the misnomer "standard deviation") is based on the belief that it is more widely understood by the general community of clinical *operators* and on the recognition that in some cases it represents the overall *SpO_2 accuracy* of *pulse oximeter equipment* better than do bias and *precision*.

NOTE 3 Additional information regarding the concepts of bias and precision can be found in ISO/IEC Guide 99 (VIM)[10] and ISO/IEC Guide 98-3 (GUM)[9].

b) The *SpO_2 accuracy, mean bias* and *precision* over the range 70 % SaO_2 to 100 % SaO_2 for each combination of *pulse oximeter probe* and *pulse oximeter monitor* as determined by a *controlled desaturation study* shall be disclosed in the *technical description.*

1) The 95 % confidence interval for characterizing the uncertainty of the estimate of *SpO_2 accuracy* shall be disclosed in the *technical description*.

2) The upper 95 % confidence limit of the observed A_{rms} shall be disclosed in the *technical description* [see 201.12.1.101.1 a)]. This is equivalent to 95 % confidence in a one-sided test.

3) Ensure that the disclosed *SpO_2 accuracy* as determined by the *controlled desaturation study* is less than the specified *SpO_2 accuracy* indicated in the *instructions for use.*

c) A linear regression (SpO_2 versus SaO_2) plot from 70 % to 100 % SaO_2 shall be generated and disclosed in the *technical description* with:

1) the *mean bias* and A_{rms}, for the pooled participant data indicated;

2) the slope;

3) the intercept; and

4) the standard estimate error of the slope.

d) A modified Bland-Altman plot[44] [i.e. (SpO_2- SaO_2) versus SaO_2][35][87] shall be provided in the *technical description* for each combination of *pulse oximeter probe* and *pulse oximeter monitor* listed in the *instructions for use* . See Figure CC.6 B for an example. This plot shall include:

NOTE 4 There is guidance or rationale for this list item contained in AA.2.9.

1) each data point plotted identifying individual participants; and

2) the upper and lower 95 % limits of agreement, using methods to account for repeated measurements.

NOTE 5 As saturation decreases, *local bias* can be amplified. This can impact the monitoring of an SpO_2 trend.

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e) A modified Bland-Altman plot[44] [i.e. (SpO_2-SaO_2) versus SaO_2] shall be provided in the *technical description* for each combination of *pulse oximeter probe* and *pulse oximeter monitor* listed in the *instructions for use*. See Figure CC.6 A for an example. This plot shall include:

1) visually distinct representation of each category of light, medium and dark as grouped by the standardized perceived skin colour scale during enrolment of the study; and

2) upper and lower 95% limits of agreement, using methods to account for repeated measurements.

f) The SpO_2 *accuracy*, *mean bias* and *precision* for each combination of *pulse oximeter probe* and *pulse oximeter monitor* shall be disclosed in the *instructions for use* over the SaO_2 subranges:

1) 70 % to 80 %;

2) 80 % to 90 %; and

3) 90 % to 100 %.

g) The standard reference for SpO_2 *accuracy* shall be SaO_2 values obtained from *CO-oximeter* analysis of simultaneously time matched arterial blood specimens.

NOTE 6 Issues to consider in simultaneous time matching of values include physiological and signal processing delays.

NOTE 7 SpO_2 *accuracy* can theoretically be determined against a *pulse oximeter* (SpO_2) *transfer standard* instead of a *CO-oximeter* (SaO_2) but this technique requires extensive *validation* of the method and has not yet (as of this writing) been shown to successfully demonstrate *essential performance* (A_{rms}).

1) The *CO-oximeter* should have a specified SaO_2 performance of 1 % (1 standard deviation) or better over the range for which the *manufacturer* makes SpO_2 *accuracy* claims.

NOTE 8 There is significant variation between types and brands of *CO-oximeters*[64]. A claim of SpO_2 *accuracy* of less than 2% is not justifiable unless a *CO-oximeter* of an *accuracy* of 0,5% or better has been used as the reference for *verification* of the claim.

2) Quality assurance, including maintenance and calibration, *procedures* for assessing *CO-oximeter* performance that are required in laboratories reporting clinical data shall be utilized for the *CO-oximeter*.

3) Particular attention shall be given to the range for which the *manufacturer* makes SpO_2 *accuracy* claims.

NOTE 9 It is not appropriate to use SaO_2 values calculated from measurements made by blood gas analysers that actually measure PaO_2 (arterial oxygen pressure) rather than SaO_2.

NOTE 10 Additional information is found in Annex DD.

Check conformity by inspection of the study report, *instructions for use* and *technical description*.

201.12.1.101.5 Data analysis for pigmentation *differential bias* determination

NOTE 1 There is guidance or rationale for this subclause contained in AA.2.10.

a) Pigmentation *differential bias* shall be determined from the data collected in the *controlled desaturation study* for the determination of SpO_2 *accuracy* (see 201.12.1.101.3).

b) Using a linear model of participant SpO_2 bias versus *ITA* at the *pulse oximeter probe* site, estimate the pigmentation *differential bias* and upper and lower 95 % confidence limits for a 100 degree change in *ITA* in each SaO_2 range.

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NOTE 2 Subclause CC.5.1 describes methods for estimating the pigmentation differential bias.

NOTE 3 Performing two one-sided tests (TOST) at 95 % can be considered equivalent to performing a two-sided test at 90 % confidence, and meets the use of a recommended 95 % confidence limit.

NOTE 4 When mean bias is modelled as linear in *ITA*, the estimate for a 100 degree change in *ITA* is the same for any two *ITA* values (e.g. -50° and 50°, that are 100 degrees apart).

c) The 95 % confidence limits of the pigmentation *differential bias* at 85 % SaO_2 (or the average in the interval of 80 % to 90 % SaO_2) shall be used in assessing performance per 201.12.1.101.2 (specification of *differential bias*).

d) Pigmentation *differential bias* shall be taken as the difference between dark and light skin (e.g. the nominal SpO_2 bias value at -50° *ITA* minus the value at +50° *ITA*).

e) The signed pigmentation differential bias shall be disclosed in graphical format in the *technical description* as follows.

   1) A first graphical format indicating each participant's SpO_2 mean bias or *local bias* over a span of SaO_2 that is centred at 85 % SaO_2 plotted versus *ITA* (see Figure CC.8).

   2) A second graphical format indicating the pigmentation *differential bias* vs SaO_2 and 95 % confidence limits across the 70 % to 100 % SaO_2 range (see Figure CC.9).

f) The mean difference of SpO_2 reading bias between participants in the dark MST categories (H,I,J) and the light categories (A,B,C) shall be disclosed in the *technical description* for observed data in the ranges of:

   1) 70 % to 80 %, SaO_2;

   2) 80 % to 90 % SaO_2; *and*

   3) 90 % to 100 % SaO_2.

Check conformity by inspection of the study report and *technical description*.

201.12.1.101.6 Requirements of the study population and collected data for the determination of SpO_2 *accuracy* and pigmentation *differential bias*

NOTE 1 There is guidance or rationale for this subclause contained in AA.2.11.

The study population and collected data of a *controlled desaturation study* data for the determination of SpO_2 *accuracy* shall include the following.

a) Include a sample size of at least 24 participants.

   NOTE 2 There is guidance or rationale for this list item contained in AA.2.11.

   1) For a *single use* or *single patient multiple use pulse oximeter probe*, use a different *pulse oximeter probe* of the same design for each participant.

   2) For a *multiple patient multiple use pulse oximeter probe*, use multiple *pulse oximeter probes* of the same design across participants.

      i) The number of *pulse oximeter probes* used shall be described in the study report.

   3) Where component variation (e.g. LED wavelength, bandage optical properties) can influence performance, these should be considered when selecting samples for testing.

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4) The study report shall detail the means by which participants were selected into the study population and included in data analysis in accordance with 201.12.1.101.6.

b) Include an average of 20 (±4) data pairs per participant with a maximum data point range of 16 data points to 30 data points per participant.

1) When more than 30 usable data pairs are collected for one participant, use the first valid 10 data pairs in each SaO_2 decade.

c) Include a distribution of data pairs, with total pairs split across each desaturation decade between 70 % to 100 % saturations, with approximately 33 %, but no less than 25 %, in each decade. The data pairs comprising each decade are obtained from participants in each of the MST and *ITA* cohort as specified in f) 7). See also f) 9).

d) 90 % or greater of participants shall provide at least one useable data pair with an SaO_2 below 85 %.

e) At least 75 % of participants shall also have provided at least one useable data pair with an SaO_2 between 70 % to 80 %.

f) Include participants with diversity of skin pigment[132] and gender as evidenced by the following.

NOTE 3 There is guidance or rationale for this list item contained in AA.2.11.

1) Include at least 33 % of each biological sex.

2) Use a standardized perceived skin colour scale to assess the investigator perceived skin tone at the forehead to assure enrolment of skin tone diversity.

3) A *standardized skin colour scale* shall be represented by printed colour swatches.

NOTE 4 There is guidance or rationale for this list item contained in AA.2.11.

4) The *standardized skin colour scale* printed colour swatches shall be *verified* using a calibrated colorimeter against target values and tolerances, prior to use.

i) To print colour swatches, professional photo-grade printers with a wide colour gamut should be used.

EXAMPLE Representative colorimeters include the CM-700d from Konica-Minolta and Colorreader Pro from Datacolor.

NOTE 5 Benchmark values for printed MST scales are shown in Table 201.103 and used as target values for printed MST. Benchmark values for printed MST scales are derived from a sample of MST colour swatches printed on professional printers in different countries.[133] The L* a* b* values in the Table 201.103 differ from the hexadecimal RGB equivalents by small amounts due to practical limitations in printing.

5) L*, a* and b* values measured with a calibrated colorimeter should be within the tolerances shown in Table 201.103 for all MST shades.

i) The L* value of shade C should be at least 3 units larger than that of shade D.

ii) The L* value of shade G should be at least 6 units larger than that of shade H.

6) A factory printed MST scale should be used.

i) The MST scale should be in accordance with the tolerances of Table 201.103.

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7) Ensure an overall study cohort is comprised of at least 25 % of the participants in each of the following MST and *ITA* categories (see Table 201.102):

NOTE 5 There is guidance or rationale for this list item contained in AA.2.11.

i) MST from A, B and C and *ITA* greater than 30;

ii) MST from D, E, F and G and *ITA* between 30 and -30; and

iii) MST from H, I and J and *ITA* less than -30 (with at least one half of *ITA* less than -30 having an *ITA* less than -50).

8) Use *objective pigmentation measurement method ITA* to assess skin pigment at the forehead to assure skin tone distribution criteria are met as specified in Table 201.102.

9) If a participant has *ITA* and MST that do not place this participant in the same category, use the *ITA* categorization.

Table 201.102 — *ITA* and MST criteria for study enrolment

Category	MST on forehead[a]	*ITA* on forehead degrees
Light	A, B, C	> 30
Medium	D, E, F, G	30 ≥ *ITA* ≥ -30
Dark	H, I, J	< -30[a]

[a] See f) 7).

10) The number of data pairs for a given SaO_2 decile (i.e. 70 % to 80 %, 80 % to 90 % and 90 % to 100 %) should be derived approximately equally from participants across skin pigment categories, ensuring that >25 % of data points come from light, medium and dark participants by MST and *ITA*.

11) Use three observers, blinded to each other's readings. In the case of disagreement, when 2 of 3 values are the same, use that value. When three values are different, use the middle value.

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Table 201.103 —MST values and tolerances

MST shade	Hexadecimal code[104]	Targets[135]			Tolerances[135]		
		L*	a*	b*	L*	a*	b*
A	F7EDE4	88,3	2,0	-0,2	5,6	3,3	8,4
B	F3E7DA	86,4	1,9	2,0	6,5	3,3	8,4
C	F6EAD0	86,8	0,8	7,8	5,9	3,9	9,6
D	EAD9BB	81,8	1,5	11,0	7,4	3,5	10,7
E	D7BD96	73,3	3,2	18,1	8,4	3,9	10,7
F	9F7D54	52,4	5,9	21,5	8,2	5,8	10,3
G	815D44	41,6	7,8	14,5	7,6	5,2	7,4
H	604234	32,5	6,9	8,2	7,3	5,3	5,7
I	3A312A	25,4	1,1	2,5	8,0	4,1	4,4
J	2A2420	21,4	0,5	0,7	9,3	3,6	4,0

12) Ensure that the observers are not colour-blind as determined by optometric or on-line testing (e.g. EnChroma2 or equivalent).

13) Use a full spectrum lamp, colour temperature of 6 500 K and a CRI of 96 % for illumination for the MST determination, if possible. Use diffuse illumination to avoid shadows on the face of the participant.

14) Measure the *ITA* values at the forehead, ventral bicep and the *pulse oximeter probe* measurement site.

NOTE 6 Forehead skin is sun exposed and ventral bicep skin is typically not sun exposed.

i) If measurement at the *pulse oximeter probe* site with a conforming colorimeter is not practical, then use a site that is most representative of the darkest skin tone of the *pulse oximeter probe* site.

ii) In the case of a fingertip *pulse oximeter probe*, use the dorsal distal phalanx, (i.e. back of the finger between the fingernail and the most distal knuckle (see Figure 201.102).

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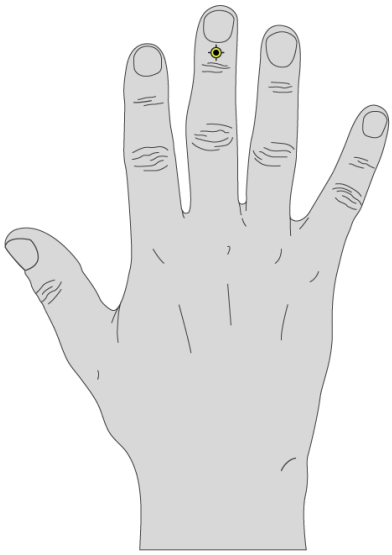

The **+** in this figure indicates the dorsal distal phalanx (DP) as the location to measure skin pigment for finger *pulse oximeter probes*. The DP is the dorsal skin between the fingernail and the dorsal interphalangeal joint.

Figure 201.102 — Dorsal distal phalanx location

15) Ensure that *ITA* colorimeter measurement specification conform with the following.

   i) *ITA* calculated from L* and b* according to Reference [52].

   ii) If available in the colorimeter, select the D65 illumination.

   iii) If a standard observer angle is available, select a standard observer equal to 2 degrees.

   iv) Use a measurement aperture between 3 mm and 8 mm.

   v) If a choice between specular reflection included or excluded is available, select the excluded setting.

g) Participants who participated in the calibration study of the *pulse oximeter equipment* under test should be excluded.

NOTE 7 There is guidance or rationale for this list item contained in AA.2.11.

1) Desaturation test facilities should make best efforts to limit the number of repeat participants in different *controlled desaturation studies* of the *pulse oximeter equipment* under test.

Check conformity by inspection of the study report.

201.12.1.102 *Accuracy* under conditions of motion

NOTE There is guidance or rationale for this subclause contained in AA.2.12.

a) If a *manufacturer* claims that the *pulse oximeter equipment* is accurate during motion, A_{rms} (*pulse oximeter equipment accuracy*), *mean bias* and *precision* specifications during motion shall be disclosed in the *instructions for use*.

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b) *Accuracy* claims shall be supported by data collected in a *controlled desaturation study* that exposes the location of the *pulse oximeter probe* (and if applicable, the *probe cable extender*) to motion typical for the *pulse oximeter probe* site.

1) Motions should include participant-driven (active) or imposed (passive) movement of the *pulse oximeter probe* monitoring location.

2) Motion should be initiated prior to and continue throughout the data sampling periods such that the sampling period occurs after system averaging has settled.

i) The *data update period* should be considered.

3) Data should be collected across the claimed saturation *accuracy* range.

4) The reference is SaO_2. See Annex DD.2.3.2 b) and c).

i) It may be drawn from a different location on the body.

5) Additional non-moving sampling periods allow a direct observation of the impact of movement.

c) A summary of the test methods and *pulse oximeter probes* used to establish the *accuracy* claims during motion shall be disclosed in the *technical description*.

1) The summary should include the average *%mod* in quiescent and motion periods during the test.

Check conformity by inspection of the study report, *instructions for use* and *technical description*.

201.12.1.103 *Accuracy* under conditions of low perfusion

a) If a *manufacturer* claims that the *pulse oximeter equipment* is accurate under conditions of low perfusion, A_{rms} (*pulse oximeter equipment accuracy*), *mean bias* and *precision* specifications under these conditions shall be disclosed in the *instructions for use*.

b) A summary of the test methods used to establish the *accuracy* claims under conditions of low perfusion shall be disclosed in the *technical description*.

1) The summary should include *%mod* as an indicator of pulsatile signal strength.

Check conformity by inspection of the study report, *instructions for use* and *technical description*.

201.12.1.104 Pulse rate *accuracy*

a) If equipped, pulse rate *accuracy* shall:

1) be disclosed in the *instructions for use*; and

2) be stated as the root-mean-square (rms) difference between paired pulse rate data recorded with the *pulse oximeter equipment* and with a reference method.

b) Pulse rate *accuracy* shall be stated either:

1) over the full claimed range of the *pulse oximeter equipment*; or

2) as separate pulse rate *accuracy* specifications over segments of that range.

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c) The reference method for the computation of pulse rate *accuracy* may be:

1) an electronic pulse simulator;

2) ECG heart rate;

3) palpated pulse;

4) thoracic auscultation;

5) a second *pulse oximeter equipment* which has been qualified by comparison to one of these references; or

6) any other method with sufficient *accuracy*.

d) The reference method for the determination of pulse rate *accuracy* shall be disclosed in the *technical description*.

Check conformity by inspection of the study report, *instructions for use* and *technical description*.

201.12.1.105 Components of the study report

a) The study shall be documented in a report.

b) For *pulse oximeter equipment* evaluated in accordance with this document, the study report shall include:

1) the contents specified for a clinical investigation report in accordance with ISO 14155:2026;

2) a copy of the clinical investigation plan in accordance with ISO 14155:2026;

3) a description of the *pulse oximeter equipment* subject to evaluation, including

i) any covered *pulse oximeter monitor*,

ii) any covered *pulse oximeter probe*, and

iii) any covered *probe extension cable;*

4) a description and rationale for any differences between the test articles and the final *pulse oximeter equipment;*

5) a description of the testing method used and its qualification (manufacturer, type and quantity of *CO-oximeters,* their calibration data or accreditation and specified *accuracy*), including whether the study was:

i) an invasive *controlled desaturation study* on healthy participants, or

ii) an invasive study on *patients,* such as in a convenience sample study;

6) for an invasive *controlled desaturation study* on healthy participants include:

i) a description of the gas delivery method to the participant,

ii) a description of the safety equipment;

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7) a description of the testing method used, including the following,

   i) *CO-Oximeter* data handling (data inclusion and removal),

   ii) method for SpO_2 or calculated SaO_2 stability determination,

   iii) if the *pulse oximeter equipment* data were collected electronically, the means used to pair the datum with each arterial sample *CO-Oximeter* reading, and

   iv) if the *pulse oximeter equipment* data were not collected electronically, detailed description of data collection methods, including criteria for unbiased determination of when each point is collected;

8) any deviations from the *procedure* indicated in this document;

9) any unusual features observed;

10) for all participants, the following,

   i) self-reported ethnicity,

   ii) age,

   iii) biological sex,

   iv) body mass,

   v) height,

   vi) Body Mass Index (BMI);

11) L* a* b* values as measured with a colorimeter for all shades of the printed colour scale of the MST *standardized skin colour scale* used for the study[133];

12) for all participants, a graphical and tabular representation of the distributions of the following:

   i) the MST as determined in 201.12.1.101.6 f) 2),

   ii) the *ITA* values at the forehead, ventral bicep and *pulse oximeter probe* site in the study as determined in 201.12.1.101.6 f) 14);

13) the percentage breakdown of each biological sex;

14) the A_{rms} (*pulse oximeter equipment accuracy*) over the range of 70 % to 100 % in accordance with 201.12.1.101.4 with an acceptance criterion of less than or equal to 3,0 % with 95 % confidence;

   i) For the range of 70 % to 100 %, a regression (SpO_2 versus SaO_2) plot with *mean bias*, A_{rms}, and number of points, linear regression slope with Y, X and the standard estimate of the error in slope.

   ii) An inverse prediction plot, that characterizes the uncertainty of the SpO_2 value for any given SpO_2 value, and provides a plot of the non-simultaneous, 95 % pointwise inverse prediction band of plausible SpO_2 values for every SpO_2 value from 70 % to100 %.

   NOTE See CC.4 for an explanation of inverse prediction plots.

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iii) For each additional range specified in accordance with 201.12.1.101.1 b) or 201.12.1.101.1 c), the A_{rms} (*pulse oximeter equipment accuracy*) over the range with an acceptance criterion of less than or equal to the specification.

iv) For each additional range specified, a regression (SpO_2 versus SaO_2) plot with *mean bias*, A_{rms}, and number of points for each range.

15) the subrange A_{rms}, (*pulse oximeter equipment accuracy*) *mean bias* and *precision* over the SaO_2 subranges in accordance with 201.12.1.101 f),

i) 70 % to 80 %,

ii) 80 % to 90 %, and

iii) 90 % to 100 %;

16) a modified Bland-Altman plot[35][44] for each combination of *pulse oximeter probe* and *pulse oximeter monitor* included in the study in accordance with 201.12.1.101.4 d);

17) a modified Bland-Altman plot[35][44] for each combination of *pulse oximeter probe* and *pulse oximeter monitor* included in the study in accordance with 201.12.1.101.4 e);

18) for the SaO_2 range of 80 % to 90 % (or at its midpoint of 85 %), the computed *differential bias* in accordance with 201.12.1.101.2 and 201.12.1.101.5;

19) line listing of all data points collected, whether or not used in the analysis, including the following information paired with each arterial sample,

i) participant #,

ii) reference SaO_2 reading,

iii) ECG/HR reading,

iv) arterial sample syringe #,

v) *pulse oximeter equipment* under test SpO_2 reading,

vi) *pulse oximeter equipment* under test pulse rate reading if applicable,

vii) assessment of stability for each arterial sample,

viii) all removed data points with justification for removal,

ix) if available, signal quality, *%mod* or perfusion information reading, and

x) if the data were collected electronically, the continuous electronic data file for the *pulse oximeter equipment* under test in human readable format;

20) if applicable, a summary of the method used to determine *accuracy* in motion in accordance with 201.12.1.102;

21) if applicable, the A_{rms} (*pulse oximeter equipment accuracy*), *mean bias* and *precision* in motion in accordance with 201.12.1.102;

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22) if applicable, a summary of the method used to determine *accuracy* in low perfusion in accordance with 201.12.1.103;

23) if applicable, the A_{rms} (*pulse oximeter equipment accuracy*), *mean bias* and *precision* in low perfusion in accordance with 201.12.1.103;

24) if applicable a summary of the method used to determine pulse rate *accuracy* in accordance with 201.12.1.104;

25) the pulse rate *accuracy* in accordance with 201.12.1.104 if applicable;

26) a dated reference to this document (the standard used for the evaluation); and

27) the dates of the testing.

Check conformity by inspection of the study report.

201.12.4 Protection against hazardous output

Additional subclauses:

201.12.4.101 *Data update period*

NOTE There is guidance or rationale for this subclause contained in AA.2.13.

a) There shall be an indication that SpO_2 or pulse rate data is not current when the *data update period* is greater than 30 s.

   1) The *data update period* time may be shorter than 30 s.

   2) A maximum *data update period* for saturation and pulse rate should be shorter than 30 s for continuous neonatal monitoring and diagnostic applications.

b) If the *pulse oximeter equipment* is equipped with an *alarm system* that detects any *physiological alarm conditions*, the *alarm system* shall provide at least a *low priority alarm condition* to indicate when the *data update period* exceeds 30 s.

c) *Pulse oximeter equipment* that is not equipped with an *alarm system* that detects any *physiological alarm condition* shall indicate when the *data update period* exceeds 30 s.

   1) The indication shall be described in the *instructions for use*.

d) If the *pulse oximeter equipment* is equipped with a *functional connection* (see 201.103.2), an indication that the *data update period* exceeds 30 s shall be included in the data stream.

Check conformity by inspection.

201.12.4.102 Signal inadequacy

NOTE 1 There is guidance or rationale for this subclause contained in AA.2.14.

a) An indicator of signal inadequacy shall be provided to the *operator* when *pulse oximeter probe* signals received by the *pulse oximeter equipment* are insufficient and may result in SpO_2 or pulse rate outside the specified *accuracy* range.

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1) *Symbol* ISO 7000-0435 (see Table 201.D.1.101, *Symbol* 2) may be used for this indication.

b) The *accompanying document* shall provide a statement that the signal inadequacy indicator cannot identify all sources of error that result in potentially incorrect SpO_2 or pulse rate.

c) Measurable real-time factors known by the *pulse oximeter equipment manufacturer* to contribute to signal inadequacy shall be used in determining when to provide indication, including:

1) DC signal level and *%mod* of the received signals for each wavelength used by the *pulse oximeter equipment* for determining SpO_2 and pulse rate values.

NOTE 2 Light attenuation by the tissue, including that by melanin in the skin, can affect DC signal levels at each wavelength differently.

NOTE 3 A non-*normalized* pulse waveform display satisfies this requirement for a signal inadequacy indicator with respect to *%mod*. A *normalized* waveform by itself does not satisfy this requirement and can mask an unreliable signal.

NOTE 4 A non-*normalized* pulse waveform display does not satisfy indication of inadequate DC signals.

NOTE 5 Other factors can also be included in determining signal inadequacy, such as sensed artifacts caused by ambient interference or *patient* movement.

NOTE 6 Not posting SpO_2 or pulse rate under inadequate signals satisfies this requirement for signal inadequacy indicator.

d) A description of the indicator, its function, and the measurable real-time factors that influence its performance used to assess inadequacy, shall be provided in the *accompanying document*.

EXAMPLE 1 Signal inadequacy indicated by a visual *information signal*, displayed message, or a *low priority alarm signal*.

1) If a non-*normalized* pulse waveform display is used, the *accompanying document* shall provide a description of visible characteristics of the non-*normalized* pulse waveform that identify inadequate *%mod*.

EXAMPLE 2 Lack of discernible pulsation or beat-to-beat periodicity in the waveform inconsistent with the *patient's* pulse rate.

e) The deviation threshold used for signal inadequacy testing shall be disclosed in the *accompany document*.

NOTE 7 The threshold for signal inadequacy can depend on several factors specific to the design the *pulse oximeter equipment*, including physical construction, electronics, and algorithm capabilities. A standardized acceptance criteria for signal quality threshold for all *pulse oximeter equipment* is not possible due to the specifics of the design. Additional information is found in Annex II.

f) If the *pulse oximeter equipment* is equipped with a *functional connection* (see 201.103.2), then signal inadequacy shall be included in the data stream (see Table GG.1).

Check conformity by inspection and inspection of the *accompanying document*.

A *pulse oximeter equipment functional tester* that provides synthetic signals may be used for checking conformity.

201.13 *Hazardous situations* and fault conditions for *ME equipment*

IEC 60601-1:2005+AMD1:2012+AMD2:2020, Clause 13 applies, except as follows:

Additional subclauses:

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201.13.101 Detection of *pulse oximeter probe* faults and *probe cable extender* faults

a) If the *pulse oximeter equipment* is equipped with an *alarm system* to detect any *physiological alarm conditions*, the *alarm system* shall provide a *technical alarm condition* to indicate when any wire in the *pulse oximeter probe* cable or *probe cable extender* is opened or shorted to any other wire in the *pulse oximeter probe* cable or *probe cable extender* that causes other than normal operation.

b) *Pulse oximeter equipment* that is not equipped with an *alarm system* that detects any *physiological alarm conditions* shall visually indicate the presence of *probe faults*.

   1) The indication shall be described in the *instructions for use*.

      EXAMPLE Indication of abnormal operation by blank display.

c) Check conformity by inspection of the *instructions for use*, if applicable, and with the following test:

   1) Disconnect the *pulse oximeter probe* from the *pulse oximeter equipment* and place it in series with a circuit with which each *pulse oximeter probe* wire can be opened or shorted to any other *pulse oximeter probe* wire. Confirm that the *pulse oximeter equipment* enters a normal operation mode before any open or short situation is initiated.

   2) Repeat for any *probe cable extender*.

   3) Confirm either that a *probe fault* is indicated by the *pulse oximeter monitor* or that the *pulse oximeter equipment* continues normal operation.

201.14 *Programmable electrical medical systems* (*PEMS*)

IEC 60601-1:2005+AMD1:2012+AMD2:2020, Clause 14 except as follows:

Addition:

201.14.101 *Cybersecurity* capabilities of *pulse oximeter equipment*

a) The design and manufacture of *pulse oximeter equipment* should contribute to safe operation related to *cybersecurity* threats.

b) *Risk control* measures as specified in IEC/TR 60601-4-5:2021, Clauses 4 to 7, should be implemented, as appropriate, with the following additions to IEC/TR 60601-4-5:2021, 4.6.3.

   1) *Essential function*: After disconnection of all data interfaces to *IT-network* connections possibly subject to *attack* (via physical or logical disconnection), if necessary, after restarting the *pulse oximeter equipment* once, all clinical functions except those related to the affected remote functionality should be in place and operate as intended.

   2) *Firecall* functions: As *pulse oximeter equipment* is used with *healthcare professional operator* supervision or at home by a *lay operator*, the user interface need not require authentication for *operators* and thus need not require *firecall* functions. If authentication for *operators* to *operator* accessible settings is in place also for the user interface, a *firecall* function should be able to overrule that *operator* authentication combined with a log entry protected against modifications by the *operator* (i.e. *responsible organization* log). *IT-network* interfaces including those for remote access typically require authentication but should not include *firecall* functions.

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3) Target *security level* SL-T: For user interfaces and for remote setting of clinical functions, the expected *security level* of SL-T, as specified in IEC/TR 60601-4-5:2021, is 1 or better. However, authentication of *operators* is not required for the user interface, only for access via data interfaces. The expected *security level* of SL-T, as specified in IEC/TR 60601-4-5:2021, is 2 or better for secure updates or restorage of software, remotely or locally.

201.15Construction of *ME equipment*

IEC 60601-1:2005+AMD1:2012+AMD2:2020, Clause 15 applies, except as follows:

Additional subclauses:

201.15.3.5.101 Additional requirements for rough handling

NOTE There is guidance or rationale for this subclause contained in AA.2.15.

201.15.3.5.101.1 Shock and vibration

NOTE 1 There is guidance or rationale for this subclause contained in AA.2.16.

a) *Pulse oximeter equipment* or its parts not intended for use during in the *emergency medical services environment* or for use in the *home healthcare environment* shall have adequate mechanical strength when subjected to mechanical stress caused by *normal use*, pushing, impact, dropping and rough handling.

   1) *Stationary equipment* is exempt from the requirements of this subclause.

NOTE 2 Requirements for shock and vibration of *pulse oximeter equipment* intended for use in the *home healthcare environment* are found in IEC 60601-1-11 and for use in the *emergency medical services environment* are found in IEC 60601-1-12. The requirements of these collateral standards are considered more stringent than the requirements in this subclause.

b) After the following tests, the *pulse oximeter equipment* shall maintain *basic safety* and *essential performance*.

c) Check conformity with the following tests:

   1) Shock test in accordance with IEC 60068-2-27:2008, using the following conditions:

      NOTE 3 This shock test equates to IEC/TR 60721-4-7:2001+ AMD1:2003, Class 7M2.

      i) test type: Type 1, or

         I) peak acceleration: 150 m/s^2 (15 g),

         II) duration: 11 ms,

         III) pulse shape: half-sine,

         IV) number of shocks: 3 shocks per direction per axis (18 total);

      ii) test type: Type 2

         I) peak acceleration: 300 m/s^2 (30 g),

         II) duration: 6 ms,

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III) pulse shape: half-sine,

IV) number of shocks: 3 shocks per direction per axis (18 total).

2) Broad-band random vibration test in accordance with IEC 60068-2-64:2008+AMD1:2019, using the following conditions:

NOTE 4 This vibration test equates to IEC/TR 60721-4-7:2001+ AMD1:2003, Classes 7M1 and 7M2.

i) acceleration amplitude:

I) 10 Hz to 100 Hz: 1,0 $(m/s^2)^2/Hz$;

II) 100 Hz to 200 Hz: –3 dB per octave;

III) 200 Hz to 2 000 Hz: 0,5 $(m/s^2)^2/Hz$;

ii) duration: 30 min per perpendicular axis (3 total).

3) Following these tests, confirm that *basic safety* and *essential performance* are maintained.

201.15.101 Mode of operation

Pulse oximeter equipment shall be suitable for *continuous operation.*

NOTE 1 Moving the *pulse oximeter probe* to a new site on the body is *normal use* and is considered *continuous operation*.

NOTE 2 Intermittent use of *pulse oximeter equipment* on one *patient* or among *patients* is *normal use* and is considered *continuous operation*.

Check conformity by inspection.

201.16 *ME systems*

IEC 60601-1:2005+AMD1:2012+AMD2:2020, Clause 16 applies.

201.17 Electromagnetic compatibility of *ME equipment* and *ME systems*

IEC 60601-1:2005+AMD1:2012+AMD2:2020, Clause 17 applies.

Additional subclauses:

201.101 *Pulse oximeter probes* and *probe cable extenders*

NOTE There is guidance or rationale for this subclause contained in AA.2.17.

201.101.1 General

a) All *pulse oximeter probes* and *probe cable extenders* shall conform with the requirements of this document, whether they are produced by the *manufacturer* of the *pulse oximeter monitor,* by another entity ("third party manufacturer" or healthcare provider) or are *remanufactured*.

b) *Manufacturers* of *remanufactured pulse oximeter probes* and *probe cable extenders* shall perform testing to ensure that all *pulse oximeter equipment* specifications are met with each model of *pulse oximeter monitor* with which the *pulse oximeter probe* or *probe cable extender* is intended to be used.

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1) The *accompanying document* of *remanufactured pulse oximeter probes* and *probe cable extenders* shall list all *pulse oximeter monitors* with which compatibility is claimed.

c) It is the responsibility of the *manufacturer* to *validate* their *processes* to ensure that any new or *remanufactured pulse oximeter probe* or *probe cable extender* conforms with the requirements of this document.

Check conformity by the tests of this document.

201.101.2 Labelling of *accessories*

a) The *model or type reference* of at least one *pulse oximeter monitor* shall be included in the *accompanying document* provided with each *pulse oximeter probe* and *probe cable extender*, conforming with 201.101.1.

b) Statements shall be included in the *accompanying document* of each *pulse oximeter probe* and *probe cable extender* to the effect that:

1) pulse oximeter probes and probe cable extenders are designed for use with specific monitors;

2) incompatible components can result in degraded performance; and

3) the operator is responsible for checking the compatibility of the pulse oximeter monitor, pulse oximeter probe and cable extender before use.

NOTE 1 Additional information is found in 201.101.1.

c) Additional information on the operation of the *pulse oximeter probe* or *probe cable extender*, its features, and factors that adversely impact the performance of the *pulse oximeter monitor* should be made available via a QR code *marked* on the *pulse oximeter monitor, pulse oximeter probe* or *probe cable extender*.

NOTE 2 Examples of means to provide the QR code at the point of use of the *pulse oximeter equipment* include: displayed on the *pulse oximeter monitor* screen; *marked* on the *enclosure*, *pulse oximeter probe*, or *probe cable extender*; and included in the *label* of *single use* or *single patient multiple use pulse oximeter probe* or *probe cable extender*.

Check conformity by inspection of the *accompanying document*.

201.102 Saturation pulse *information signal*

If a variable-pitch auditory *information signal* is provided by the *pulse oximeter monitor* to indicate the detection of a pulse and the relative SpO_2 level, the pitch change shall follow the SpO_2 reading (e.g. the pitch decreases as the SpO_2 reading decreases).

Check conformity by functional testing.

201.103 *Functional connection*

201.103.1 General

Basic safety and *essential performance* shall be maintained during failure of equipment connected to or the disruptions of connections to *functional connection* of *pulse oximeter equipment*.

Check conformity by functional testing.

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201.103.2 Connection to electronic health record or other *ME equipment*

NOTE There is guidance or rationale for this subclause contained in AA.2.18.

a) *Pulse oximeter equipment* should be equipped with a *functional connection* that permits data transmission from the *pulse oximeter equipment* to an electronic health record.

b) If equipped with an electronic data interface that is intended to provide access to non-proprietary *pulse oximeter equipment* data,

   1) the transmission shall be in accordance with Annex GG.

   2) the data transmission should be capable of being provided with a *network/data coupling* in accordance with AAMI 2700-1:2019[19].

   3) for *pulse oximeter equipment* that uses an electronic data interface as a *manufacturer* specific interface or an internal data interface between specific communication participants, the *pulse oximeter equipment* may deviate from the requirements of Annex GG.

   EXAMPLE Modules communicating with a host device that do not have an *operator* accessible port.

201.103.3 Connection to a *distributed alarm system*

a) For *pulse oximeter equipment* that is equipped with an *alarm system* that detects a *physiological alarm condition*, the *alarm system* should be equipped with a *functional connection* that permits connection to a *distributed alarm system*.

b) The data transmission should be capable of being provided with a *network/data coupling* in accordance with AAMI 2700-1:2019[19].

202 Electromagnetic disturbances – Requirements and tests

IEC 60601-1-2:2014+AMD1:2020 applies except as follows:

202.4.3.1 Configurations

Amendment (replace the second dash of 4.3.1 with):

– setting the SpO_2 within the calibrated range to be at least 5 % different from that of a noise-induced value and less than (100 % minus the SpO_2 *accuracy* of the *pulse oximeter equipment*). This shall apply to *immunity* testing of *pulse oximeter equipment.*

  NOTE 100 The noise-induced value can be a value e.g. where $R = 1$ or R is the ratio of the gain from the IR channel to the gain from the red channel. Other noise-induced values have been observed.

– setting the pulse rate to be different from that of the noise-induced signal frequency and within the specified range of the pulse rate display.

aa) The SpO_2 and pulse rate signal may be derived from a *patient* simulator for these tests.

202.5.2.2.1 Requirements applicable to all *ME equipment* and *ME systems*

Amendment (add note to list element b):

NOTE 100 The requirements of this document are not considered deviations or allowances.

Additional subclause:

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202.8.1.101 Additional general requirements

a) Under the *immunity test levels* specified in IEC 60601-1-2:2014+AMD1:2020, 8.9, *pulse oximeter equipment* shall be able to provide *basic safety* and *essential performance*.

b) The following degradations, if associated with *basic safety* or *essential performance* shall not be allowed:

   1) component failures;

   2) changes in programmable parameters or settings;

   3) reset to default settings; and

   4) change of operating mode.

c) During transient phenomena (i.e. electrostatic discharge, electrical fast transients/bursts, surges, electrical transient conduction along supply lines and proximity magnetic fields), the *pulse oximeter equipment* shall meet the *immunity* pass/fail criteria for *basic s*afety. Within 30 s after the transient electromagnetic phenomena are discontinued, the *pulse oximeter equipment* shall resume normal operation without *operator* intervention, without loss of any *operator settings* or stored data and shall provide *basic safety* and *essential performance*.

202.8.2 *Patient* physiological simulation

Amendment (add between the first and second paragraph):

aa) During *immunity* testing, the *pulse oximeter equipment* shall be tested at an SpO_2 reading within the calibrated range that is at least 5 % different from that of a noise-induced value and less than (100 % minus the SpO_2 *accuracy* of the *pulse oximeter equipment*).

NOTE 100 The noise-induced value can be a value e.g. where R = 1 or R is the ratio of the gain from the IR channel to the gain from the red channel. Other noise-induced values have been observed.

bb) The pulse rate shall be different from that of the noise-induced signal frequency and within the specified range of the pulse rate display.

cc) The SpO_2 and pulse rate signal may be derived from a *patient* simulator for these tests.

206 Usability

IEC 60601-1-6:2010+AMD1:2013+AMD2:2020 applies except as follows:

For *pulse oximeter equipment*, the following shall be considered *primary operating functions*:

a) setting the *operator*-adjustable controls:

   1) setting *alarm limits*;

   2) inactivating *alarm signals*; and

   3) switching between different modes;

b) observing monitored parameters;

c) applying the *pulse oximeter probe* to the *patient*;

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d) starting the *pulse oximeter equipment* from power off; and

e) connecting and disconnecting the *distributed alarm system*, if provided.

208 General requirements, tests and guidance for alarm systems in medical electrical equipment and medical electrical systems

IEC 60601-1-8:2006+AMD1:2012+AMD2:2020 applies except as follows:

Additional subclauses:

208.6.1.2.101 Additional requirements for *alarm condition* priority

NOTE 1 There is guidance or rationale for this subclause contained in AA.2.19.

If the *pulse oximeter equipment* is equipped with an *alarm system* that detects a *physiological alarm condition*, the *alarm system* shall provide at least a *medium priority alarm condition* for low SpO_2 level.

NOTE 2 A *medium priority* high SpO_2 level *alarm condition* can enhance *patient* safety for certain clinical applications (e.g. neonatal monitoring).

Check conformity by inspection.

208.6.5.4.101 Additional requirements for *default alarm preset*

NOTE There is guidance or rationale for this subclause contained in AA.2.20.

a) If the *pulse oximeter monitor* is equipped with an *alarm system* to detect a low SpO_2 level *physiological alarm condition*, the *alarm limit* in the *manufacturer*-configured *alarm preset* for the SpO_2 level *physiological alarm condition* shall not be less than 85 % SpO_2[54][116].

b) Unless the low SpO_2 *alarm limit* is displayed continuously, the low SpO_2 *alarm limit* of any *operator*-configured *alarm preset* shall not be less than the low SpO_2 *alarm limit* stored in the *default alarm preset*.

Check conformity by functional testing.

208.6.8.5.101 Additional requirements for *alarm signal* inactivation states, indication and access

The *manufacturer*-configured default *audio paused* or *alarm paused* interval of *pulse oximeter equipment* shall not exceed 2 min.

Check conformity by functional testing.

211 Requirements for medical electrical equipment and medical electrical systems used in the home healthcare environment

IEC 60601-1-11:2015+AMD1:2020 applies except as follows:

Additional subclauses:

The tests of IEC 60601-1-11:2015+AMD1:2020, Clause 10, and IEC 60601-1:2005+AMD1:2012+AMD2:2020, 15.3, shall be performed on the same sample of the *pulse oximeter equipment* following any *remanufacturing* established for this equipment.

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212 Requirements for medical electrical equipment and medical electrical systems used in the emergency medical services environment

IEC 60601-1-12:2014+AMD1:2020 applies except as follows:

Additional subclauses:

The tests of IEC 60601-1-12:2014+AMD1:2020, Clause 10, and IEC 60601-1:2005 +AMD1:2012+AMD2:2020, 15.3, shall be performed on the same sample of the *pulse oximeter equipment* following any *remanufacturing* established for this equipment.

Annexes of IEC 60601-1:2005+AMD1:2012+AMD2:2020 apply, except as follows:

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Annex C
(informative)

Guide to *marking* and labelling requirements for *ME equipment* and *ME systems*

IEC 60601-1:2005+AMD1:2012+AMD2:2020, Annex C applies, except as follows:

Addition:

201.C.1 *Marking* on the outside of *ME equipment*, *ME systems* or their parts

Additional requirements for *marking* on the outside of *pulse oximeter equipment* or their parts or *accessories* are found in Table 201.C.101.

Table 201.C.101 — *Marking* on the outside of *pulse oximeter equipment* or its parts or *accessories*

Description of *marking*	Subclause
For *accessories* supplied separately, indication of any limitations or adverse effects of the *accessory* on the *basic safety* or *essential performance* of the *pulse oximeter monitor*, if applicable	201.7.2.4.101 a) 2)
For *accessories* supplied separately, the requirements of ISO 20417:2026, 6.1.1 c)	201.7.2.4.101 a) 1)
If applicable for a *remanufactured pulse oximeter probe*, so indicated	201.7.2.101 b)
If applicable, IP classification of carrying case	201.7.2.9.101 b) 2)
If applicable, keep dry or ISO 15223-1:2021, *Symbol* 5.3.4	201.7.2.9.101 d)
If not provided with low *SpO_2 limit alarm condition*, so indicated	201.7.2.101 a)
IP classification of *enclosure*	201.7.2.9.101 a) 201.7.2.9.101 b) 1)
Mandatory action *safety sign*: follow *instructions for use*	201.7.2.3
Requirements of ISO 20417	201.7.1.101
Units of measure of oxygen saturation	201.7.4.3 aa)
Units of measure of pulse rate	201.7.4.3 bb)

201.C.2 *Accompanying documents*, general

Additional requirements for general information to be included in the *accompanying documents* of *pulse oximeter equipment* or its parts or *accessories* are found in Table 201.C.102.

Table 201.C.102 — *Accompanying documents*, general

Description of disclosure	Subclause
Description of signal inadequacy indicator, its function and the measurable real-time factors that influence its performance used to assess inadequacy	201.12.4.102 d)
Deviation threshold used for signal inadequacy testing	201.12.4.102 e)
For a non-*normalized* waveform display, a description of visible characteristics of the non-*normalized* pulse waveform that identify inadequate *%mod*	201.12.4.102 d) 1)
For *pulse oximeter probes* and *probe cable extenders*, a compatible *pulse oximeter monitor*	201.101.2 a)

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Description of disclosure	Subclause
For *pulse oximeter probes* and *probe cable extenders*, probes are designed for use with specific monitors	201.101.2 b) 1)
For *pulse oximeter probes* and *probe cable extenders*, the *operator* is responsible for checking compatibility prior to use	201.101.2 b) 3)
For *pulse oximeter probes* or *probe cable extenders*, incompatible components can result in degraded performance	201.101.2 b) 2)
For *remanufactured pulse oximeter probes* and *probe cable extenders*, list of compatible *pulse oximeter monitors*	201.101.1 b) 1)
Requirements of ISO 20417	201.7.1.101
Statement that the signal inadequacy indicator cannot identify all sources of error that result in potentially incorrect SpO_2 or pulse rate	201.12.4.102 b)

201.C.3 *Accompanying documents, instructions for use*

Additional requirements for information to be included in the *instructions for use* of *pulse oximeter equipment,* its parts or *accessories* are found in Table 201.C.103.

Table 201.C.103 — *Accompanying documents, instructions for use*

Description of disclosure	Subclause
Accuracy specifications in low perfusion, if applicable	201.12.1.103 a)
Accuracy specifications under motion, if applicable	201.12.1.102 a)
Declared SpO_2 *accuracy* over the range 70 % SaO_2 to 100 % SaO_2 for each *pulse oximeter probe*	201.12.1.101.1 a) 1)
Description with graphic indicating proper orientation and placement of the *pulse oximeter probe* and what to avoid	201.7.9.2.9.101 a)
Disclosure indicating to the effect that because pulse oximeter equipment measurements are statistically distributed, only about two-thirds of pulse oximeter equipment measurements in a larger population can can be expected to fall within ± A_{rms} and about 95 % falling within ±2 times A_{rms} of the value measured by a CO-oximeter	201.12.1.101.1 a) 4)
Disclosure indicating to the effect that the A_{rms} value is determined in a controlled desaturation study using specific probes and monitor combinations that is performed on healthy participants and represents the best-case scenario with real-world clinical inaccuracy expected to be worse than the A_{rms} value disclosed from the controlled desaturation study	201.12.1.101.1 a) 5)
Displayed ranges of pulse rate	201.7.9.2.1.101 e)
Displayed ranges of SpO_2	201.7.9.2.1.101 d)
For *accessories* supplied separately where *marking* the *accessory* is not practicable, the requirements of 201.7.2.4.101	201.7.2.4.101 b)
For each *pulse oximeter monitor*, list of compatible *pulse oximeter probes* and *probe cable extenders*	201.7.9.2.1.101 g)
For each *pulse oximeter probe*, range of peak wavelengths and optical output power and its utility	201.7.9.2.1.101 b)
For *probe cable extenders*, any reasonably foreseeable adverse effect of any recommended *accessory* on the *basic safety* or *essential performance* of the *pulse oximeter monitor*	201.7.9.2.14.101 b) 2)
For *probe cable extenders*, the list of compatible *pulse oximeter monitors* and *pulse oximeter probes*	201.7.9.2.14.101 b)
For *pulse oximeter equipment* not provided with an *alarm system* that includes the capability to detect *physiological alarm conditions*, means of indicating a *probe fault*	201.13.101 b) 1)
For *pulse oximeter probes* or *probe cable extenders*, probes are designed for use with specific monitors	201.7.9.2.2.101 a)
For *pulse oximeter probes* or *probe cable extenders*, the responsible organization or operator is responsible for checking compatibility prior to use to prevent *patient* injury or inaccurate readings	201.7.9.2.2.101 b)

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Description of disclosure	Subclause
For *pulse oximeter probes*, the list of compatible *pulse oximeter monitors* and *probe cable extenders*	201.7.9.2.14.101 a)
If no *SpO_2* and pulse rate *alarm conditions*, so indicated	201.7.9.2.1.101 f)
If not equipped with an *alarm system* that includes the capability to detect *physiological alarm conditions*, description of the indication that the *data update period* exceeds 30 s	201.12.4.101 c) 1)
If the *applied part* temperature can exceed 41 °C, instructions emphasizing the importance of proper application and any changes in the recommended maximum application time	201.7.9.2.9.101 f)
If the *applied part* temperature can exceed 41 °C, temperature settings greater than 41 °C shall not be used on *patients* with susceptible skin	201.11.1.2.2 bb) 3)
If the *applied part* temperature can exceed 41 °C, the sequence of *operator* actions needed to activate	201.11.1.2.2 bb) 1) ii)
Interference known to influence the *SpO_2 accuracy*	201.12.1.101.3 f) 2)
IP classification and its meaning for *enclosure* and, if applicable, carrying case	201.7.9.2.9.101 d)
Operation of *pulse oximeter equipment* following *supply mains* interruption longer than 30 s	201.11.8.101.3
Pulse rate *accuracy*, if applicable	201.12.1.104 a)
Recommended maximum application time for each type of *pulse oximeter probe* at a single site	201.7.9.2.9.101 c)
Requirements of ISO 20417	201.7.1.101
Specified combinations of *pulse oximeter equipment*	201.4.102 b)
SpO_2 accuracy, mean bias and *precision* for each combination of *pulse oximeter probe* and *pulse oximeter monitor* indicated in the instructions for use over the *SaO_2* range of 70 % to 80 %	201.12.1.101.4 f) 1)
SpO_2 accuracy, mean bias and *precision* for each combination of *pulse oximeter probe* and *pulse oximeter monitor* indicated in the instructions for use over the *SaO_2* range of 80 % to 90 %	201.12.1.101.4 f) 2)
SpO_2 accuracy, mean bias and *precision* for each combination of *pulse oximeter probe* and *pulse oximeter monitor* indicated in the instructions for use over the *SaO_2* range of 90 % to 100 %	201.12.1.101.4 f) 3)
Statement that *Pulse oximeter equipment* is not typically capable of nor intended as a means for the measurement of blood flow or blood volume	201.7.9.2.9.101 e)
Statement that the signal inadequacy indicator cannot identify all sources of error that result in potentially incorrect *SpO_2* or pulse rate	201.7.9.2.9.101 b)
That the *pulse oximeter equipment* displays *functional oxygen saturation*	201.7.9.2.1.101 a)
The effect of *data update period*, data averaging, *alarm condition* and *alarm signal generation delay* on displayed and transmitted *SpO_2* and pulse rate values	201.7.9.2.1.101 c)
Warning to the effect that artificial acrylic nails, gel-based manicures, nail jewellery, specific colours of nail polish, varied nail colour patterns, long nails and bruised nail beds interfere with the optical pathway and can cause inaccurate readings	201.7.9.2.2.101 d)
Warning to the effect that misapplication of a *pulse oximeter probe* with excessive pressure for prolonged periods can induce pressure injury	201.7.9.2.2.101 c)

201.C.4 *Accompanying documents, technical description*

Additional requirements for the *technical description* of *pulse oximeter equipment* or its parts or *accessories* are found in Table 201.C.104.

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Table 201.C.104 — *Accompanying documents, technical description*

Description of disclosure	Subclause
95 % confidence interval for characterizing the uncertainty of the estimate of *SpO_2 accuracy*	201.12.1.101.4 b) 1)
Functional tester cannot be used to assess *accuracy*	201.7.9.3.1.101 a)
If the *applied part* temperature can exceed 41 °C, the method used to measure *applied part* temperature	201.11.1.2.2 bb) 5)
Linear regression (*SpO_2* versus *SaO_2*) plot from 70 % to 100 % *SaO_2*	201.12.1.101.4 c)
Mean difference of *SpO_2* reading bias between participants in the dark MST categories (H,I,J) and the light categories (A,B,C) over the range 70 % to 80 %, *SaO_2*	201.12.1.101.5 f) 1)
Mean difference of *SpO_2* reading bias between participants in the dark MST categories (H,I,J) and the light categories (A,B,C) over the range 80 % to 90 % *SaO_2*	201.12.1.101.5 f) 2)
Mean difference of *SpO_2* reading bias between participants in the dark MST categories (H,I,J) and the light categories (A,B,C) over the range 90 % to 100 % *SaO_2*	201.12.1.101.5 f) 3)
Modified Bland-Altman plot identifying individual participants	201.12.1.101.4 d)
Modified Bland-Altman plot identifying participants by standardized perceived skin colour scale	201.12.1.101.4 e)
Reference method used for pulse rate *accuracy*, if applicable	201.12.1.104 d)
Requirements of ISO 20417	201.7.1.101
Signed pigmentation *differential bias* in graphical format 70 % to 100 % *SaO_2* with 95 % confidence limits	201.12.1.101.5 e) 2)
Signed pigmentation *differential bias* in graphical format centred at 85 % *Sa*	201.12.1.101.5 e) 1)
SpO_2 accuracy, mean bias and *precision* over the range 70 % *SaO_2* to 100 % *SaO_2* for each combination of *pulse oximeter probe* and *pulse oximeter monitor*	201.12.1.101.4 b)
Summary of methods used to establish *SpO_2 accuracy*	201.12.1.101.3 g)
Summary of methods used to establish *SpO_2 accuracy* during motion, if applicable	201.12.1.102 c)
Summary of methods used to establish *SpO_2 accuracy* in low perfusion, if applicable	201.12.1.103 b)
Upper 95 % confidence limit of the observed A_{rms}	201.12.1.101.4 b) 2)

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Annex D
(informative)

Symbols on *marking*

IEC 60601-1:2005+AMD1:2012+AMD2:2020, Annex D applies, except as follows:

Addition:

Additional requirements for the *symbols* on *marking* of *pulse oximeter equipment* or its parts or *accessories* are found in Table 201.D.101.

Table 201.D.1.101 — Additional *symbols* on *marking*

No	*Symbol*	Reference	Title and description
1		IEC 60878:2022 ISO 7000-0626 (*Symbol* 5.3.4 ISO 15223-1:2021)	Keep away from rain (Keep dry) To indicate the transport package which shall be kept away from rain and be kept in dry conditions.
2		IEC 60878:2022 ISO 7000-0435	Assistance; query (Malfunction) To indicate malfunction or to identify the control by which the *operator* can ask for assistance (help button).

Additional Annexes:

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Annex AA
(informative)

Particular guidance and rationale

AA.1 General guidance

This Annex provides rationale for some requirements of this document and is intended for those who are familiar with the subject of this document but who have not participated in its development. An understanding of the rationales underlying these requirements is considered to be essential for their proper application. Furthermore, as clinical practice and technology change, it is believed that a rationale facilitates any revision of this document necessitated by those developments.

AA.2 Rationale for particular clauses and subclauses

AA.2.1 — 201.1.1 — Scope

Pulse oximetry facilitates *patient* care by providing an approximation of arterial haemoglobin saturation with oxygen and allows for the possibility of early detection of the potential complications associated with *patient* hypoxaemia and hyperoxaemia.

Current technology requires an adequate concentration of haemoglobin, a pulsatile change in blood flow, and light transmission through a tissue bed to approximate the in vivo haemoglobin oxygen saturation. *Pulse oximeter equipment* is not typically capable of functioning effectively during cardiopulmonary bypass or at extreme low-flow states and is not, at present, intended as a means for the measurement of blood flow or blood volume.

Given these limitations, *pulse oximeter equipment* does not provide precise measurements of arterial haemoglobin saturation. The presently marketed in vivo *pulse oximeter equipment* is not a replacement for measurement of blood samples by in vitro optical oximeters. The values derived from pulse oximetry are not a measurement of blood or solid-tissue oxygenation. Pulse oximetry provides no direct indication of oxygen delivery to tissue, or of tissue oxygen consumption[85].

This Annex provides a brief account of the history, uses and limitations of pulse oximetry and some recommendations towards the goal of reducing the *risk* to *patients* due to over-reliance on the values of oxygen saturation and pulse rate as well as the *risks* resulting from a 'signal-not-available' situation.

Pulse oximeter equipment is widely used by healthcare providers as an indirect monitor of arterial blood oxygen saturation (SaO_2) along with the traditional four vital signs (i.e. pulse rate, respiratory rate, temperature and blood pressure). It is an important indicator of the *patient's* condition. SaO_2 is an estimate of how much oxygen the haemoglobin actually contains compared to how much it can maximally contain, expressed as a percentage: 95 % to 100 % is "normal" for most young healthy people, and under 90 % is considered low and can require treatment depending on the situation. Direct SaO_2 measurements can be performed on a sample of blood drawn from an artery and, although more expensive and invasive, are considered the gold standard measurement when blood CO-oximetry is measured. CO-oximetry directly measures SaO_2. Pulse oximetry estimates of SaO_2 are displayed as SpO_2, and use an indirect non-invasive optical technology to estimate SaO_2.

Pulse oximetry, invented in 1974, was made available as a research tool in 1977[101] however wider clinical use did not occur until the mid-1980's[100] with various technological improvements. The *pulse oximeter equipment* uses the pulse waveform and a *ratio of ratios* to cancel out the impact from non-arterial blood flow[51]. Early clinical use of *pulse oximeter equipment*, which guided this documents requirements, focused on acute and critical care medicine including anaesthesia for surgical *procedures*, where continuous monitoring

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for early detection and treatment of hypoxemia was expected to significantly decrease *risk* of *harm* to *patients*. With advances in technology and increased availability of *pulse oximeter equipment*, the scope of use has grown to include a much broader range of medical applications including screening or titration for supplemental oxygen[114], screening for sleep apnoea[98] and screening for congenital cardiac defects[95]. At the same time, standards criteria focus on the original use case scenario and do not include the *risk* of clinical decision-making. Thus, clinicians and other users need to be aware of their limitations for optimal use. Lack of this understanding has the potential to lead to systemic error in clinical decision making.

Published literature has raised questions about disparities in pulse oximetry performance in study participants and *patients* with darker skin pigmentation leading to a higher *risk* of missed diagnosis of hypoxemia, delays in treatment eligibility decisions, and worse outcomes. In some of these studies skin pigmentation was measured and in others, pigmentation was inferred by self-reporting of race which served as a proxy for pigmentation. The possible reasons for measurement disparities and methods to address them were given serious consideration during review of this document. A confounding factor has been that methods used to assess skin pigmentation including self-reporting and using existing subjective methods have known limitations[110]. Pending ongoing scientific investigations to address this issue, requirements for *pulse oximeter equipment verification* testing have been, and are being, modified to be substantially inclusive of people with a wide range of skin pigmentations. The number of participants in those studies have been increased to help improve overall accuracy of study data. Expansions of the requirements for study participants in *pulse oximeter equipment verification* testing is where the reporting and analysis of skin pigmentation is having the largest impact on pulse oximetry[109].

At the time of publication, questions remain about the sources and extent of confounding factors contributing to bias in *pulse oximeter equipment* performance. Multiple ongoing studies aim to address these uncertainties. The committees closely monitor emerging evidence and issue updates as new data becomes available. Currently, leading theories regarding the source of oximetry bias include the concentration of melanin, non-melanin chromophores and the detected *percent modulation* of tissue being measured by the probe. This interplay of factors is not yet well understood and evidence to drive decision making is not yet publicly available.

Therefore, this third edition newly requires that *pulse oximeter equipment* be tested in a larger and better characterized diverse population. This third edition requires that *pulse oximeter equipment* now be tested in a population with a wide range of skin tones (201.12.1.101.6). It additionally requires that *differential bias* across the skin tone cohorts of a *controlled desaturation study* across oxygenation ranges be reported in the *accompanying documents* (201.12.1.105).

For optimal use moving forward, the *manufacturers* should provide the clinical user with as much information as possible, such as error ranges and other limitations in an easily understandable format. Clinicians and users should be aware of the potential and specific limitations of *pulse oximetry equipment* for their clinical situation.

As such, the committees have tried to identify useful information that the *manufacturer* should convey to the user of safe operation for *pulse oximeter equipment*. Most of these requirements are noted within Table 201.C.103.

Whilst not yet able to be standardized, expressed clinical preference for ideal functionality includes an indication of estimated confidence limits for displayed SpO_2 values, taking into account the *accuracy* as documented in optimal situations (e.g. laboratory testing in healthy volunteers) as well as currently detected signal variability, signal intensity and modulation. A simpler substitute for this can be a warning on the user interface indicating the 95 % confidence interval as calculated from the laboratory testing, and warnings of impaired precision under scenarios known to impact *accuracy*, such as but not limited to:

— darkly pigmented skin;

— low values of oxygen saturation;

— motion; and

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— perceived and actual perfusion at the sensor site.

Equipment used in research applications is often experimental or intended primarily for non-medical uses. Imposition of the requirements of this document or equipment is not warranted for non-medical uses.

AA.2.2 — 201.3.256 — *Remanufacturing*

The term *remanufacturing* was chosen, instead of terms such as the deprecated reprocessing or refurbishing, because the committees wished to include known use cases, which are broad. Therefore, any activity beyond the *manufacturer's instructions for use* that can result in subsequent reuse is considered *remanufacturing*. This includes an end user or other *manufacturer* cleaning and then reusing a *single use pulse oximeter probe*, as well as using components of *single use pulse oximeter probe* as the raw material for a *remanufacturing process* from a used condition to create a "new" *pulse oximeter probe* for its *intended use*.

AA.2.3 — 201.4.3.101 – Additional requirements for *essential performance*

Sufficient *accuracies* of SpO_2 and pulse rate are necessary for *pulse oximeter equipment* to be suitable for its intended purpose and are required to prevent adverse *patient* events, as is expected by the *essential principles of safety and performance*[7]. When limit *alarm conditions* are provided, *operators* rely on the proper operation of limit *alarm conditions* to alert them to take appropriate actions based on the condition of the *patient*.

Pulse oximeter equipment is expected to maintain these capabilities or indicate to the *operator* that it cannot perform these tasks.

AA.2.4 — 201.7.9.3.1.101 – Additional general requirements

The appropriate application of *functional testers* has been misunderstood by some *operators* or *responsible organizations*. See Annex EE for a discussion of this issue.

AA.2.5 — 201.11.6.5.101 — Additional requirements for ingress of water or particulate matter into *ME equipment* or *ME system*

Fluids commonly found in the care environment include saline, blood and body fluids. Maintaining *basic safety* and *essential performance* following reasonably foreseeable encounters with fluids protects *operators* and *patients* from unacceptable *risks*.

AA.2.6 — 201.12.1.101 — *SpO_2 accuracy* of *pulse oximeter equipment*

SpO_2 accuracy is affected by the combination of the *pulse oximeter monitor*, any cable, the *pulse oximeter probe* and human tissue. Figure EE.6 shows an example of a *pulse oximeter probe* that degrades *SpO_2 accuracy* because of its variable optical influence among different test participants.

A study of *five CO-oximeters*, compared pairwise, from different *manufacturers* using data from 12 participants undergoing a desaturation protocol found that as saturation levels were decreased from normoxic level of 97 % to hypoxic levels of 85 % and 75 % the variance of the SaO_2 measurement increased significantly[64].

Two types of tests in which human participants are used, one for the *verification* of *SpO_2 accuracy* of *pulse oximeter equipment* and the other for *pulse oximeter equipment* development, either type can be performed in the laboratory or the intended environment of use.

a) Invasive testing: the *SpO_2 accuracy* of *pulse oximeter equipment* is measured by comparing SpO_2 readings of the *pulse oximeter equipment* to values of SaO_2 determined from arterial specimens analysed by a *CO-oximeter* in accordance with 201.12.1.101.3. Two types of individuals can participate in invasive studies:

   — healthy adult volunteers who consent to induced hypoxia and arterial blood sampling as part of the experimental *procedure* (see DD.2); and

   — *patients* in whom arterial blood samples are available for analysis (see DD.3).

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b) Non-invasive evaluation testing: the *SpO_2* performance of *pulse oximeter equipment* can be assessed by comparing *SpO_2* readings to values obtained using a *transfer standard*, because its calibration is directly traceable to a *CO-oximeter*.

Some previously published documents appear to mandate that *pulse oximeter equipment* be calibrated directly against in vitro blood analysis using *CO-oximeters*. This annex presents two such published documents and explains when the committees think in vitro analysis is required and when it is not.

The Clinical and Laboratory Standards Institute (CLSI) Pulse Oximetry Guideline[24] says:

> "7.2 To validate pulse oximeter equipment readings, incorporate or assess agreement between *SpO_2* and arterial oxyhaemoglobin saturation (*SaO_2*) obtained by direct measurement, these measurements should be initially performed simultaneously and then periodically re-evaluated in relation to the patient's clinical state."

The committees read the AARC position[29][30][31][32] as a clinical practice guideline that does not address the issue as to how the original calibration of *pulse oximeter equipment* should be established. This committees conclude that the AARC's guideline is appropriate in the clinical context. For a variety of reasons, *pulse oximeter equipment* readings on individual *patients* can differ from *CO-oximeter* readings, so that it is always appropriate in clinical use to consider confirming *SpO_2* readings using a more accurate measurement. This statement is consistent with the point of view expressed in AA.1.

The FDA's *Guidance Document: Pulse Oximeters Premarket Notification Submissions*[28] addresses the issue of the *manufacturer's* performance *verification*. It says, in Section 4.1.1:

> "We recommend you follow 201.12.1.101 and Annex EE.2, Procedure for invasive laboratory testing on healthy volunteers, of ISO 80601-2-61:2011[1] or equivalent method to validate the *SpO_2 accuracy* specifications of your pulse oximeter system by comparing each value from your system and a simultaneous value from CO-oximetry of an arterial blood sample."

The committees have taken the position that the use of *transfer standard* is permissible for use by a *manufacturer* in developing *pulse oximeter equipment* calibration. When data are taken on multiple *patients*, so that statistics can be developed, the committees feel that it is appropriate to use *transfer standard* that have been calibrated against *CO-oximetry* for the calibration of other *pulse oximeter equipment*. In the clinical laboratory conditions under which such calibrations are conducted, many of the known sources of error in pulse oximetry are virtually eliminated. Examples of such sources of error are low perfusion, EMI, motion, nail polish, henna dye, tattoos, *pulse oximeter probe* mis-positioning and ambient light. The efficacy of the calibration *process* using *transfer standard* is demonstrated by the fact that the testing of *pulse oximeter probes* of a given type over many years has consistently produced identical results and has resulted in decades of safe and increasing utilization of pulse oximetry. Regardless of the methods used for establishing *pulse oximeter equipment* calibration, performing *verification* of *accuracy* per 201.12.1.101.1 requires a *controlled desaturation study* with direct comparison to *SaO_2* using *CO-oximetry.*

AA.2.7 — 201.12.1.101.1 — Specification of *SpO_2 accuracy*

There was considerable discussion about the minimum acceptable *SpO_2 accuracy* specification of *pulse oximeter equipment*. Ideally, *pulse oximeter equipment* delivers high *SpO_2 accuracy* (A_{rms} <1 %) with all *pulse oximeter probes* and application sites. However, due to well-known limitations in current pulse oximetry technology, that level of *SpO_2 accuracy* is not routinely achievable[39][51].

Therefore, the committees had to consider the following question: "What is the minimum acceptable *SpO_2 accuracy* for safe and effective use of *pulse oximeter equipment*?"

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1 ISO 80601-2-61:2011 is a previous edition of this document.

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Due to the diverse applications of *pulse oximeter equipment*, minimum performance requirements are not universal. Two general categories of use can be described as monitoring and diagnosis.

— Monitoring can be described as the use of trends or *alarm signals* to facilitate the early detection of saturation or pulse rate changes.

— Diagnosis — or diagnostic use — can be described as measurement of SpO_2 as an estimate of SaO_2 to facilitate diagnosis or guide therapy.

Diagnostic applications, for example medical care in acute and critical care locations (OR, ICU, PACU, transport etc.) and respiratory and sleep lab diagnostic studies, congestive cardiac failure diagnosis, clinical trials, extreme environment healthcare use (battlefield and first responder trauma care etc.), usually require higher SpO_2 *accuracy*. Regardless of the specified SpO_2 *accuracy*, inherent limitations in SpO_2 *accuracy* can necessitate arterial blood sample analysis.

Based on clinical experience and the historical use of *pulse oximeter equipment*, SpO_2 *accuracy* less than 3 % is acceptable for many monitoring applications. With advances in technology and increased availability of *pulse oximeter equipment*, the scope of use has grown to include a much broader range of applications. This expanded range of applications has required design changes to suit specific environments and also recognition of limitations associated with each use type. Clinicians on the committees expressed concern that *pulse oximeter equipment* specified with SpO_2 inaccuracy in excess of 3,0 % can lead to mistreatment in clinical practice. Even though greater SpO_2 *accuracy* is usually more desirable, and frequently attainable, this figure represents a clinically acceptable trade-off between lower SpO_2 *accuracy* and greater flexibility in *pulse oximeter probe* placement and performance.

The committees agreed that it is important to provide a uniform basis for comparing different *pulse oximeter equipment.* That is why this document requires that SpO_2 *accuracy* be specified over the 70 % to 100 % SaO_2 range, as well as why SpO_2 *accuracy* is permitted to be specified over other ranges (e.g. 1 % over the range 90 % to 100 % SaO_2).

The SpO_2 *accuracy* of *pulse oximeter equipment* partially depends on the SaO_2 of the *patient*. Currently designed *pulse oximeter equipment* is generally more accurate at SaO_2 levels above 90 % than they are below 80 %. Limiting the span over which the SpO_2 *accuracy* is stated more realistically communicates the true performance in the range of interest. The span for *pulse oximeter equipment* that specifies SpO_2 *accuracy* below 65 % SaO_2 is limited to 20 % to avoid biasing the low saturation SpO_2 *accuracy* (by averaging in the better performance of the higher ranges).

AA.2.8 — 201.12.1.101.3 — Data collection for determination of SpO_2 *accuracy* and pigmentation *differential bias*

During a *controlled desaturation study*, it is often difficult to achieve a target SaO_2, particularly at the lower end of the SaO_2 range. Attempts should be made at least to achieve a measured SaO_2 within 3 % SpO_2 of the stated range of SpO_2 *accuracy*, including at least 24 participants and at least 480 data points.

The SpO_2 *accuracy* of *pulse oximeter equipment* depends strongly on the optical interaction of the *pulse oximeter probe's* emitted and collected light and the *patient's* blood-perfused tissues. The correlation of the measured pulsatile change in light transmission through blood-perfused tissues and the underlying arterial oxygen saturation depends, among other things, on the spectral content of the *pulse oximeter probe's* emitted light and interaction of the *pulse oximeter probe* optics and the skin surface. Since these complex wavelength-dependent interactions are not assessed nor reproduced by *pulse oximeter equipment functional testers* and simulators, such equipment cannot possibly characterize or *validate* the true *accuracy* of the *pulse oximeter probe/pulse oximeter monitor* combinations. *Functional testers* can be used to verify the functionality of *pulse oximeter monitors* and the electrical integrity of the *pulse oximeter probes*. (see also Annex EE)

Over the last 30 years, public and private hypoxia laboratories have conducted more than 10k hypoxia studies with zero serious adverse events from either the hypoxia *procedure* or the arterial lines. Based on these data, the *risk* of even minor adverse events is less than 0,03 %. Data from these hypoxia laboratories corroborate

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this finding. Studies conducted by Ikeda, MacLeod et al.[79] measuring cerebral venous blood from the jugular bulb showed no abnormally low brain saturations; studies by Bickler and Severinghaus[41] evaluating neurologic function during hypoxia using pupillary dilation and visual tracking demonstrated the absence of any neurologic dysfunction. Many of the adult volunteers in the above hypoxia studies returned for additional hypoxia studies. All of these adult volunteers reported normal function.

Based on evidence presented, the committees concluded that more detail in the recommended methods [201.12.1.101.3 c)] regarding use of hypoxia or arterial lines in healthy adult volunteers supports continued safety. These are:

— Time in low O_2 saturation plateaus (i.e. 60 % to 70 %, 70 % to 80 %, and 80 % to 90 %) should be limited to the minimum amount of time required to obtain stable test/reference data, and as tolerated by the participant, as indicated in [DD.2.3.2 o)].

— Inclusion: Age 18 y to 50 y, ASA category 1[33] and positive Allen's Test.

— Exclusion: pregnancy.

— Monitoring and observation with ECG, heart rate, $EtCO_2$, respiratory rate and FiO_2[121].

— Arterial lines should be placed by a clinician using jurisdictional guidelines for arterial puncture.

— Arterial lines should be placed by a clinician using guidelines (e.g. Centers for Disease Control or CDC) for infection prevention.

— Aftercare should include 5 min to15 min pressure, pressure bandage >1 h, instructions for limited activity and Principal Investigator's contact information.

The committees conclude the *residual risk* inherent in a controlled hypoxia study and arterial line placement in healthy adult volunteers, after suitable mitigation by following these additional *procedures*, can be reduced to a non-significant level.

AA.2.9 — 201.12.1.101.4 — Data analysis for determination of *SpO_2 accuracy*

CO-oximeters have an inherent inaccuracy that influences *SpO_2 accuracy* assessment[64][65][126][127]. To reduce *pulse oximeter equipment* inaccuracy, the inaccuracy of the reference *CO-oximeter's* measurement of *SaO_2* needs to be controlled.

The committees are not aware of a practical or traceable *procedure* that allows a *manufacturer* or *responsible organization* to verify *SaO_2 accuracy* of a *CO-oximeter*. To minimize the influence of the *CO-oximeter* inaccuracy in the A_{rms} measurement, careful attention should be paid to ensure that the *CO-oximeter* is performing within its specified performance capability. *Verification* of correct operation by use of the *CO-oximeter manufacturer's* recommended maintenance *procedures* is necessary, but is not sufficient to ensure a traceable, accurate measurement. Further quality assurance *procedures* for verifying *CO-oximeter accuracy* are needed.

EXAMPLE 1 CLSI[22]

EXAMPLE 2 College of American Pathologists[53]

— d)

The modified Bland-Altman plot was selected over the Bland-Altman plot because the uncertainty in the measurement of *SaO_2* is substantially smaller than that of *SpO_2*.

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AA.2.10 — 201.12.1.101.5 — Data analysis for pigmentation *differential bias* determination

There was considerable discussion around the best approach to measure *differential bias* due to skin pigmentation and its maximum acceptable value(s). The use of *ITA* to objectively quantify skin pigmentation is a significant step forward to characterizing *pulse oximeter equipment* performance with respect to skin pigmentation. Differential bias was defined as the difference in SpO_2 reading bias (expectation of $SpO_2 - SaO_2$) associated with a skin pigmentation difference conferred by a 100-degree difference in *ITA* value at the probe site [see 201.12.1.101.6 f) 14) ii)]. Determining the maximum acceptable *differential bias* and the SaO_2 span for its evaluation was more challenging.

To assist in developing the new testing requirements, the committees analysed *controlled desaturation study* data collected at the UCSF hypoxia laboratory from 34 de-identified finger-based *pulse oximeter equipment* included in the OpenOximetry dataset[63]. Findings indicated that the magnitude of *differential bias* varied greatly among these tested *pulse oximeter equipment* and was SaO_2-dependent in most cases. The largest absolute value of *differential bias* commonly occurred at lower saturations, typically with positive values, and trends to be much smaller in the 95 % to 100 % saturation range. The tightest confidence intervals in their estimates were generally located between 80 % and 85 % SaO_2 and the widest intervals were located at the ends of the saturation range, i.e. 70 % and 100 % SaO_2.

To help in determining a maximum acceptable threshold for this new assessment, overall *accuracy* and bias performance of each of the 34 *pulse oximeter equipment* was ranked independently by many committees members, ranging from very good (1) to unacceptable (4). Rankings were based on subjective reviews of the *pulse oximeter equipment* data represented graphically with trend plots (SaO_2 vs SpO_2), bias plots ($SpO_2 - SaO_2$ vs SaO_2), and pigmentation *differential bias* plots at each saturation from 70 % to 100 % SaO_2. Ranking data was compiled and compared to the respective calculated A_{rms} and pigmentation *differential bias* values of *pulse oximeter equipment* considered at various SaO_2 levels.

The comparisons indicated that A_{rms} had the best sensitivity and specificity to differentiate between "good" and "poor" performing *pulse oximeter equipment*. Pigmentation *differential bias* at the lower saturations also showed good sensitivity and specificity, however this capability decreased meaningfully when assessing *differential bias* at higher saturations. Considering that this bias was generally largest at lower saturations, these results made sense. However, as many clinical decisions are made at higher saturations, the committees did not want to focus only on low saturations that can be perceived to be too far from the point of typical use.

In addition, *differential bias* confidence intervals were found to increase at both ends of the 70 % to 100 % saturation range and were larger in *pulse oximeter equipment* that exhibited a high degree of variability across participants within a particular *ITA* cohort, even when the overall behaviour was otherwise unremarkable. Rankings were found to be more highly correlated to *differential bias* at its confidence limits than to the point estimates across SaO_2 ranges.

Careful consideration of the above concepts steered the committees toward assessing pigmentation *differential bias* using the SpO_2 *mean bias* in the 80 % to 90 % SaO_2 span (equivalently, the *local bias* at the span midpoint of 85 % SaO_2) at the 95 % confidence limits. While assessing the performance in two spans was considered (< 85 % and > 85 % SaO_2), the single middle span centred at 85 % SaO_2 was considered simpler and balances the signal (magnitude of the *differential bias*) with the variability of the method (tightest confidence intervals).

The committees do not consider this evaluation to be a direct statement of skin tone dependent SpO_2 bias at 85 % SaO_2 but rather an indicator of the overall SpO_2 sensitivity to skin pigmentation of the *pulse oximeter equipment*, as the performance in this SaO_2 span shows good correlation to the bias in adjacent ranges. Recognizing that there remains interest in disclosing the observed *differential bias* performance at higher saturations as well, this third edition of this document requires providing a graph of the measured values and confidence intervals at each SaO_2 across the full 70 % to 100 % SaO_2 [see 201.12.1.101.5 e) 2), and Figure CC.9].

The use of two metrics for assessing *pulse oximeter equipment* performance in *controlled desaturation studies* — A_{rms} for overall *accuracy* (bias & *precision*) and absolute *differential bias* at the 95 % confidence limit to

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characterize sensitivity to skin pigmentation — greatly improved distinguishing the ranked best performing *pulse oximeter equipment* datasets from the others.

At the time of publication, there was insufficient published evidence to support specific acceptance criteria for *differential bias*. Much debate and effort were spent assessing what they might be and how they might be represented in this document, recognizing the need to be appropriate for all *pulse oximeter equipment* and independent of probe site. All the tested finger-based *pulse oximeter equipment* in the OpenOximetry dataset subjectively ranked as "good" demonstrated A_{rms} values < 3,0 % at the 95 % confidence limit and *differential bias* < 2,5 % at the 95 % confidence limit (a few were < 1,0 %). Only one of the "poor" performing *pulse oximeter equipment* met both these thresholds, though only marginally. Tightening the threshold below 2 % began to eliminate *pulse oximeter equipment* deemed acceptable.

The committees recognize that smaller levels of allowed *differential bias* is highly desirable. However, data from *pulse oximeter equipment* intended for other probe sites (e.g. forehead, ear) are not available at the time of this publication. This information is needed to help establish an appropriate universal threshold for *differential bias*. Without understanding the consequences of a tight acceptance threshold, the availability of such *pulse oximeter probes* can be jeopardized. The committees, therefore, felt it was premature to establish acceptance criteria appropriate for all *pulse oximeter probe* types. Accordingly, the committees decided to recommend the absolute *differential bias*, as described above, should be < 2,5 % at the 95 % confidence limit and midpoint of the 80 % to 90 % SaO_2 span, with a plan to revisit the threshold and requirements after additional experience and evidence becomes available.

New testing and disclosure requirements, alongside the tightening of the A_{rms} requirement relative to prior editions, are helpful in identifying poorly performing *pulse oximeter equipment*. As *differential bias* is currently a disclosure requirement, clinicians and purchasers should review this information and determine its suitability for their purposes.

To better understand confounding factors contributing to bias in *pulse oximeter equipment* performance, and understand the impact of new testing requirements, the committees strongly encourage *manufacturers* and researchers to contribute to open-access data sharing efforts, such as the repository hosted at https://openoximetry.org/data-repository.

In conclusion, the committees anticipate that *differential bias* performance becomes a requirement (i.e. shall be less than 2,5 %), and that the threshold can potentially be tightened to less than 2,0 % at the 95 % confidence limit at the midpoint of the 80 % to 90 % SaO_2 span in the next edition or amendments. *Manufacturers* are strongly encouraged to update their *pulse oximeter equipment* to meet this recommendation as quickly as possible, pending an update of this from a disclosure requirement to a performance requirement.

AA.2.11 — 201.12.1.101.6 — Requirements of the study population and collected data for the determination of *SpO_2 accuracy* and pigmentation *differential bias*

The following parameters should be considered.

- Number and source of participants
  - The study should be powered with a sufficient number of participants to attain the statistical significance necessary to demonstrate a specified *SpO_2 accuracy* and *differential bias*, see 201.12.1.101.1 and 201.12.1.101.2.
  - Participants should be healthy adult volunteers.
  - For the broadest application to the largest cohort of *patients*, the participants should vary in their physical characteristics to the greatest extent possible.

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NOTE The characteristics of the participants can be limited due to safety reasons or availability (e.g. only female participants being available to *validate* a paediatric finger *pulse oximeter probe* due to their meeting the criteria for finger size).

— Participant inclusion and exclusion criteria including sample size.

— The study protocol should specify the inclusion/exclusion criteria.

— Participants participate in the study on a voluntary basis.

— All participants should be in good health at the time of the study. Unless specified otherwise in the protocol, the following values can be applied: COHb <3 %, MetHb <2 %, *ctHb* >10 g/dl; these values are not intended to be a comprehensive determination of "good health".

— Inclusion criteria should serve the purpose of the study. (Examples are not intended to be comprehensive.)

— Inclusion: Age 18 y-50 y, ASA category 1, positive Allen's Test.

EXAMPLE 1 Both male and female participants.

EXAMPLE 2 Specific finger size.

EXAMPLE 3 Healthy adult participants capable of undergoing controlled hypoxaemia to the levels called for in the protocol with minimal medical *risk*.

— Exclusion: pregnancy

EXAMPLE 4 Smokers or individuals exposed to high levels of carbon monoxide that result in elevated carboxyhaemoglobin levels, unless specific dyshaemoglobins are called for in the study protocol.

EXAMPLE 5 Individuals subject to conditions that result in elevated levels of methaemoglobin, unless specific dyshaemoglobins are called for in the study protocol.

EXAMPLE 6 Participants who would be placed at undue medical *risk* associated with any *procedures* called for in the protocol (e.g. arterial cannulation or hypoxia).

— Criteria for study termination

— Study protocol should specify circumstances or participant response to the *procedure* that becomes grounds for study termination.

EXAMPLE 7 The participant is discovered to meet one of the pre-specified exclusion criteria (e.g. elevated methaemoglobin level).

— a)

The committees evaluated the results of several analyses of sample size and *differential bias* rationale that were based on accumulated *controlled desaturation study* data[63]. The decision is to use a sample size of ≥24 in *controlled desaturation studies* and absolute *differential bias* recommendation criteria of < 2,5 % with 95 % confidence for the SaO_2 range of 80 % to 90 % (or equivalently at its midpoint of 85 %).

A_{rms} sample size analysis follows.

As expected, power increases in diminishing increments (approximately as 1/sqrt(n)) as the sample size goes from 12 to 36 participants in the participant pool. In one of the analyses, bootstrapping[43] of well performing

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and poorly performing *pulse oximeter equipment* was conducted using 3 % A_{rms} acceptance criteria. Participant data sets were ranked by their respective *accuracies* and sampled accordingly to simulate varying performance levels. The probability to detect that a true A_{rms} is 3,24 % (poor performance) or greater was compared in five different sample sizes, 12, 18, 24, 30 and 150. The lower sample sizes are realistic, while 150 represents an upper limit. Results showed that there is a difference in probability detection, of 12 %, between a sample size of 12 and 30; a difference in probability detection of 6 %, between a sample size of 18 and 30; and a difference of 2 % between a sample size of 24 and 30. A sample size of 24 properly fails *pulse oximeter equipment* that has an A_{rms} of 3,24 %, 90 % of the time. See Table AA.1.

Table AA.1 — Probability of detection per sample size

Minimum probability of detecting an RMS of 3,24 % or greater comparison of 5 sample sizes	
Sample size	**Probability of detection**
150	99,9 %
30	92 %
24	90 %
18	86 %
12	80 %

Pulse oximeter equipment that meets *accuracy* requirements in 201.12.1.101.1 has an observed A_{rms} ≤3,0 % in a *controlled desaturation study*. Figure AA.1 shows the approximate probability of obtaining an observed A_{rms} less than or equal to 3,0 % (the statistical power curve), as a function of the true A_{rms}, with varying sample sizes as simulated from the bootstrap distributions with decreasing targeted performance levels. For each performance level, "true A_{rms}" on the x axis is calculated as the weighted mean A_{rms} from the respective distributions.

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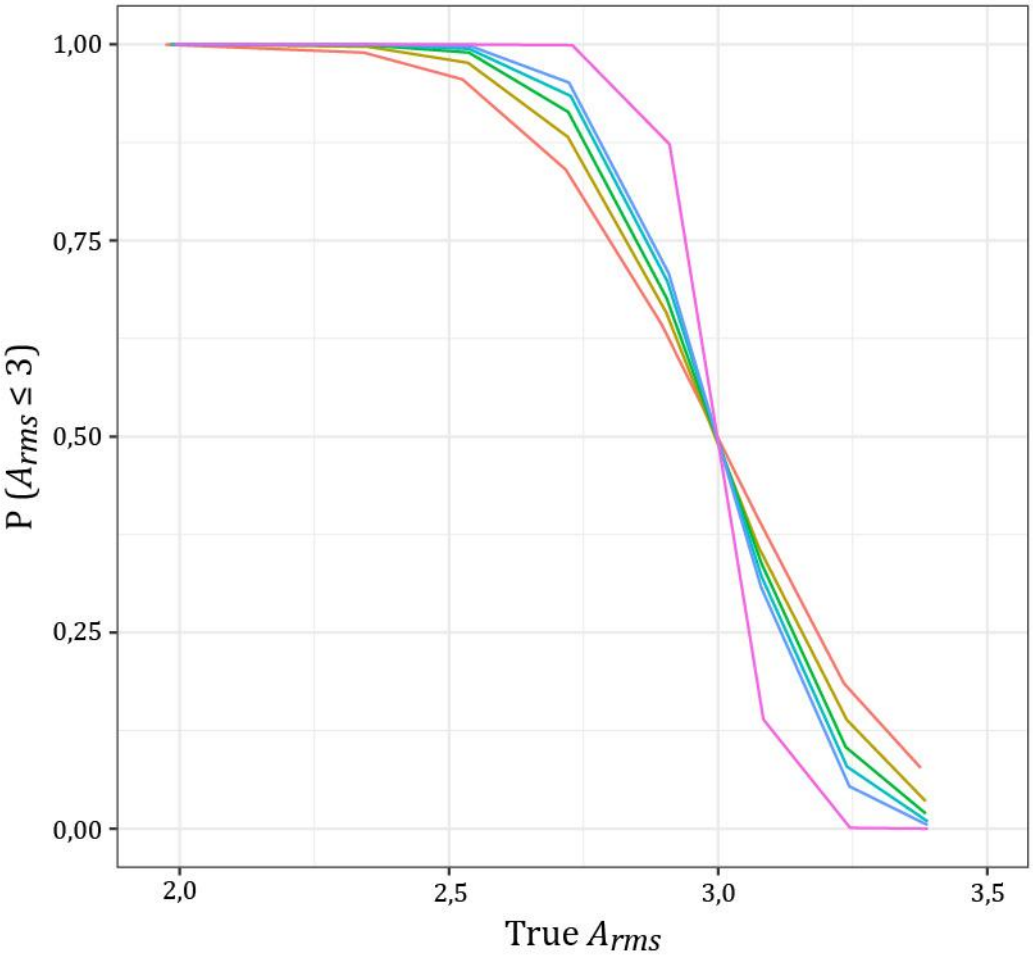

Key

$P(A_{rms} \leq 3)$	is the probability of obtaining an observed A_{rms} less than or equal to 3,0 % using *transfer standard*
True A_{rms}	is the true A_{rms} with a *CO-oximeter* reference
	is the sample size of 12
	is the sample size of 18
	is the sample size of 24
	is the sample size of 30
	is the sample size of 36
	is the sample size of 100

Figure AA.1 — Probability of obtaining an observed A_{rms} for fingerclip vs reference *CO-oximeter* less than or equal to 3,0 %, as a function of the true A_{rms}, by sample size

The diminishing returns (change in power) as sample size increases. Statistical power is usually a function of 1/sqrt(n), so these diminishing returns are expected.

The committees considered the balance between power and feasibility of test implementation. The committees consider the sample size of 24 with a power of 90 % to provide the most reasonable balance.

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— f)

Additional research on the effects of disease, sex, age, pigmentation, and BMI on pulse oximetry is needed. Until the relationship between these factors and pulse oximetry behaviour is well understood it is not possible to disentangle technical performance from physiological factors impacting *SpO_2 accuracy*.

Combined with self-reported ethnicity and a *standardized skin colour scale*, colourimetry /spectroscopy is a valuable scientific tool to assess the variability of skin colours tested in a particular clinical *controlled desaturation study*. The composition of the *controlled desaturation study* test cohort is key to having equitable pulse oximetry performance. Representative *accuracy* across diverse populations can then be presented as part of the study report for determination of *SpO_2 accuracy* as this *accuracy* is partially dependent on the participants composition and the execution of clinical *controlled desaturation study*.

The current theory regarding the source of oximetry bias is thought to be the concentration of melanin in tissue being illuminated. It is also possible that several factors such as non-melanin chromophores and detected signal strength can play pivotal roles in pulse oximetry *accuracy* and bias. This interplay of factors is not yet well understood, however, the documentation of skin colour in the study report for determination of *SpO_2 accuracy* is an important tool in assessing how representative a particular *controlled desaturation study* was, and in describing pulse oximetry behaviour across diverse populations in the clinical setting.

Skin is known to be variable across a single person. The effects of disease, sun exposure and age can change or vary these colours. Skin pigmentation should also not be conflated with race or ethnicity. The rationale for using different skin *colour measurement sites* is to:

— measure skin pigment at skin *colour measurement sites* which have minimal sun exposure, are not prone to redness or erythema and have little to no hair (constitutive);

— measure skin colour at skin *colour measurement sites*, which have average sun exposure, are not prone to redness or erythema and have little to no hair (facultative); and

— measure the pigmentation at the site of *pulse oximeter probe* placement as the closest approximation of skin pigment that can have impact on the *pulse oximeter equipment's* performance.

The commonly used *pulse oximeter probe* skin *colour measurement sites* can exhibit a narrower variety of skin pigment than non-*pulse oximeter probe* sites, such the ventral bicep regions, and sites such as the forehead.

In clinical use it is often not practical to use a *PPM* at a pulse oximetry measurement site due to the size or accessibility to the selected measurement equipment, especially for paediatric populations. Sites such as the forehead and ventral bicep often provide a large enough skin surface area to perform colorimetry/spectroscopy reliably. In infants the sternum can be an acceptable skin *colour measurement site*.

— f) 3)

Race/ethnicity and skin tone are overlapping, but distinct concepts, as race is a social construct. Within and across each ethno-racial category there is variation in skin tone. Research strongly suggests that health disparities and other social inequalities are significantly associated with skin tone. These inequalities are related, at least in part, to colourism, which is a system of discrimination based on differential treatment related to one's skin tone; and is based primarily on how people appear to others (with the forehead being the most common location people perceive when assessing other's skin tones) in social interaction. The addition of diversity requirements in laboratory *SpO_2 accuracy* testing is aimed at mitigating the impacts of colourism and improve the health equity of *pulse oximeter equipment.*

Pulse oximetry *accuracy* is based on the laboratory testing of a few representative samples of individuals. These small pools of individuals attempt to represent the variety found within the human species. To ensure that these pools are as representative as they can reasonably be, requirements are laid out to record the degree of representation.

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Ideally, colorimetry, or if possible, spectroscopy should be used to enrol an adequately diverse study population with regard to skin pigmentations. Self-reported ethnicity and other diversity considerations such as age, gender, etc. should be reported separately. Colorimetry and spectroscopy have limitations and cannot capture aspects of how human skin is perceived[111][140] Things such as skin undertones and to some extent variation in a* can be lacking in, or under-represented by, *ITA*. Requiring a subjective pigmentation measurement be performed by several, trained investigator(s) in a laboratory study, can be beneficial to ensure adequate diversity in the test pool as well as to provide data that can be easily communicated to *lay* audiences[92][132].

— f) 7)

The scope and purpose of the colorimetry efforts described in this document are to provide an objective skin colour measurement for purposes of assessing the impact skin pigmentation has on *pulse oximeter equipment accuracy* and how performance of *pulse oximeter equipment* can vary across different skin pigmentations[134]. The purpose is not to provide an international standard for skin colour measurement. The use of a subjective *PMM* is recommended as a tool to visualize the diversity of enrolled participants that is complementary to an objective *PMM*.

Objective colour measurements are assumed to be presented using the L*a*b* colour coordinate model where L* represents lightness, a* represents the red/green value and b* represents the blue/yellow value. Reflectance spectrophotometry is a *PMM* that generates a reflectance spectrum, which can be converted to L*a*b* coordinates (once assumptions are made about observer angle and illuminant), and then L* and b* can be used to generate *individual typology angle* (*ITA*). Other forms of colorimetry that generate *ITA's*, and that can be reliably mapped as L*a*b* coordinates, can also be considered. Individual L*a*b* coordinates can then be mapped onto the selected representative skin colour scale such as the MST scale and placed into bins.

Subjective pigmentation measurement for the purposes of helping to ensure adequate diversity in the test pool and to provide data that can be easily communicated to *lay* audiences can be done using the Monk Skin Tone Scale (MST scale)[67]. The Fitzpatrick Scale (or FST) is arguably one of the most commonly used and well-known subjective pigmentation measures in the world, but it has come under heavy criticism from experts for inadequately capturing darker skin tones[108][110] which makes the FST a problematic choice to ensure adequate diversity in test pools. It is worth noting that the FST was not originally designed to be a subjective pigmentation measure, but instead is part of a questionnaire to determine the appropriate dose of oral methoxsalen for treating psoriasis using photochemotherapy in Caucasians[60].

The MST scale was explicitly designed to quantify how skin colour is perceived by human beings around the world and has many distinct advantages over other existing subjective pigmentation measures. Specifically, contrary to other subjective scales (e.g. Fitzpatrick Scale, Von Luschan, Massey-Martin, PERLA) the MST scale is well-specified (i.e. has explicitly stated L*, a*, b* values), which allows it to serve as a standardized point of reference for *pigmentation measurement methods* and communication. The MST scale has undergone extensive global *validation* with peer-reviewed publications that demonstrate that the MST scale is perceived as inclusive and representative (significantly more so than the Fitzpatrick Scale) and that it has high levels of annotator/rater consensus, globally[72][122]. Notably, the MST scale has already been widely adopted in the technology industry (e.g. Google and Meta/Facebook) for similar purposes (e.g. ensuring adequate skin tone diversity in datasets and mitigating skin tone related biases in AI & Machine Learning). The MST scale has been evaluated in the context of pulse oximetry: recent studies even finds a stronger correlation between MST scale skin tone ratings and pulse oximetry bias than between colorimeter skin tone ratings and pulse oximetry bias (the MST had a stronger correlation with pulse oximetry bias than other subjective pigmentation measures, such as the FST and Von Luschan)[70],[92].

The MST scale was explicitly developed for use online and with display screens, which is quite likely an important future modality of subjective pigmentation measurement (e.g. for online recruitment of test participants). Standardized print outs of the MST scale are in development and are expected to be available soon - further ensuring that the MST scale can serve as a well-specified and standardized point of reference for both measuring and communicating about skin tone with *lay* people and experts alike.

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When selecting a colorimeter and its settings consider that: certain choices for parameters involved in colorimetry are not critical (i.e. which standard observer position is chosen). If an observer position value is advised by colorimetry equipment manufacturer, this is merely to improve the comparability of studies.

Colorimeters used as *PMM's* should be reflective devices that carry their own light source (i.e. have a flashlamp of some kind). Colorimeters that are 'passive' colorimeters should not be used as a *PMM* (i.e. sensors that measure colour of a light emitting object such as a monitor). Reflectance spectrophotometers can be used as colorimeters since the reflectance spectrum is easily translated into L*a*b* values. If colorimetry equipment offers the ability to select a standard observer (2, 8 or 10 degree) the 10-degree observer is preferred. This is advised to allow better comparison of studies, not because a 10-degree observer is 'better' than colour values for a 2-degree or 8-degree observer equipment. If colorimetry equipment offers the ability to select a measurement aperture, the aperture should be near 3mm. A larger aperture of 10 mm can provide better averaging of inhomogeneous skin colour (e.g. from freckles) but is considered too large for some locations, such as the smallest fingers and neonatal *pulse oximeter probe* sites, whose proportions are smaller and differently shaped than adult smallest fingers. To average out any skin inhomogeneities it is advised to do several measurements, ideally not less than 3, near/around the same location and use the averages of the corresponding computed *ITA* values.

Viewing angle of the equipment versus the standard observer angle (e.g. 8-degree viewing angle for the Konica Minolta equipment) while available xyz curves are for standard observer angles 2-degree and 10-degree viewing angles. Specular reflectance is caused to some degree by oil or moisture on the skin. When taking colorimetry measurements, specular reflectance should be excluded (SCE) where possible. Specular reflectance included (SCI) is acceptable if SCE is not available.

The committees considered two perspectives regarding which site to use for collecting the *ITA* value for determination of *differential bias*. One was to use the *ITA* value at the ventral bicep, the other was to determine the *ITA* value at the *pulse oximeter probe* site, or at a site where the tissue was illuminated by the emitter (i.e. participates in the absorption and scattering).

When measuring *ITA* for use in the calculation of *differential bias*, the committees determined the *pulse oximeter probe* site is the most relevant location, unless measurement at that site is not practical in which case a site that is most representative of the darkest skin tone of the *pulse oximeter probe* site is measured. When a finger *pulse oximeter probe* is used, the dorsal distal phalanx (DP) is the site for determining the *ITA* value (see Figure 201.102). Supporting information for use of this site is two-fold. Skin adjacent to, but not in the light path of the emitters and detector of finger *pulse oximeter probes*, including the dorsal DP, can have a significant effect on the SpO_2 reading. Further, information collected from intensive care *patients* at UCSF Open Oximetry Project (unpublished) shows that skin of the dorsal DP has a similar *ITA* range and distribution as the forehead, though this can vary by context.

A commonly held concept is that the only pigmented area affecting the SpO_2 reading is that which is directly in-line between the emitters and detector. However, it is known that scattering and reflection of photons is the source that causes the entire fingertip to glow. Light is reflected from all adjacent skin and finally reaches the detector, thus, influencing the *pulse oximeter equipment* reading. This condition is illustrated with graphs in Figures EE.5 and Figure EE.6.

Figure EE.5 shows the influence of blue coloured bandage material in the adhesive *pulse oximeter probe*. In this plot, the blue *pulse oximeter probe* material exhibited similar light absorption properties as melanin in very deeply pigmented skin, namely much stronger absorbance in the red (660 nm) than in the near infrared (850 nm to 940 nm).

The blue material replaced part of the bandage material in a standard production *pulse oximeter probe*. The material did not cover the emitters or detector, thus allowing an unobstructed light path.

Plots in Figure EE.6 show results from hypoxia trials in 5 light skinned individuals wearing the *pulse oximeter probe* with blue bandage material. The first plot shows a high bias in participants with blue bandage *pulse oximeter probes*. In that plot, one is a standard production *pulse oximeter probe* without *the* blue material; that *pulse oximeter probe* is on the line of identity with no bias. The second plot shows performance of multiple

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pulse oximeter probes in the same cohort of individuals wearing standard production *pulse oximeter probes*. In this plot, there is virtually no bias. The results indicated that a high SpO_2 bias was created by the skewing blue material that was on the surface of the skin adjacent to the emitters and detector. This was very similar to the way darkly pigmented skin that is adjacent to the emitters and detector absorbs light.

This indicated that skin adjacent to the emitter/detector strongly influences the SpO_2 calculation and is a valid site for measurement of *pulse oximeter probe* site pigmentation.

— g)

Limiting repeated participants is intended to prevent *verification* of final product using the same data set as the development data. It is generally not possible to track participants across studies due to privacy constraints.

AA.2.12 — 201.12.1.102 *Accuracy* under conditions of motion

Pulse oximetry can be challenged by *patient* motion when it disrupts the optical signals used for determining SpO_2. There are numerous interfering factors involved, including but not limited to (1) movement of the *pulse oximeter probe* relative to the *patient's* skin and (2) changes in the local amount of venous and arterial blood within the tissue caused by inertia or varying coupled applied forces at or near the *pulse oximeter probe* site. For systems that include a *pulse oximeter probe* cable, movement of the cable can further adversely affect signals.

When testing motion conditions, the following should be considered:

— Periodic vs aperiodic, continuous vs intermittent, and regular vs chaotic amplitude

— "Active" (*patient* activity-induced disruptions) and "passive" (transport bumps and vibrations) motion noise and scenarios in which each has value

— Methods to quantify motion can include, but are not limited to, pressure pads to record *pulse oximeter probe* movement or signal comparison to the quiescent periods

Some *pulse oximeter equipment functional testers* provide synthetic signals that are intended to mimic a set of disruptions caused by *patient* movement. Playing these signals into *pulse oximeter equipment* directly or through a connected *pulse oximeter probe* can help assess the system's processing algorithms in the presence of added noise in a reproducible manner. However the impact of movement and applied forces at the *pulse oximeter probe*-tissue interface and *cable extender* are critically missing. Similarly, replaying actual pulse oximeter signals recorded from moving human participants can be unique to the *pulse oximeter probe/cable extender/pulse oximeter monitor* used and cannot represent signals from the test equipment system; nor can it be an objective test if the same recorded signals were used in developing the *pulse oximeter equipment*. *Manufacturers* are welcome to supplement their development or testing with *pulse oximeter equipment functional testers*, however such testing alone is insufficient for *verifying accuracy* performance under conditions of motion.

Accordingly, the committees believe prospective human testing of the combined *pulse oximeter probe/pulse oximeter monitor* is required to assure that results represent real-world use.

AA.2.13 — 201.12.4.101 — *Data update period*

Pulse oximeter equipment is required to provide an indication that the displayed SpO_2 value is not current if the *data update period* of SpO_2 exceeds 30 s. Subclause 201.7.9.2.1.101 c) includes a requirement to disclose the *data update period* in the *accompanying documents*. However, there is no requirement that limits the duration of the *data update period*. The additional requirement that "there shall be an indication that the displayed value is not current" was added by the committees based on potentially significant delays that can occur between an event that activates an *alarm condition*, and the actual generation of the *alarm signals*. The displayed SpO_2 value does not reflect changes in the measured SpO_2 value until completion of each update

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period. If an event that activates an *alarm condition*, such as *patient* desaturation, occurs just after the display is updated, a significant delay can occur between the event and the generation of the *alarm signals*. This can create a *hazardous situation* for the *patient* if the *data update period* is long.

To mitigate this potentially *hazardous situation*, the committees believe it is important for the *pulse oximeter equipment* to provide an indication to the *operator* when the displayed SpO_2 value has not been updated in the last 30 s, and as such, can be invalid. This provides the *operator* timely information to assess the *patient's* condition and take appropriate action, if necessary.

AA.2.14 — 201.12.4.102 — Signal inadequacy

SpO_2 *accuracy* can degrade under various physiological and environmental conditions. Clinicians need to be alerted to this possible degradation of performance to avoid unsafe conditions. The plethysmographic display can provide information relevant to this condition.

Many factors contribute to degradation of signal adequacy with potential loss of *accuracy*. Changes of the plethysmogram can be sensitive to noise and changes in *%mod*, but plethysmographic changes are not specific to factors that degrade *accuracy* versus factors that corrupt the plethysmogram but do not degrade *accuracy*. These factors can include, but are not limited to:

— signal strength;

— noise frequency and amplitude;

— source of noise;

— plethysmographic morphology;

— ambient light intensity; and

— *pulse oximeter probe* positioning and alignment.

It is beneficial to provide a means for assessment of signal adequacy as it relates to general performance, including confidence in measurement *accuracy*. Although this is best accomplished by a comprehensive real-time assessment of signal adequacy and a visual indication of said status, it can also be partially accomplished in a clinically acceptable manner (e.g. with an appropriately scaled plethysmographic display).

A non-scaled pulse plethysmographic display can lack the resolution to reveal clinically significant changes in pulsatile signal strength in the low range[36][37][60]. Therefore, scaling of the plethysmographic display to increase resolution in the low signal-strength range can enhance the utility of the plethysmogram for assessing signal adequacy with respect to *%mod*.

The non-pulsatile signal strength (the DC component of the time varying signal) is an important parameter to include when determining signal adequacy, especially as it relates to signal levels relative the *pulse oximeter* noise floor and digitizing resolution. Many factors can influence DC signal strength such as the *pulse oximeter probe* location and skin pigmentation. Melanin absorption is stronger for shorter wavelengths compared to longer wavelengths and can create an imbalance in the DC signal levels between the red and infrared signals[80][81][119][139]. Relying only on the appearance of the plethysmogram can hide these poor signal conditions and highlights the importance providing a signal inadequacy indicator that includes both pulsatile and non-pulsatile components of signals at each of the wavelengths used for determining SpO_2 and pulse rate.

The threshold for providing an indication of signal inadequacy should be determined by the *manufacturer* of *pulse oximeter equipment*. The threshold can depend on many factors relating the unique design of the *pulse oximeter*. Hence, a universal threshold cannot be provided for all *pulse oximeter equipment* and is *manufacturer* specified. The indicator can consist of several components that inform the *operator* about the inadequacy in the signal. In addition to a simple indicator, additional status information from the *pulse oximeter monitor* can be displayed, for example: low *%mod*, motion artifacts, or ambient light.

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At a minimum the *pulse oximeter* can be assessed over a range of *%mod* and DC signal levels (including an DC signal level imbalance between the wavelengths, to replicate the absorption due to melanin). A *functional tester* can be used to generate reproducible and repeatable signals at various levels to assist in developing or testing an indicator of signal inadequacy. When the displayed SpO_2 or pulse rate deviates beyond a specified threshold from the set-point of the *functional tester*, an indication should be provided to the user.

AA.2.15 — 201.15.3.5.101 — Additional requirements for rough handling

ME equipment, including *pulse oximeter equipment,* in *normal use,* used for professional transport of a *patient* outside a professional healthcare facility likely is subjected to these mechanical stresses (e.g. vibration, shock, bump, and drop) and can randomly be subjected to additional stresses. Therefore, *ME equipment* intended to be used for professional transport of a *patient* outside a professional healthcare facility needs to be robust enough to withstand the mechanical strength testing described by IEC 60721-3-7:2026, class 7M3. IEC 60721-3-7:2026 indicates that in addition to the conditions covered by class 7M2, class 7M3 applies to use at and direct transfer between locations with significant vibrations, or with high-level shocks. Rough handling and transfer of *ME equipment* is expected in these environments.

There are no established generalized test programmes that exactly reproduce the variety of vibration and shock conditions that *ME equipment* can meet when installed in a range of land vehicles and aircraft. Therefore, the dynamic tests specified in this subclause have been chosen because *ME equipment* tested to these levels are likely to withstand the dynamic disturbances routinely seen during use in the range of vehicles and aircraft (including helicopters) likely to be used to transport *patients*.

The use of *ME equipment* in road ambulances, fixed-wing and rotary wing aircraft, naval vessels, etc. can require additional tests and *verification* of safety when used in these different environments.

For free-fall testing described in IEC 60068-2-31:2008, the committees used the rationale for the various levels to gauge the severity of the test based on Table AA.2 of this rationale. The category of the test level chosen for *portable ME equipment* was *portable* cases. The committees agreed that *pulse oximeter equipment* should be required to meet a level of drop testing for the professional transport environment. The committees also agreed that much *pulse oximeter equipment* is likely to be supplied with a protective or carrying case for use in transport environments. The committees agreed that it is an adequate test for *portable ME equipment* to be dropped while in their carrying cases, as this is most like the real-world environment. For *mobile ME equipment*, a less severe level was chosen since wheeled *ME equipment* is typically heavier.

In selecting the requirements, the committees reviewed other sources for material related to these tests (e.g. FDA Reviewers Guidance[28] for premarket notification submissions) but found the best fit was with IEC 60721-3-7:2026 and its companion IEC/TR 60721-4-7:2001. This document mapped well to the requirements indicated in Table AA.1 of IEC/TR 60721-4-7:2001. The aforementioned document specifies 3 classes of mechanical conditions: 7M1, 7M2 and 7M3. The committees found that classes 7M2 and 7M3 best represent the conditions seen during transport of a *patient* within a professional healthcare facility and professional transport of a *patient* outside a professional healthcare facility, respectively. The committees agreed that different tests and test levels should be applied to *ME equipment* intended for use in a professional healthcare facility versus *ME equipment* intended for use during professional transport of a *patient* outside the professional healthcare facility.

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Table AA.2 — Qualitative assessment of *pulse oximeter equipment* shock and vibration environment

***ME equipment* category**	**Location**											
	Standard environments						**Transport vehicles**					
	Home			**Healthcare facility**			**Wheels**			**Wings/Rotary**		
Mobile	D1	S1	V1	D1	S2	V1	D1	S3	V2	D1	S3	V3
Portable	D1	S2	V0	D1	S2	V1	D1	S3	V2	D1	S3	V3
Hand-held	D3	S1	V0	D3	S2	V1	D3	S3	V2	D3	S3	V3
Stationary	None			None			Not applicable					
S = shock; V = vibration; D = drop Rating: 0 = no test, 1 = least severe or 7M1[a]; 2 = moderate severity or 7M2; 3 = most severe or 7M3												
[a] The 7Mx designations are specified in IEC 60721-3-7:2026.												

Verifying that the *ME equipment* is functioning within the *manufacturer's* specifications while the vibration (random and sinusoidal) tests are being conducted was not believed necessary. This line of thought was considered, and it was decided that the test done in this manner is overly burdensome and only adds a minimum additional level of safety to the *ME equipment* that does not outweigh the costs. Verifying proper functioning after completion of the tests is believed adequate.

AA.2.16 — 201.15.3.5.101.1 — Shock and vibration

ME equipment, including *pulse oximeter equipment*, in *normal use*, used within a professional healthcare facility likely is subjected to these mechanical stresses (e.g. vibration, shock) and can randomly be subjected to additional stresses. Therefore, *ME equipment* intended to be used in the professional healthcare facility needs to be robust enough to withstand the vibration and shock testing described by IEC 60721-3-7:2026 level 7M2. IEC 60721-3-7:2026 indicates that this class applies to use at, and direct transfer between, locations with only low-level vibrations, or with medium-level shocks. Careful handling and transfer of products is expected in these environments.

The committees chose the duration of 10 min for the random vibration test even though the duration recommended by IEC 60721-3-7:2026 is 30 min to maintain testing compatibility with ISO 80601-2-61:2011. As was indicated in the previous edition, the committees increased the duration to 30 min.

AA.2.17 — 201.101 — *Pulse oximeter probes* and *probe cable extenders*

Pulse oximeter probes and *probe cable extenders* are as important in establishing the safety and *accuracy* of the complete *pulse oximeter equipment* as is the *pulse oximeter monitor* itself. Subclause 201.101 establishes that the *manufacturer* of the *pulse oximeter probe* or *probe cable extender* (including a *manufacturer* of a *remanufactured pulse oximeter probe* or *probe cable extender*) is responsible for the separately testable properties (e.g. biocompatibility) of their components as well as for the affected combined properties. This *manufacturer* is responsible for testing the affected combined properties for their *pulse oximeter probe* or *probe cable extender* when used with any *pulse oximeter equipment* they have specified as compatible. The affected combined properties include at least *accuracy*, EMC, electrical safety, and protection against excessive temperature at the *pulse oximeter probe*-tissue interface. As an example of a possible effect of *remanufacturing* on biocompatibility, some surface-cleaning agents can result in impregnation of the material with solvent that, if not sufficiently removed by subsequent *processing*, can cause a chemical burn when that *process* is not described (and therefore *validated*) in the *accompanying documents*.

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AA.2.18 — 201.103.2 — Connection to electronic health record or other *ME equipment*

The standardization of a minimum set of parameters and settings allows greater interoperability between *pulse oximeter equipment* and *ME systems*, thus enabling new applications and paradigms that can increase *patient safety* and improve *patient* care.

The Society for Technology in Anesthesia (STA) proposed the following horizontal guidance: [137]

A medical device electronic data interface (EDI) shall be capable of communicating the following.

— Medical device identification data including *manufacturer*, model number, device serial number and software/firmware version number.

— All data available for display to the *operator*, including numeric values, waveforms and *alarm conditions*.

— The mode of operation and the state of *operator*-configurable equipment settings (e.g. signal filters, signal averaging time, *alarm limits*).

— Medical device clock time and last clock time update, time zone.

— *Patient* ID if stored in the medical device.

This document applies these general requirements to *pulse oximeter equipment*.

Pulse oximeter equipment should be equipped with a *functional connection* that permits integration into an integrated clinical environment. This connection is usually accomplished with an electronic data interface. The type of use drives the requirements. The electronic data interface of a *pulse oximeter equipment* can be used to provide *patient* data for the electronic health record. It can also be used to connect different *ME equipment* to achieve a new medical function. Electronic data that is used to archive physiological data that has been averaged over several seconds or minutes for later review has different requirements than a real-time electronic data interface that is intended to be used for a closed loop system[12].

Some regulations require "state-of-the-art." The committees conclude that existing *pulse oximeter equipment* that do not conform with the data interface requirements per Annex GG of this document should not preclude marketing existing *pulse oximeter equipment* that have otherwise adequate clinical functionality despite limitations in communication capabilities. Loss of access to this otherwise acceptable *pulse oximeter equipment* can undermine the goals of increasing access to acceptable *pulse oximeter equipment*. The committees have the intention to not only ensure safe data interfaces with this subclause but also want to stimulate clinical care innovation with regard to interoperability. Therefore, the requirements for the electronic data interface in this clause (and as detailed out in Annex GG), should not be misunderstood to render existing *pulse oximeter equipment* to be no longer "state-of-the-art." The committees wished to differentiate between performance related requirements and data interface requirements. The committees did not intend that the data interface requirements be construed as "state-of-the-art" and therefore, impact the *manufacturer's* ability to continue to place existing *pulse oximeter equipment* on the market. Such consequences impair innovation instead of fostering innovation regarding electronic data interfaces. The committees have limited the requirements in this subclause and in Annex GG to new generic interfaces that are intended for open interoperability. Internal system interfaces or *manufacturer* specific interfaces (e.g. between a bedside *patient* monitor and a central station provided by the same *manufacturer*) as well as data interface between specified communication participants are exempted from the requirements of Annex GG.

Nevertheless, Annex GG provides guidance regarding the importance of certain data elements to help *manufacturers* to design electronic data interfaces to support innovations in clinical care.

The transmission of *pulse oximeter equipment* data to other *ME equipment*, *ME systems* or health software systems for purposes including decision support, control and data logging is problematic due to missing data elements or the use of proprietary interfaces and protocols. It is the intent of this document, as a safety and performance standard, to specify a minimum set of measured parameters, equipment identification

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parameters and equipment settings that should be available for transmission if the *pulse oximeter equipment's* electronic data interface is intended to be incorporated into an integrated clinical environment (ICE).

The ISO/IEEE 11073 family of standards specifies an information model for *pulse oximeter equipment* and includes a device specialization document (ISO/IEEE 11073-10404:2022). Another approach divides the healthcare space into domains. One such domain, the IHE Patient Care Device (PCD)[2] domain through the use of existing standards such as HL7 and clinical language vocabularies such as LOINC, is described as providing a framework for integrating medical devices into the healthcare enterprise. AAMI 2700-1:2019 describes the need for an "ICE" model to take into consideration interactions with the connected *ME equipment*, workflow, and *patient* state to support the coordinated use of medical devices for improved *patient safety*[19].

Pulse oximeter equipment in clinical use has provided parameters, identification data and settings through the *functional connection.* However, that data has been transmitted primarily using proprietary interfaces and protocols. To help foster interoperability of *pulse oximeter equipment* in the medical device ecosystem, increased standardization of this interface is desirable. This document has sought logically to categorize the data that can be transmitted or received as parameters, identification data, settings data, configuration data, specification data, service monitoring data and *alarm system*-related data. In addition to these categories and types of data, *manufacturers* are encouraged to leverage the *functional connection* to allow the *pulse oximeter equipment* greater capabilities, including the use of intelligent algorithms that can reside in the *pulse oximeter equipment* and which can adjust their algorithm or display settings based upon information received externally. This includes, for example, the location, status and data from other sensors, such as the sphygmomanometer to indicate inflation if the cuff and *pulse oximeter sensor* are on the same limb.

While this is a desirable context information, the reality shows that the burden cannot put on the caregivers to provide this information to the system. Automated and failure prove detection of that actual application sites of cuffs and pulse oximeters is currently not available and can require further technology developments.

With the proliferation of *pulse oximeter equipment* with varying performance and features, it is becoming increasing important for clinical care to determine the *pulse oximeter equipment's* suitability for a particular clinical application. At the moment, this is solely determined by caregivers based on their knowledge of the equipment and requirements of the application. Given that the requirements for an application such as closed-loop control (whether autonomous or with a clinician providing the titration) depends on *pulse oximeter equipment* specifics, such as averaging time, *accuracy* of attached *pulse oximeter probe*, time response and delays in the *pulse oximeter equipment*, the determination of applicability of the *pulse oximeter equipment* can be challenging for the average caregiver. However, if the *pulse oximeter equipment* provides this information through the *functional connection,* the determination of applicability can be made by querying the equipment settings, configuration and specifications. The committees are aware that these are specific requirements for a currently limited number of applications that are using pulse oximeters. Therefore, the committees have concluded that these requirements do not need to be applied to all *pulse oximeter equipment* with a *functional connection.*

AA.2.19 — 208.6.1.2.101 — Additional requirements for *alarm condition* priority

The language in the previous versions of this document is similar, except the introductory phrase is "If intended for continuous monitoring…" This language led to extended discussions among experts and their advisors as to just what were the circumstances in which low SpO_2 attended monitoring" are sufficiently ambiguous to require extensive clarification, and can be interpreted to include sleep studies, which do not require *physiological alarm conditions* at all. The committees finally agreed that *operators* and *responsible organizations* should know when they require a *pulse oximeter monitor* to have *physiological alarm conditions*, so that a useful contribution of this document is to ensure that *pulse oximeter monitors* having no *physiological alarm conditions* are *labelled* appropriately [see 201.7.2.101 a)and 201.7.9.2.1.101 f)], and that if such *alarm*

2 Available at https://wiki.ihe.net/index.php/Patient_Care_Device.

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conditions are included, there is an *alarm condition* for the parameter that is usually most important (i.e. low SpO_2).

Some *pulse oximeter monitors* can have *technical alarm conditions* for *pulse oximeter equipment*-related variables, such as low battery, but no *physiological alarm conditions*. Such *pulse oximeter monitors* are not required to have a low SpO_2 level *alarm condition*.

AA.2.20 — 208.6.5.4.101 – Additional requirements for *default alarm preset*

An 85 % SpO_2 is a generally accepted lower *alarm limit* for most clinical situations; however lower *alarm limits* can be desirable in particular clinical conditions. The *operator* is permitted to set lower *alarm limits* during *normal use*.

In selecting 85 % SpO_2 as the minimum *manufacturer*-configured default *alarm limit* for the low SpO_2 level *alarm condition*, the committees made a compromise between two clinical requirements. One requirement was that *pulse oximeter equipment* should act as an early indicator of distress in a *patient* with relatively normal oxygenation. In this situation, it can be good clinical practice to select a default *alarm limit* above the "knee" of the oxyhaemoglobin dissociation curve that provides as much margin of safety as is practical. The second requirement is to avoid frequent *alarm signals* not necessarily requiring clinical intervention, which can "desensitize" caregivers to *alarm signals* (i.e. cause 'alarm fatigue'). In this case, one can argue for a default *alarm limit* low enough to guarantee that most *alarm conditions* are meaningful by anyone's measure. It was acknowledged that in both clinical situations, many, if not most, *operators* were likely to rely on the default low SpO_2 *alarm limit*.

Another factor the committees considered is that many examples of *pulse oximeter equipment* intended for continuous monitoring allow *responsible organization*-configured or *operator*-configured default *alarm limits* and that for specific monitoring settings, default *alarm limits* that were more closely tailored to the needs of the *patient* and *operator* in that setting can be selected. Given these considerations, a lower limit of 85 % SpO_2 for the *manufacturer*-configured default *alarm limit* was felt to be an acceptable compromise that best met both clinical requirements.

AA.2.21 — DD.3 – *Procedure* for testing on *patients*

The objective of laboratory testing is to control an environment such that there is only one variable being tested, in this case SaO_2. Real-world, or in hospital testing (sometimes referred to as convenience sampling), compounds the analysis of uncertainty. This is why SpO_2 *accuracy* is typically determined in the laboratory setting.

Intentionally desaturating potentially critically ill hospitalized *patients* is unethical, unsafe and is confounded by an array of pathologies and comorbidities, priority should always be given to *patient* care in the hospital setting. Ethical behaviour as mandated by the Hippocratic oath, prevents the protocolized matching of CO-oximetry and pulse oximetry measurements, as seen in laboratory desaturation studies. To allow comparisons, in-hospital convenience sampling pulse oximetry studies should be limited to recording SpO_2 values and taking blood samples only during periods of extended haemodynamic and SpO_2 stability, both for the safety of *patients*, and to allow for an approximation of *accuracy* in the study[110]. A real-world evidence study should report results using a Bland-Altman plot, as compared to a *controlled desaturation study* where a modified Bland Alman plot is required. This is because the correlation coefficient in real world study is not known with confidence and is likely less than 0,9.

Due to differences in the hospital and laboratory settings, tight controls over pulse oximetry and CO-oximetry *accuracy* impacting factors often is not possible. Thus, although informative to the clinical performance of a *pulse oximeter equipment*, for sick and at-risk populations, *pulse oximeter equipment accuracy* studies should be inferred from laboratory data in healthy adults. Observational, in-hospital convenience sampling studies should be informative to *accuracy* but not considered definitive due to these limitations.

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Annex BB
(informative)

Skin temperature at the *pulse oximeter probe*

BB.1 Summary

A literature review relating to temperature requirements leads to the conclusion that it is appropriate and conservative to retain the 41 °C limit for infants (*patients* up to 1 year of age) and to apply the limits of 42 °C for 8 h and 43 °C for 4 h for older *patients*.

BB.2 Literature review

The committees have taken the use of *external* heat to produce a 35 °C surface temperature, in the absence of strong peripheral circulation, as being worst case. Although strong local perfusion can lead to a skin temperature of 35 °C or above, forced convective heat transfer by blood increases the effective thermal conductivity of the skin. Thus, if the 35 °C temperature is endogenously produced, a given heat input from the *pulse oximeter probe* produces less temperature rise. Additionally, use of external heat in the care of *patients* can routinely reach 35 °C (e.g. incubator, radiant warmer), so an ambient environment of 35 °C for testing *pulse oximeter pulse* thermal performance is appropriate.

In this document, the committees have adopted the FDA's 35 °C guidance for the test environment and made explicit an interpretation that "ambient" temperature, as used in the FDA guidance[28], can be taken as local skin temperature when the *pulse oximeter probe* is not energized. Heat generated by the light-emitting diodes of a *pulse oximeter probe* primarily dissipates through the skin of the *patient*, not to the surrounding air. Thus, the *patient's* skin temperature (without the *pulse oximeter probe*) is much more important in determining the temperature to which the *pulse oximeter probe*/skin interface eventually rises than is the temperature of the surrounding air. It is therefore appropriate for skin temperature, rather than air temperature, to be specified.

The same 35 °C maximum skin temperature appears in this document for neonates as for adults. 35 °C is a sufficient maximum, even though infant incubators can be adjusted to raise abdominal skin temperature as high as 37 °C. In the absence of strong local perfusion, the skin of the extremities is several degrees cooler than the skin of the abdomen, as indicated in the following literature:

– Templeman and Bell[129] showed mean heel temperatures near 33 °C, when abdominal temperature was regulated in the 36 °C to 37 °C range, in both air-heated incubators and radiant warmers;

– Malin and Baumgart[96] showed, in a radiant warmer environment, mean heel temperatures were 4,5 °C below mean rectal temperature when the abdominal wall temperature was 35,5 °C, but only about 2 °C below at 37,5 °C;

– Topper and Stewart[130] studying the use of heated water pads to supplement radiant warmers, found back and abdomen temperatures were nearly equal, but mean foot temperature was about 2,1 °C lower (heating pad on) and 2,6 °C lower (heating pad off);

– Seguin[123] studied the distorting effects on incubator servo control of heated transcutaneous sensors. During the control phase, with the transcutaneous sensor not in use, mean foot temperature was 33,4 °C, for an oesophageal temperature of 36,9 °C. This work was with radiant warmers, servo-controlled for an abdominal skin probe temperature of 36,5 °C to 37 °C;

– Harpin *et al.*[71] studying the responses of newborns to overheating, in air-heated incubators, showed a consistent pattern in which hand temperature was 1,5 °C to 5 °C below rectal temperature when the baby

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was at the low end of the "thermoneutral" range, to about 0,5 °C below rectal temperature when the baby was overheated. The authors interpreted the higher hand temperatures as consistent with stronger local circulation.

– Greenhalgh, *et al.*[68] studied *patients* scheduled for removal of redundant skin (abdominoplasty, breast reduction surgery). *Pulse oximeter probes* were applied and left in place for 8 h (or less if significant pain was noted) and set at 42,5 °C, 43 °C, 43,5 °C, and 44 °C. They found that *pulse oximeter probes* were safe up to a temperature of 43 °C for at least 8 h on well-perfused skin.

Additional published reviews and case reports confirm these findings[6][38][46][49][50][82][105][113][115].

There is little experimental evidence supporting the possibility that the natural damage-repair mechanism of skin is weaker when circulation is poor and whether that can lead to a lower threshold temperature for thermal injury[137][138]. An early direct experiment[72] done on pigs showed no effect of local perfusion on injury threshold. More recent experiments, also on pigs[78][86], showed that in the presence of high local pressure (100 mmHg) over a large area (51 mm to 57 mm diameter) it is hard to determine a threshold temperature for injury. Greater injury occurred at 35 °C than at 25 °C, but some injury occurred at 25 °C. Any recommended safe temperature threshold for *pulse oximeter probes* should be accompanied by the usual caution that *pulse oximeter probes* need to be applied so as to avoid excessive pressure[91]. Given this precaution, the recommended temperature thresholds appear safe in view of the most pessimistic literature values. In this way, the effects of poor perfusion that probably existed in some of the experimental participants who were studied have been included.

Table BB.1 shows the committee's best estimates of the safe skin temperature thresholds implied by each of many reports in the clinical literature. The inconsistencies among these reports arise from at least two causes.

– All the available data for neonates come from studies of transcutaneous blood gas monitoring, in which the observed variable is usually the temperature of the transcutaneous sensor core. Skin temperature is an uncontrolled variable, which the committees have estimated as being 1 °C below the transcutaneous sensor core temperature, but which can actually vary more widely[50][58][76][83].

– Important variables, including the *accuracy* of temperature measurements and the varying physiology of *patients*, were not addressed consistently in many of these experiments.

To interpret each report, the threshold safe temperature was taken to be the level at which no blisters were observed. Erythema, which can imply heat-induced hyperaemia, or can imply thermal damage to part of the thickness of the epidermis (commonly called a first-degree burn), was taken as marginally acceptable, since recovery from simple reddened skin is typically rapid. Blisters are unambiguously recognizable as injuries and imply damage to basal cells in the epidermis (a second-degree burn). If the duration of exposure was less than 8 h, the committees arrived at the safe 8 h temperature using the rule of thumb of Moritz and Henriques[72][104] that doubling exposure time reduces the safe temperature by 1 °C.

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Table BB.1 — *Pulse oximeter probe* safe application time and source

Reference	Safe skin temperature for *n* h	Safe skin temperature for 8 h
Neonates		
Boyle, 1980[46]	43 °C for 4 h to 7 h	>42 °C
Bucher, 1986[48]	41 °C for 24 h	>42 °C
Cabal, 1981[50]	42,5 °C for 4 h	>41,5 °C
Eberhard, 1975[57]	41 °C for up to 84 h	>42 °C
Eberhard, 1976[58]	43 °C for 4 h "eliminate[d] the risk of blister formation *almost* entirely". 42 °C was "tolerated well [for] up to 24 h."	42 °C
Fanconi, 1996[59]	41 °C for up to 24 h, in the absence of eugenol	>41 °C
Golden, 1981[66]	<42 °C for 2 h	<40 °C
Huch, 1981[76][77]	44 °C for 1 h (appears to be a purposely conservative guess. No data presented)	41 °C
Laptook, 1981[90]	43 °C for 4 h	42 °C
Löfgren, 1983[93]	<43 °C for 8 h	42 °C
Monaco, 1981[102]	43 °C, 3 h to 4 h	42 °C
Rimdeika, 2005[117]	≤42°C, burns noted after 15 min contact with water warmer	≤ 42°C,
Schachinger, 1983[120]	<43 °C, 2 h. No original data presented	<41 °C
Venus, 1981[131]	44 °C, up to 6 h	43 °C
Intermediate ages		
Poler, 1992[113]	43 °C for period of application of pulse oximeter	43 °C
Adults		
Greenhalgh, 2004[68]	43 °C for 8 h	43 °C
Manzinger, 1990[97]	Rats, not humans. Water baths at 60 °C, 75 °C, and 90 °C, for 4 s, 10 s, or 15 s	Results generally support Moritz
Moncrief, 1979[103]	44 °C for 6 h (this is a review article, not an experimental report, and is possibly based on Moritz[73][105]	>43 °C
Moritz, 1947[73][105]	44 °C for 5 h	>43 °C
Poler, 1992[113]	43 °C for period of application of pulse oximeter	43 °C
Vyas, 1988[136]	43 °C for 8 h	43 °C
Wienert, 1983[138]	<43 °C for 8 h	<43 °C

The literature references fall, for the most part, into two cohorts. There are many citations of work with transcutaneous monitors, which apply for the most part to neonates. Another group of documents represent burn threshold studies with adult volunteers. Only a few references apply to participants in the intermediate age cohort.

Reviewing the estimates in Table BB.1 led to the following conclusions:

– 42 °C should be safe for infants (including neonates), but there are enough conflicting results to warrant caution. For this reason, it is recommended that the traditional 41 °C limit for infant applications not be increased and that the default setting of 41 °C be retained.

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– 43 °C for 8 h should be safe for adults, but there have been few studies since the classic work of Moritz *et al.;* and the results of Wienert *et al.* suggest caution. For that reason, it was concluded that the justifiable limit for adults is 42 °C for 8 h, and (using Moritz's rule), 43 °C for 4 h.

It is appropriate and conservative to retain the 41 °C limit for infants (*patients* up to 1 year of age) and to apply the limits of 42 °C for 8 h and 43 °C for 4 h for older *patients*, based on the observation that dermal circulation is immature before 1 y of age[114] and that in other structural respects the skin is adult-like by this age[118].

BB.3 Test methods

This document does not require a particular method of measuring the skin temperature beneath the *pulse oximeter probe*. There are many different widely known and accepted methods of measuring surface temperatures. Different *pulse oximeter probe manufacturers* have evolved their own methods of measuring temperature, using either human test participants or thermo-mechanical simulators. It is impractical today to find a single universally acceptable test method, and the excellent thermal safety record of pulse oximetry suggests that such a method is not necessary. *Pulse oximeter probe* designers who wish to take advantage of the higher temperatures should keep the following cautions in mind.

– Measurement tolerances are required to be evaluated carefully. The *manufacturer* should know the true *accuracy* of temperature measurement when designing *pulse oximeter probes* for use at temperatures above 41 °C since a higher temperature reduces the margin of safety.

– Temperature sensors are required to be small enough so as not to distort the measurement. The largest temperature sensors that have been found acceptable have characteristic dimensions near 0,5 mm (e.g. the bead of a thermocouple welded from 0,25 mm wire). Often still smaller temperature sensors are used.

– The temperature sensor is required to not reduce the measured peak temperature by conducting a significant amount of heat away from the measurement region. Thus, it is usually inappropriate to use the copper-constantan type T thermocouples that are common in medical investigation since the high thermal conductivity of the copper wire can cause a falsely low temperature measurement.

– Experimental methods are required to be adequate to ensure that recommended temperature limits are met under "reasonable worst case" conditions. As an example, reasonable worst case for neonatal *pulse oximeter probes* can include the following conditions.

  – The *patient* has poor peripheral circulation. There is therefore little forced-convection heat transfer by blood to increase the effective thermal conductivity of surface tissue.
  – The light-emitting diodes (LEDs) in the *pulse oximeter probe* are driven at the maximum current which the *pulse oximeter monitor* is capable of providing during normal operation (this condition can occur in a *patient* with very dark skin, thick skin, or large tissue light path, e.g. thick foot or wrist).
  – An active heat source is in use to raise the baby's abdominal skin temperature artificially to 37 °C.

It is not the intention to require that every model of *pulse oximeter probe* be tested directly on "worst-case" *patients*. The *manufacturer* should select evaluation methods for the thermal performance of the *pulse oximeter probe* that led to confident prediction of thermal safety on "worst-case" *patients*.

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Annex CC
(informative)

Determination of *accuracy*, pigmentation *differential bias,* sample size, and study design considerations

CC.1 General

This annex discusses both the formulae used to evaluate the quality of *pulse oximeter equipment* measurements, and the names that are assigned to those formulae.

It has been common for the *SpO_2 accuracy* specifications of *pulse oximeter equipment* to be stated in terms such as "±2 %, one standard deviation." In this document, the committees have chosen a different name for the recommended *SpO_2 accuracy* measure, while retaining essentially the same formula (a value of n - 1 is replaced with n) that has been in common use. The committees recommend definitions of *local bias*, *mean bias*, and *precision* that are consistent with common engineering usage, but slightly different from the meanings of these terms, as they have sometimes been used in the pulse oximetry literature. The reasons for the recommendations are explained in this annex. The committees also discuss the term "ambiguity," which was introduced by Severinghaus et al.[126], and explain the committees conclude that the term *accuracy* can perform a similar function.

Beyond *accuracy* there are other parameters defined such as pigmentation *differential bias* with the methods for estimating and presenting the graphical depiction described in CC.5. Additionally, statistical guidance for the controlled hypoxia study design, including the calculation of study sample size and evaluating requirements tested are discussed.

CC.2 *Accuracy*, bias and *precision*

CC.2.1 Concepts

The terms accuracy, bias and precision have all been used in a variety of ways. *The compilation of ASTM standard definitions* (ASTM, 7th ed., 1990)[21] assembles 11 definitions of accuracy, 9 of bias, and 19 of precision, all taken from ASTM documents. The committees have chosen specific definitions that are consistent with the general definitions appearing in ASTM E456[20]. These definitions, with their associated notes, are as follows:

— accuracy: the closeness of agreement between a test result and an accepted reference value

NOTE 1 The term accuracy, when applied to a set of test results, involves a combination of a random component and of a common systematic error or bias component.

— bias: the difference between the expectation of the test results and an accepted reference value

NOTE 2 Bias is the total systematic error as contrasted to random error. There can be one or more systematic error components contributing to the bias. A larger systematic difference from the accepted reference value is reflected by a larger bias value.

NOTE 3 Expectation is a statistical term which can be interpreted approximately as the mean of the values that would be obtained if the measurement were made many times.

— precision: the closeness of agreement between independent test results obtained under stipulated conditions

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NOTE 4 Precision depends on random errors and does not relate to the true value or the specified value.

NOTE 5 Precision is usually expressed in terms of imprecision and computed as a standard deviation of the test results. Less precision is reflected by a larger standard deviation.

NOTE 6 "Independent test results" means results obtained in a manner not influenced by any previous result on the same or similar test object. Quantitative measures of precision depend critically on the stipulated conditions. Repeatability and reproducibility conditions are particular sets of extreme stipulated conditions.

Based on these concepts, the committees developed the following terms and definitions:

— *accuracy* (see 201.3.202);

— *local bias* (see 201.3.236);

— *mean bias* (see 201.3.238); and

— *precision* (see 201.3.248)

CC.2.2 Effects of offset and linearity errors

'The committees' choice of definitions was influenced by considering three synthesized data sets, which were possibly derived from a *controlled desaturation study*, and that are shown in Figures CC.1 to CC.3. The horizontal axis in each of these figures represents oxygen saturation readings ($S_{\mathrm{R}i}$) taken from a reference equipment (e.g. a *CO-oximeter*), and the vertical axis represents oxygen saturation readings (SpO_{2i}) from the *pulse oximeter equipment* under test. Reference lines shown on the charts are the line of identity (at which test and reference equipment give equal readings) and the other two lines represent deviations of ±2 % from the line of identity.

The three figures differ only in the nature of the simple modifications made to one basic data set:

– Figure CC.1, the base case, was created so that a regression line fitted to the data falls almost perfectly on the line of identity (slope is 1,00 and mean offset is 0).

– Figure CC.2 was created from Figure CC.1 by adding a constant 1,5 unit offset to each y value.

– Figure CC.3 was created from Figure CC.1 by adding an x-dependent error to each value: $y(x) = 0{,}1x - 8{,}6523$, so that the added error is zero near the centre of the graph, positive at the right, and negative at the left. The adjustment formula was chosen to give zero mean additional error.

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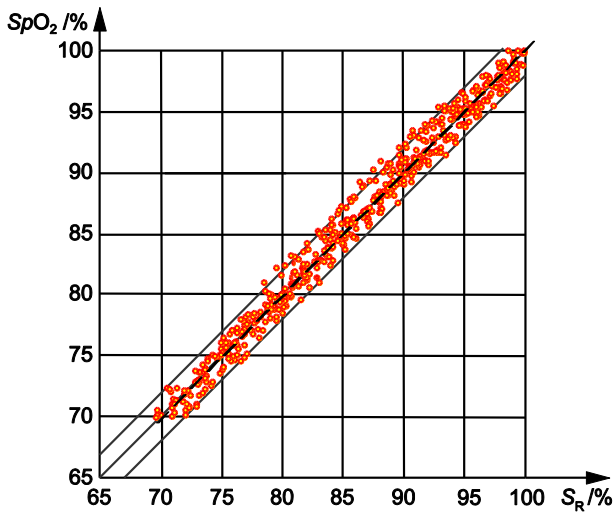

Test *pulse oximeter probe SpO₂* as a function of reference S_R

Negligible *mean bias* (0,02 %)

Regression line slope = 1,000

s_{res} = 1,034 % $B_S = 0$

A_{rms} = 1,033 % P_S = 1,033

Trend line formula: $y = 1{,}000\ 2\ x + 0{,}02$

Figure CC.1 — Synthesized calibration data (base case)

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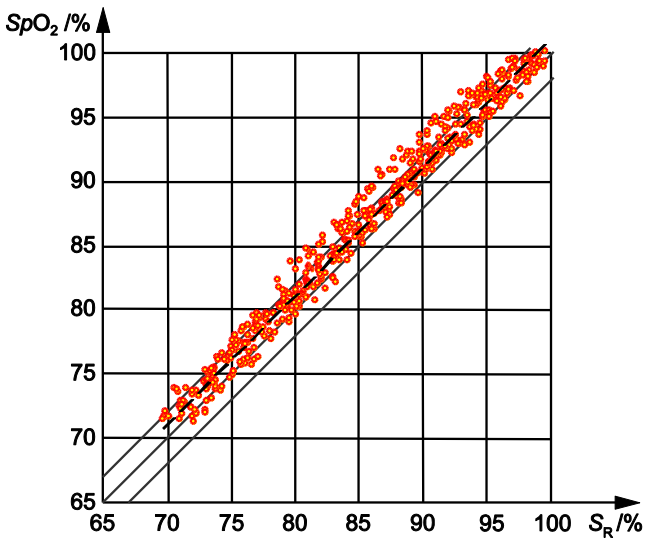

Test *pulse oximeter probe SpO₂* as a function of reference S_R

Mean bias 1,5 %

Regression line slope is still 1,000

$s_{res} = 1{,}035$ % $B_S = 1{,}5$

$A_{rms} = 1{,}823$ % $P_S = 1{,}033$

Trend line formula: $y = 1{,}000\ 2\ x + 1{,}48$

Figure CC.2 — Constant offset has been added to base case

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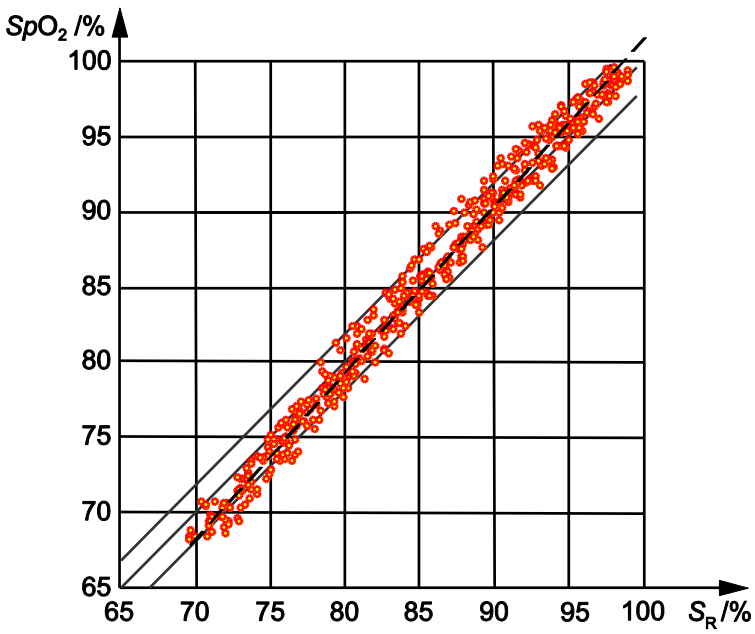

Test *pulse oximeter probe SpO₂* as a function of reference S_R

Negligible *mean bias* (0,001 %)

Regression line slope is now 1,100

$s_{res} = 1{,}034\ \%$ $B_S = 0$

$A_{rms} = 1{,}332\ \%$ $P_S = 1{,}333$

Trend line formula: $y = 1{,}100\ 2\ x - 8{,}67$

Figure CC.3 — Tilt has been added to base case

CC.2.3 Bias (see Figures CC.4 and CC.5)

CC.2.3.1 General

Local bias (indicated here by *b*) at a given value of *x*, is the difference between the *y*-value of the regression line at that coordinate and the *y*-value of the line of identity, as indicated in Formula (CC.1) and graphically represented in Figure CC.4.

$$b_i = SpO_{2\mathrm{fit},i} - S_{\mathrm{R}i} \qquad \text{(CC.1)}$$

where

i is 1 … n

Mean bias is a single number (indicated here by *B*), representing the whole data set. It is the mean difference of the test and reference values, preserving sign, as indicated in Formula (CC.2) and graphically represented in Figure CC.5;

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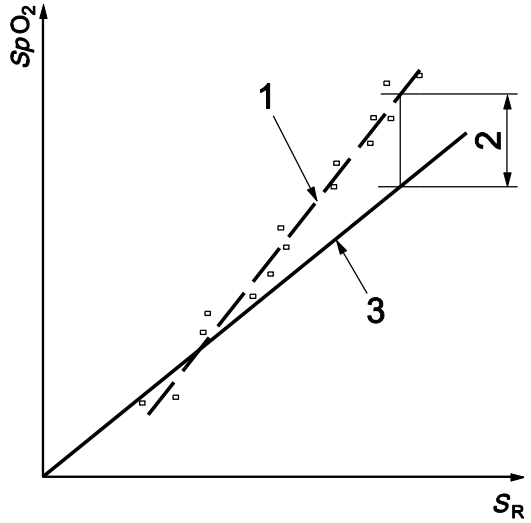
$$B = \frac{\sum_{i=1}^{n} (SpO_{2i} - S_{Ri})}{n} \qquad \text{(CC.2)}$$



Key

1 regression line
2 *local bias*
3 line of identity

SpO_2 is the *pulse oximeter monitor's* indicated oxygen saturation

S_R is the reference oxygen saturation

Figure CC.4 — Graphical representation for the definition of *local bias* (Test *pulse oximeter probe* SpO_2 as a function of reference S_R)

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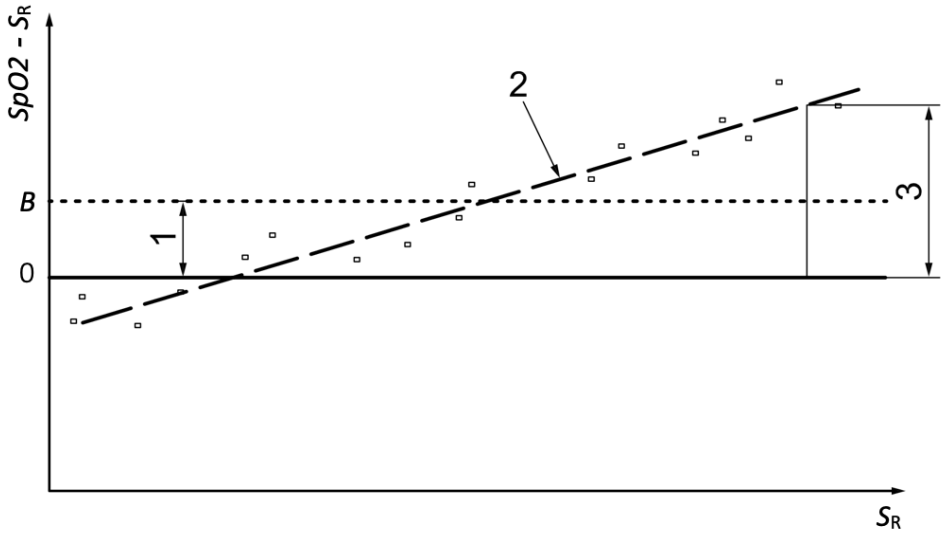

Key

1 *mean bias, B*
2 regression line
3 *local bias*

SpO_2 is the *pulse oximeter monitor's* indicated saturation

S_R is the reference oxygen saturation

Figure CC.5 — Graphical representation for the definition of *local bias* and *mean bias* (Test probe SpO_2 as a function of reference S_R)

When defined in this way, *mean bias, B,* is, as it should be, the average of all *local bias* values, as shown in Formula (CC.3).

$$B = \frac{\sum_{i=1}^{n}(SpO_{2i} - S_{Ri})}{n} = \frac{\sum_{i=1}^{n}\left[(SpO_{2i} - SpO_{2fit,i}) + (SpO_{2fit,i} - S_{Ri})\right]}{n} = 0 + \frac{\sum_{i=1}^{n} b_i}{n} \qquad \text{(CC.3)}$$

The zero term on the right-hand side results from the regression that defines SpO_{2fit}, and the second term simply recognizes the definition of *b* in the first paragraph.

Figures CC.1 and CC.3 both exhibit a *mean bias* of zero, while Figure CC.2 has a *mean bias* of 1,5 units. The value of *local bias* is everywhere zero in Figure CC.1, consistently 1,5 units in Figure CC.2 and in Figure CC.3 follows the formula $b = 0{,}100\,2\,x - 8{,}67$.

CC.2.3.3 Influence of skin adjacent to emitters and detectors

A commonly held concept is that the only pigmented area affecting the SpO_2 reading is that which is directly in-line between the emitters and detector.

However, it is known that scattering and reflection of photons cause the area in proximity of the emitter to glow (e.g. the fingertip with a finger-based *pulse oximeter probe*). Light illuminates the entire area and is reflected from all adjacent skin and tissues before finally reaching the detector, thus, influencing the pulse oximeter reading. This behaviour is illustrated with graphs in Figure EE.5 and Figure EE.6, noting that the blue bandage material used in that demonstration exhibits a differential absorbance of red and near IR light like

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that of deeply pigmented skin causing a falsely positive SpO_2 bias. This indicated that skin adjacent to the emitter/detector strongly influences the SpO_2 calculation and is a valid site for measurement representative of *pulse oximeter probe* site pigmentation.

CC.2.4 *Precision*

Figure CC.3 represents a case that sometimes occurs in pulse oximetry, especially when a new model of *pulse oximeter probe* is being developed for use with the calibration curves that are built into an existing *pulse oximeter monitor*. The fact that there is a variable offset between test and reference values in this data set implies that it is useful to make a distinction between *local bias* and *mean bias*. Data sets can have more complex dependencies of bias on S_R, but this example suffices to show what happens to various data-characterization formulae when *local bias* varies with saturation.

The committees support defining *precision* as the standard deviation of the residuals (s_{res}), given by Formula (CC.4, and 201.3.248)[44].

$$s_{res} = \sqrt{\frac{\sum_{i=1}^{n}\left(SpO_{2i} - SpO_{2fit,i}\right)^2}{(n-2)}} \qquad \text{(CC.4)}$$

where

n is the number of data pairs in the sample;

$SpO_{2i} - SpO_{2fit,i}$ is the difference between the ith SpO_2 datum and the value of the fitted curve corresponding to the ith reference value.

S_{Ri}. s_{res} can intuitively be recognized as the scatter of data points about the best-fit calibration curve. It is a measure of the expected scatter in multiple measurements made with the same *pulse oximeter equipment* at a given oxygen saturation, while considering both variations among *patients* and the repeatability of the *ME equipment* electronics and software.

NOTE In Figures CC.1, CC.2, and CC.3, s_{res} has a consistent value near 1,034 %. All three data sets have the same scatter of data points with respect to the best-fit regression line, and the nearly identical values of s_{res} reflect that fact. The presence of bias in two of these figures has no effect on this document's measure of *precision*.

CC.2.5 *Accuracy*

As suggested by the definition that appears in ASTM E 456-96[20], the committees want *accuracy* to represent a combination of the systematic and random components of error. The definition which has long been used by many *manufacturers* is the root-mean-square (rms) difference between measured values (SpO_{2i}) and reference values (S_{Ri}), as given by Formula (CC.5).

$$A_{rms} = \sqrt{\frac{\sum_{i=1}^{n}\left(SpO_{2i} - S_{Ri}\right)^2}{n}} \qquad \text{(CC.5)}$$

The committees believe that most *manufacturers*, when stating *pulse oximeter equipment* SpO_2 *accuracy* as a "standard deviation," have actually been computing A_{rms}. At least one *manufacturer* has internally used the abbreviation SDI, meaning "standard deviation with respect to the line of identity." This is a misnomer since A_{rms} is not a standard deviation. What is important is that the measure itself is useful. Engineers recognize A_{rms}

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as being very similar to the common measurement "rms error." It is a way of averaging the absolute values of errors over the full measurement range.

NOTE The use of n in the denominator of the expression for A_{rms} is rather than $n - 1$, which is used if A_{rms} were a standard deviation. The difference in the numerical value is typically trivial. The appearance of $n - 1$ in the definition of standard deviation arises from the fact that only $n - 1$ of the samples that comprise the standard deviation can be freely chosen (statisticians say that there are "$n - 1$ degrees of freedom"). The nth sample is constrained in value because the definition of standard deviation includes the difference from a mean, implying that the nth sample is chosen so that the mean has the known value. There is no such constraint on the calculation of A_{rms} because the expression does not include any predetermined parameter, such as a mean.

Understanding that A_{rms} is not a standard deviation is important in avoiding error in calculating *SpO_2 accuracy* of *pulse oximetry equipment*[126]. If one were to create a spreadsheet column containing all the differences, $SpO_{2i} - S_{Ri}$, and instruct the spreadsheet software to calculate the standard deviation of the data, the result is not be A_{rms} (in fact, as noted below, it is P_S, a measure of precision developed by Severinghaus *et al.*[24]). Standard deviation, for any variable x, is indicated in Formula (CC.6).

$$s_x = \sqrt{\frac{\sum_{i=1}^{n}\left(x_i - \overline{x}\right)^2}{n-1}} \qquad \text{(CC.6)}$$

where

$\overline{x}$ is the mean of all the values of x_i.

Comparing this to the expression for A_{rms}, one can see that in A_{rms} there is no subtraction of a mean value. A_{rms} does *not* measure scatter about a mean value. It measures the difference between test values and reference values. The numerical differences between A_{rms} and P_S can be seen in the captions of Figures CC.1 to CC.3.

A_{rms} is affected both by random scatter and by *mean bias* and *local bias*.

In Figure CC.1, because *local bias* is negligible over the entire range (resulting in negligible *mean bias* as well), A_{rms} = 1,033 %, which is nearly equal to s_{res}. The fact that A_{rms} and s_{res} are close to 1 % is consistent with the visual observation that most data in Figure CC.1 lie within the ±2 % reference lines on the graph. In a normal distribution, 95 % of observations are expected to lie within upper and lower standard deviations from the mean.

In Figure CC.2, with a consistently large offset (i.e. a constant *local bias* resulting in a non-zero *mean bias*) A_{rms} has increased to 1,823 %.

In Figure CC.3, with *zero mean bias* but a varying *local bias*, A_{rms} has the intermediate value of 1,332 %. Because the *local bias* in Figure CC.3 is almost everywhere less in absolute magnitude than the constant offset in Figure CC.2, it is appropriate that the measure of overall *SpO_2 accuracy* be lower in Figure CC.3 than in Figure CC.2 (i.e. Figure CC.3 exhibits better *SpO_2 accuracy* than Figure CC.2).

CC.2.6 Analysis

Now it is appropriate to discuss the relationship between the definitions used above and the terms used by two respected sources that have been influential in the clinical literature of pulse oximetry. Bland and Altman[35][44][45] campaigned effectively against the misuse of correlation coefficients in comparing two methods of measurement, and introduced a useful graphical method of examining the data from comparison experiments. Severinghaus *et al.*[127] introduced definitions of bias and precision that were based on the Bland and Altman method, and also defined the new term, ambiguity, as the sum of precision and bias.

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In the following paragraphs, the variables B_S and P_S are used for the definitions of bias and precision that were used by Severinghaus[126]. He defined bias as the mean difference of the test and reference values, preserving sign as indicated in Formula (CC.7).

$$B_S = \frac{\sum_{i=1}^{n} (SpO_{2i} - S_{Ri})}{n} \qquad \text{(CC.7)}$$

By no coincidence, this is identical to this document's definition of *mean bias*. The committees have adopted Severinghaus's language for the definition, with the additional recognition that *pulse oximeter equipment* calibration studies sometimes exhibit variation of bias with saturation, so that it is useful to distinguish between *local bias* and *mean bias*.

Severinghaus *et al.* defined precision as the "standard deviation of the bias" as indicated in Formula (CC.8);

$$P_S = \sqrt{\frac{\sum_{i=1}^{n} \left(SpO_{2i} - S_{Ri} - B_S\right)^2}{n-1}} \qquad \text{(CC.8)}$$

This measure is different from this document's recommended definition of *precision*. One perspective is that P_S is the root-mean-square (rms) deviation of differences from *mean bias*, while s_{res} is the rms deviation of differences from *local bias*. Recall that s_{res} was the same in Figures CC.1 through CC.3. Compare what happens to P_S in these three cases:

— in Figure CC.1, $P_S = 1{,}033$ (identical to s_{res});

— in Figure CC.2, $P_S = 1{,}033$ (in this case, P_S has the desirable property of a "precision" measure, of responding to scatter about the regression line but not responding to the constant offset reflected in the non-zero value of *mean bias*);

— in Figure CC.3, $P_S = 1{,}333$ (P_S has increased to match A_{rms}. Because *local bias* is variable, it causes an increase in P_S, even though the random component of error, as measured by s_{res}, has not changed.)

Bland and Altman[35], in discussing the example in their Figure 2, say "…there is no obvious relation between the difference and the mean. Under these circumstances one can summarize the lack of agreement by calculating the bias, estimated by the mean difference $\bar{d}$ and the standard deviation of the differences (s)."

Thus, Bland and Altman's $\bar{d}$ is equivalent to John Severinghaus' B_S, and their s is equivalent to his P_S. Bland and Altman noted the utility of the standard deviation of differences appears when there is no obvious relation between the difference and the mean. As Figure CC.3 illustrates, the presence of variable local offset makes it preferable to use a different measure of random error.

Finally, consider the term ambiguity, which Severinghaus *et al.*[126] introduced, as the sum of bias and precision:

$$A_S = B_S + P_S \qquad \text{(CC.9)}$$

The value of this term as a figure of merit is that it combines in single number components of both systematic and random error. It can be shown that the recommended *SpO_2 accuracy* measure, A_{rms}, has a similar property, so that it should not be necessary to use both A_{rms} and ambiguity in analysing the results of a particular experiment. The proof begins with the mathematical identity indicated in Formula (CC.10).

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$$\sum_{i=1}^{n} x_i^2 = \sum_{i=1}^{n} \left(x_i - \bar{x}\right)^2 + \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n} \tag{CC.10}$$

For x_i, use the difference, $SpO_{2i} - S_{Ri}$. Expansion and substitution lead to a demonstration that

$$A_{\mathrm{rms}} \approx \sqrt{{P_{\mathrm{S}}}^2 + {B_{\mathrm{S}}}^2} \tag{CC.11}$$

Some readers can find it convenient to use this formula as a route to computing A_{rms}. If the differences between test *pulse oximeter equipment* and reference *CO-oximeter* readings are entered in one column of a spreadsheet, B_{S} is the mean of that column and P_{S} is its standard deviation.

CC.2.7 Calculation of the A_{rms} confidence interval

Conventional statistical methods exist to calculate the confidence interval for a *mean bias* or standard deviation; however, the root mean square error (RMSE) is the combination of bias and variance, not following a normal distribution. RMSE is not a linear statistic and traditional formulas for confidence intervals don't easily apply. More advanced statistical methods for RMSE, particularly, *controlled desaturation study* data that contains repeated measures, have been developed for calculating a confidence interval[106]. Alternatively, it has been common practice to use bootstrap methods[43] (nonlinear approach).

The bootstrap needs to respect the data structure. For study data containing repeated measures, bootstrap by participants and take all measurements from those selected for the bootstrap sample (i.e. clustered bootstrap, resampled with replacement).

To properly apply the bootstrap method to the controlled desaturation data, which contains participant-dependent bias and *precision* (repeated measures), a clustered bootstrap is used. Stratifying participant selection by skin pigmentation cohort is not necessary.

CC.3 Modified Bland-Altman plot

Subclauses 201.12.1.101.4 d) and e) require providing modified Bland-Altman plots[45][87] of the data collected in the *controlled desaturation study*.

Historically, *pulse oximeter equipment* performance was summarized solely by the A_{rms} value observed in the *controlled desaturation study*. A single metric alone, however, does not communicate additional characteristics that are important for which clinicians need to be aware. The committees discussed additional ways in which the overall performance seen in the *controlled desaturation study* can be more effectively communicated, such as graphically presenting the scatter of reading errors (SpO_2 – SaO_2), participant variation, both of which commonly differ between high and low SaO_2, and general behaviours across skin pigmentation.

Providing this information as an error plot complements the linear regression plot [SpO_2 versus SaO_2, 201.12.1.101.4 c)] by zooming-in to provide better visibility of SpO_2 reading variation across the saturation range and participants. The committees chose to use a modified Bland-Altman plot, which considers SaO_2 the independent variable[87], that further includes a regression line to better reflect any SaO_2-dependent reading bias, and identifies independently the data associated with the light, medium, and dark skin pigmentation cohorts.

The following elements are required by 201.12.1.101.4 d) and e).

— Identification of the individual participants and the participant specific data points.

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— Visually distinct representation of each category of light, medium and dark as grouped by the standardized perceived skin colour scale during enrolment of the study.

— The study population regression line.

— Indicate the upper and lower 95 % limits of agreement, calculated as 1,96 multiplied by the standard deviation of the data about the *mean bias* line, using methods to account for repeated measurements.

There is latitude in how this information is presented graphically including.

— Colours selected to delineate skin pigmentation cohorts.

— Manner used to identify individual participants.

— Use of multiple graphs to accommodate the required elements. For example, one graph that identifies individual participants and another that identifies light, medium and dark cohorts.

— When sample sizes are larger, providing separate plots for the light, medium, and dark cohorts can be considered.

— Optionally, individual participant regression lines can be included.

— Commonly, most of the data lies in the central region of the plot; distinguishing amongst this data isn't particularly insightful. Deemphasizing these points visually can improve clarity of the remaining data.

Figures CC.6 A and B provide examples. The Figure CC.6 A distinguishes the skin pigmentation cohorts, however, not the individual participants, thus it requires a separate graph to provide that information. The example in Figure CC.6 B combines all the requirements in a single graph.

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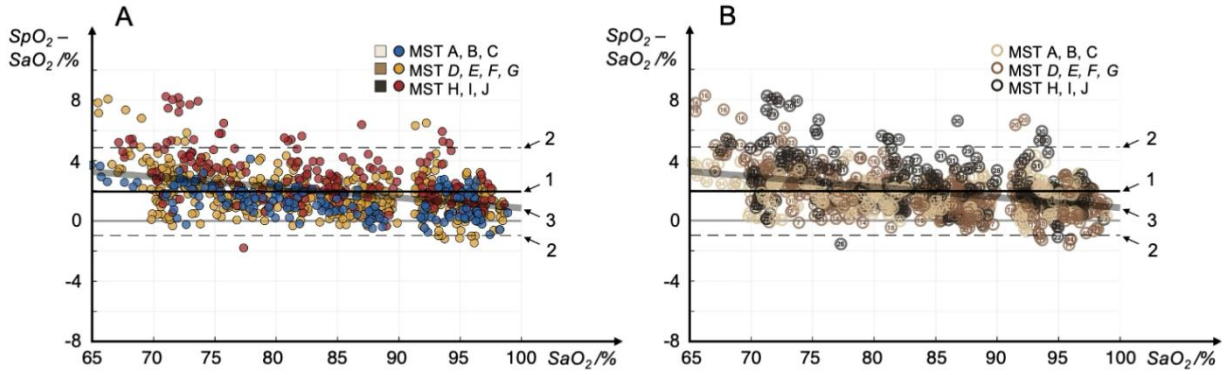

Key

1 is the *mean bias* line for the pooled data

2 indicates upper and lower Limits of Agreement (*mean bias* ±1,96 s_x) where s_x refers to the standard deviation of the data about the *mean bias*, accounting for repeated measures[45]

3 is the linear regression line for the pooled data

MST refers to Monk Skin Tone cohort

Figure A data is coloured by participants' light, medium, and dark MST cohort; adjacent boxes in the legend depict the associated progression in skin tones.

Figure B labels points with participant number; colours indicate light, medium, and dark MST cohort

Points indicate the pooled SpO_2-SaO_2 differences vs SaO_2 from the *controlled desaturation study*

Figure CC.6 Two examples of modified Bland-Altman graphs of the *controlled desaturation study* data, delineating participants, and bias by skin tone

These plots are not intended to quantify *differential bias*, but instead to indicate the pigmentation-dependent behaviour in relation to the overall scatter of the data. The *pulse oximeter equipment* in this example exhibits a degree of both saturation-dependent bias as indicated in the sloped regression line, and *differential bias* as apparent by the offset between clusters of data of the lightest and darkest MST cohorts.

CC.4 Inverse prediction plot

The 95 % inverse prediction band should be based on inverting a statistical model, e.g. a regression, of SpO_2 given SaO_2. If other covariates besides SaO_2 are included in the model, compute the inverse prediction band for SaO_2 at the means of continuous covariates and at the proportions of categorical covariates. The inverse prediction band should account for the correlation of SpO_2 repeated measures within single study participants[55][69][84]. Inverse prediction is also called inverse estimation or the calibration problem.

CC.5 Pigmentation *differential bias*

Subclause 201.12.1.101.5 and Clause AA.2.10 include guidance and rationale for evaluating pigmentation *differential bias* to characterize the tested *pulse oximeter equipment* against accepted thresholds in bias disparity across skin pigment. This subclause provides considerations and methods that can be used to determine its value and confidence interval from the collected data. Independent of the method used, it should be statistically valid, provide an adequate fit to the data, and results should be communicated effectively.

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CC.5.1 Data analysis methods

The committees discussed various methods to analyse data for assessing pigmentation *differential bias*, including both linear and non-linear models. A review of participant bias (SpO_2 - SaO_2) versus skin pigmentation (*ITA*) data from 34 *pulse oximeter equipment* in *controlled desaturation studies* conducted at UCSF (see AA.2.10 and 201.12.1.101.5) suggests that the relationship is predominantly linear. Without being prescriptive, nor limiting other approaches, three linear methods were evaluated that range from simple to more complex:

a) A Linear Regression (LR) approach: perform a linear regression of the per-participant mean SpO_2-SaO_2 bias in a span of interest against the per-participant *ITA.* Summarize each participant's bias as the mean of their SpO_2-SaO_2 differences within the targeted SaO_2 span. Then perform a linear regression (LR) of this participant-level bias estimate on the participant-level *ITA* value. That is, considering a targeted span of 80 % to 90 % SaO_2 (i.e, one that is centred at a singular target value of 85 % SaO_2), the LR method linearly regresses the per-participant mean of SpO_2 - SaO_2 over that span on the per-participant *ITA*. The estimated differential bias over a 100-degree span of *ITA* is -100 times the resulting regression slope.

b) Modified LR (MLR) approach: MLR is like the LR approach, though replaces the mean of the participant's SpO_2-SaO_2 differences in the target span with their per-participant SpO_2 *local bias* at the span midpoint. This improves the bias estimate when the participant's SaO_2 data is not distributed uniformly or SpO_2 is highly scattered within the span. The linear-regression-based bias is calculated by linearly regressing each participant's SpO_2 on SaO_2 (over the entire 70 % to 100 % SaO_2 range) and estimating the *local bias* using the predicted SpO_2 at 85 % SaO_2. Then a linear regression of per-participant bias versus per-participant *ITA* is run. The estimated *differential bias* over a 100-degree span of *ITA* is -100 times the resulting regression slope.

c) Linear Mixed Effects (LME) approach: The LME model uses all of the paired (SpO_2, SaO_2) measurements over the entire 70 % to 100 % SaO_2 range to estimate the expected value of SpO_2 as a function of SaO_2, *ITA*, and their interaction, with random participant level effects included in the model to account for correlation between repeated measurements on each participant. The estimated *differential bias* over a 100-degree span of *ITA* is -100 times the regression slope at 85 % SaO_2. Experience in implementing the LME model suggests that (a) random per participant intercept and random per participant SaO_2 slope effect should be modelled as correlated, not independent; (b) the *ITA* to SaO_2 interaction effect should be modelled as not random, but fixed, because otherwise model fit can be unstable; and (c) centring SaO_2 at 85 % SaO_2 helps ensure convergence.

For any model used, it is important to note the proper sign of the *differential bias* such that a positive value indicates a higher reading of SpO_2 in highly pigmented skin compared to less pigmented skin.

The half-width of the confidence interval of the calculated *differential bias* computed by the LR and MLR approaches is 100 times the standard error of the *ITA* regression slope times the inverse of the standard normal cumulative distribution function evaluated at the desired probability (i.e. 95 % for a 90 % confidence interval).

Alternatively, the *differential bias* confidence interval can be determined using a stratified cluster bootstrap method. Each bootstrap sample is obtained by randomly sampling participants with replacement for creating the study pool, and including all the SpO_2, SaO_2 data pairs from the sampled participants (as opposed to also randomly sampling with replacement each sampled participant's data pairs). Stratified bootstrap sampling refers to randomly sampling with replacement the same number of participants within each skin pigmentation group as observed in the actual sample, thus ensuring that each bootstrap sample follows the same requirements used in conducting the study [see 201.12.1.101.6 f) 7)].

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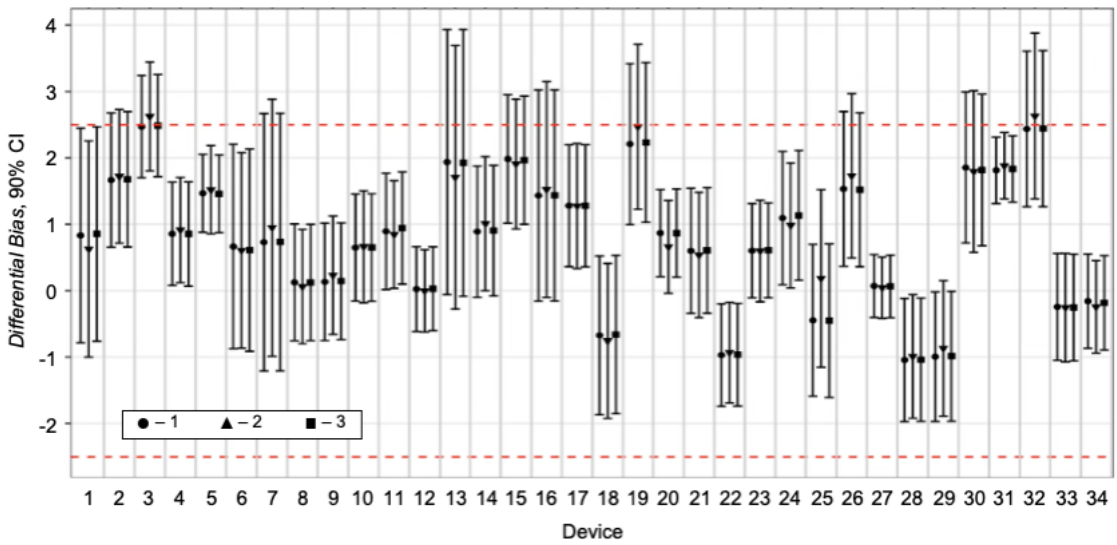
For statistical packages that include the capability to run an LME model, *differential bias* is expressed as a linear contrast in the model parameters, namely, 100 times the quantity *ITA* slope plus *ITA*-SaO_2 interaction coefficient times SaO_2 value (i.e. 85 %). The standard error of the maximum likelihood estimate of this linear contrast is obtained using the Delta Method. The half-width of the confidence interval is the standard error times the appropriate percentile (e.g. 95 %) of either the standard normal distribution or the t-distribution with degrees of freedom approximated using Satterthwaite's Method.

The Two One-Sided Test (TOST) approach to equivalence testing is to compare the limits of the 90 % confidence interval to the upper and lower acceptance bounds. As can be seen in Figure CC.7, a plot of results obtained from each approach (LR, MLR, LME), the three analysis methods yield the same equivalence conclusions for all 34 *pulse oximeter equipment*. Theoretically, one can show the MLR and LME methods for estimating *differential bias* at an SaO_2 value are in fact estimating the same quantity. The two estimates also have similar variances when (1) *differential bias* is estimated at the mean SaO_2 value, and (2) the number of pairs of (SpO_2, SaO_2) are similar for each participant. By design, desaturation studies have similar numbers of (SpO_2, SaO_2) pairs per participant and have a mean SaO_2 value of about 85 %, the value at which *differential bias* is calculated in Figure CC.7. Thus, the nearly identical results for the LME and MLR approaches is not surprising.



Key

1 Denotes the estimates made using the Linear Mixed Effects model (LME)
2 Denotes the estimates made using the Linear Regression model (LR)
3 Denotes the estimates made using the Modified Linear Regression model (MLR)

Device refers to the assessed *pulse oximeter equipment* respective data from the *controlled desaturation study*

Differential Bias is the 100° *ITA* difference of SpO_2 reading bias in units of %

90 % CI refers to the error bars indicating the 90 % confidence interval of the estimates

The dashed red lines indicate signed *differential bias* of ±2.5 %

Figure CC.7 — Point estimates and 90 % confidence intervals of *differential bias* from LME, LR, and MLR analysis of 34 unlabelled *pulse oximeter equipment* at 85 % SaO_2

Regardless of the method used to determine pigmentation *differential bias*, 201.12.1.101.5 e) 1) requires providing a graph of per-participant bias versus per-participant *ITA* evaluated at the span centred at 85 % SaO_2. The example graph shown in Figure CC.8 includes simulated per-participant bias values and the linear

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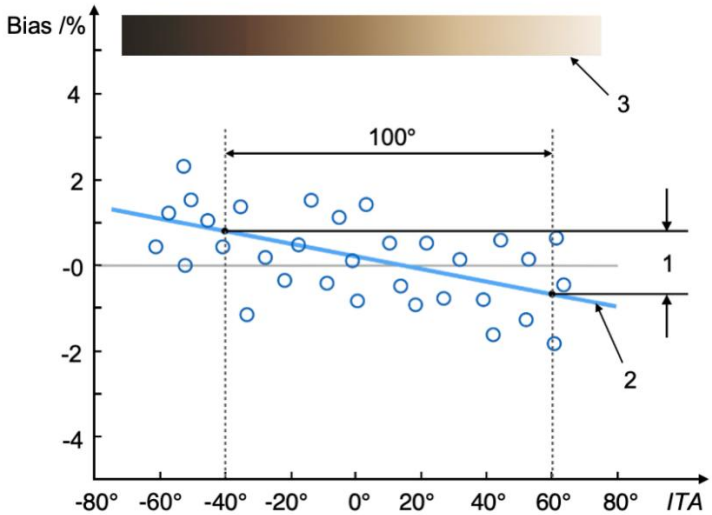
regression. The indication of 100-degree separation and the magnitude of *differential bias* is provided here for reference only. *ITA* correlates well to melanin concentration in skin, however, it is not a widely relatable measure to most readers. The colour bar at the top of Figure CC.8 provides a visualization to help put the data in context. Alternatively, for example, the data points themselves can be shaded by the participant's MST cohort to convey this information. Note that *ITA* and MST do not align precisely.



Key

1 is the magnitude of the *differential bias* in %
2 is the linear regression line
3 is a depiction of the approximate skin tone associated with the *ITA* scale
ITA is in units of degrees (°)
Bias is the local SpO_2–SaO_2 difference in a particular SaO_2 span (or its midpoint)

NOTE In the graph, there is 1 point for each simulated participant in the assessed SaO_2 range, as observed in the *controlled desaturation study*.

Figure CC.8 Example graph of simulated per-participant bias versus per-participant *ITA* within a particular SaO_2 span

201.12.1.101.1.5 e) 2) calls for providing a graph of the pigmentation *differential bias* vs. SaO_2. The *process* for determining *differential bias* in CC.5 can be repeated over multiple SaO_2 span-midpoints to describe what the performance is over the entire saturation range of 70 % to 100 %. An example graph is shown in Figure CC.9, where the open circles represent the pigmentation *differential bias* at the respective saturations. The solid lines are the upper 95 % and lower 95 % confidence limits of the pigmentation *differential bias* per SaO_2 value.

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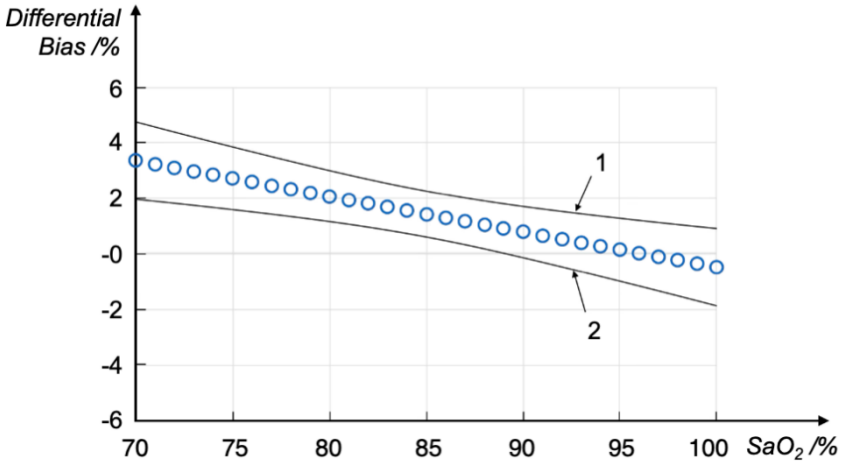

Key

1 is the 95 % upper confidence limit
2 is the 95 % lower confidence limit
Differential Bias is the 100° *ITA* difference of SpO_2 reading bias in units of %

NOTE In the graph, there is one point and associated confidence limit for each integer value of SaO_2 across the 70 % to 100 % SaO_2 span for the data collected in the *controlled desaturation study*.

Figure CC.9 — Example graph of the observed pigmentation *differential bias* versus SaO_2

CC.5.2 Equivalence Testing

A test for acceptable pigmentation *differential bias* is in essence a test for "equivalence" in that too much differential bias in either direction indicates poorly performing *pulse oximeter equipment*. Equivalence testing is most easily accomplished using the so-called "two one-sided tests (TOST)" approach. The hypothesis for pigmentation *differential bias* should therefore be re-stated as a test of equivalence, requiring that the *differential bias* be within a set of lower and upper bounds (e.g. ±2.0 %). Note that the TOST approach to demonstrating equivalence can be carried out by constructing a 90 % (not 95 %) confidence interval for the appropriate parameter or contrast and determining whether it lies entirely within the acceptance bounds.

For the *ITA* analysis, the *differential bias* requirement is stated in terms of the difference in bias over a 100-degree range (e.g. 50° and –50°). If *differential bias* is linear in *ITA*, as is assumed in the LR and MLR approaches, then this requirement is exactly equivalent to a requirement on the *ITA* slope. For example, the requirement of a differential bias range of -2,0 % and 2,0 %, is exactly equivalent to an *ITA* slope of -0,02 and 0,02 (i.e. the estimated *differential bias* over a 100-degree span of *ITA* is -100 times the *ITA* regression slope). When using the TOST approach, this is carried out by constructing a 90 % confidence interval for the slope and showing that the confidence interval is within the (-0,02, 0,02) interval. For the LME approach, *differential bias* is expressed as a linear contrast in the *ITA* slope and the *ITA*-SaO_2 interaction coefficient, thus a TOST is applied to this contrast.

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CC.5.3 Determination of sample size for the *controlled desaturation study*

CC.5.3.1 Accuracy (A_{rms})

To determine the number of participants needed to demonstrate meeting an A_{rms} < 3.0 % with 95 % confidence and 80 % power, several approaches are offered:

a) If sufficient data of size is available from prior controlled desaturation studies in which the Arms estimate is less than 3,0 % then determine the sample size for the stratified cluster bootstrap analysis by trial and error as follows:

   1) Choose a starting sample size *n* (e.g. *n*=24).
   2) Obtain *B* stratified cluster bootstrap samples of size *n* based on the prior dataset (which typically has size that differs from *n*).
   3) Obtain the stratified cluster bootstrap 95 % upper confidence limit on A_{rms} for each of these *B* datasets.
   4) The power to show A_{rms} less than 3,0 equals the number of bootstrap instances in which the 95 % confidence limit was less than 3,0, divided by *B* and then multiplied by 100.
   5) Increase or decrease *n* and repeat steps 1) to 5), if applicable.

b) If prior data is available that is insufficient for populating datasets within each of the skin pigmentation cohorts, then determine sample size as in a) using unstratified cluster bootstrap sampling that is independent of *ITA* or *MST* to estimate the number of participants needed to size the study.

c) Follow an analytical approach for sample size estimates that account for repeated measures.

d) Start with the minimum number participants required [201.12.1.101.6 a)] and utilize an adaptive design to achieve the targeted confidence limits and power. During the course of the study, the adaptive design determines the sample size that achieves these targets at which point the study can be stopped.

CC.5.3.2 Pigmentation *differential bias*

To determine the number of participants needed to demonstrate a *differential bias* less than 2,5 % with 95 % confidence and 80 % power, the general approaches outlined in CC.5.3.1 for A_{rms} remain applicable when adapted accordingly.

The plots in Figure CC.10 each show a series of power curves from an R statistical program language implementation of the MLR approach using estimates of *differential bias* (-100 multiplied by the slope), standard error of the *ITA* distribution, and standard error of the regression slope from the 34 unlabelled *pulse oximeter equipment*, and comparing to a requirement of less than 2,5 % at SaO_2 = 85 % using a TOST.

Each curve in Figure CC.10 indicates the power to statistically demonstrate *differential bias* less than 2,5 % as a function of the number of participants in the study -- the probability that the 90 % confidence interval for the regression slope is contained within the values ±0.025.

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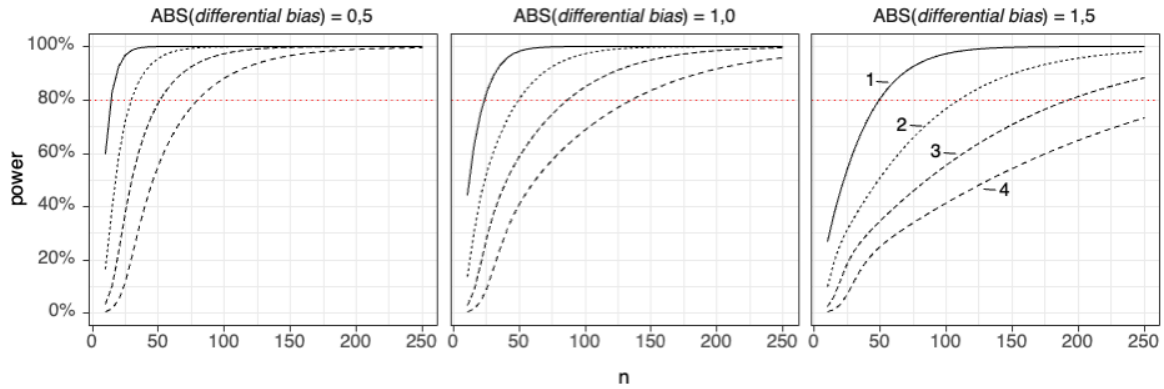

Key

1 represents a *differential bias* standard error of 1,0
2 represents a *differential bias* standard error of 1,5
3 represents a *differential bias* standard error of 2,0
4 represents a *differential bias* standard error of 2,5
n is the number of participants
power is the statistical power
ABS refers to the absolute value

NOTE The three panels assume different true *differential biases*. Intercepts of the curves with the dotted red line indicate the respective number of participants associated with reaching an 80 % power.

Figure CC.10 Power curves for the participant-level *ITA* vs participant-level bias regression analysis method (MLR).

CC.6 Adaptive clinical study design

Manufactures cannot know prospectively all the parameters necessary to estimate the sample size to demonstrate performance of A_{rms} and pigmentation *differential bias* with desired statistical confidence [see 201.12.1.101 and DD.2]. Assuming a study needs only the minimum number of enrolled participants can prove to be insufficient, or early prototype data behaviours can lead to an overestimation of study size. Studies designed with flexibility can accommodate additional participants when needed, as well as provide intermediate information to discontinue study enrollment. Such flexibility, however, should be conducted appropriately, using accepted adaptive design principles and must be defined prospectively[61][62].

Adaptive study designs include those that allow for adjusting sample size given interim data. Some designs can have multiple interim looks, potentially stopping early when meeting pre-defined criteria for overwhelming efficacy. Designs can build in stopping for futility when interim data suggest acceptance criteria are unlikely to be met at any reasonable sample size. This reduces costs and limits exposure to participants when there is very low chance of being successful.

There are numerous adaptive designs available to achieve these goals. One example is the “promising zone” approach[99] that uses conditional power to re-estimate sample size when interim data indicate that the *pulse oximeter equipment* effect of interest (e.g. a device effect) is in a specified range of power; other sample re-estimation methods are available as well[128]. Adaptive designs for potentially stopping studies early for efficacy or futility at interim looks include, for example, group sequential[88][107][112], stochastic curtailment

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methods that monitor either predictive power[40] or conditional power[89], and information monitoring methods[47]. These examples are not meant to be limiting, but only a brief introduction to a much larger and well-studied field.

CC.7 Computational Tools

Code bases and tools are planned to become available online for calculating these and other performance metrics associated with the controlled desaturation study. Those tools can be found at in pulse oximetry statistical tools directory at:
https://www.iso.org/committee/52012.html?view=documents

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Annex DD
(normative)

Method for invasive studies for evaluating and documenting *SpO_2 accuracy* in human participants

DD.1 General

This annex is provided for evaluating and documenting the *SpO_2 accuracy* of *pulse oximeter equipment*. The methods described in this annex are applicable to both new *pulse oximeter equipment* and modified *pulse oximeter equipment* or parts whenever human testing using invasive methods are required. Optimal conditions are used to demonstrate maximum *accuracy* capability by isolating the *pulse oximeter equipment* from confounding conditions, for example, environmental factors, low perfusion, movement, incorrect *pulse oximeter probe* placement, dis-haemoglobins, and nail polish. This method can also be used to assess impact to *accuracy* during specifically targeted and specified non-optimal conditions such as varying environmental factors, participant movement, or low pulse-amplitude states (low perfusion). Subclause 201.12.1.101.3 b) 2) requires that a study conducted to evaluate the *SpO_2 accuracy* of *pulse oximeter equipment* conforms with ISO 14155:2026.

The *SpO_2 accuracy* of *pulse oximeter equipment* is measured by comparing *SpO_2* readings of the *pulse oximeter equipment* to values of *SaO_2* determined from arterial specimens analysed by a *CO-oximeter* in accordance with 201.12.1.101.3. a) Two types of individuals can participate in invasive studies:

– healthy adult volunteers who consent to induced hypoxia and arterial blood sampling as part of the experimental *procedure* (see DD.2); and

– *patients* in whom arterial blood samples are available for analysis (see DD.3).

This annex describes testing methods for assessing the *SpO_2 accuracy* of *pulse oximeter equipment*. It does not prescribe medical practice, proper safety *procedures* or institutional review board (IRB) or ethics committee (EC) *processes*.

DD.2 *Procedure* for invasive laboratory testing on healthy adult volunteers

DD.2.1 Purpose of an invasive *controlled desaturation study*

The general purpose of invasive *controlled desaturation studies* is to verify the *SpO_2 accuracy* of *pulse oximeter equipment* in comparison to "gold-standard" measurements of *functional oxygenation* (i.e. *SaO_2* as analysed by a *CO-oximeter*). This is achieved through paired observations of *SpO_2* and *SaO_2* values over the specified *SpO_2 accuracy* range (e.g. 70 % to 100 % *SaO_2*) of the *pulse oximeter equipment* on a cohort of healthy adult volunteers. The fraction of inspired oxygen (F_iO_2) delivered to test participants is varied to achieve a series of targeted steady-state saturation periods. Arterial blood samples are periodically taken from an indwelling arterial catheter for use in the comparison.

The methods described in this annex involve *procedures* that need to be supervised by qualified personnel. Participants have an artery cannulated and then are exposed to inspired oxygen concentrations lower than room air. Accordingly, this study method always requires protocol approval by an IRB or EC, including informed consent of the participants. See ISO 14155:2026.

DD.2.2 Scope of an invasive *controlled desaturation study*

This invasive *controlled desaturation study* method is used to *verify* the *SpO_2 accuracy* of *pulse oximeter equipment* under well-controlled, optimal laboratory conditions on healthy adult participants[40]. This method

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can be used during specifically non-optimal conditions such as participant movement or low pulse-amplitude states.

DD.2.3 Methods

DD.2.3.1 Apparatus

The following shall be used.

DD.2.3.1.1 ***CO-oximeter*** for measuring SaO_2 per the *CO-oximeter manufacturer*-recommended *procedures* and supplies.

DD.2.3.1.2 **Materials** for arterial catheterization and blood sampling.

DD.2.3.1.3 **Means** for recording SpO_2 values, which can be manual or automated.

DD.2.3.1.4 ***Pulse oximeter equipment*** to be tested. See also DD.2.3.4 c).

DD.2.3.1.5 **Means** for delivering a medical-grade oxygen-nitrogen mixture of varying FiO_2 levels to the participant (e.g. pre-mixed high-pressure cylinders or a gas-mixing device).

DD.2.3.2 *Procedure*

Perform an invasive *controlled desaturation study* by doing the following.

a) Specify the criteria and methods for determining SpO_2 stability at the *pulse oximeter probe* site in the protocol.

b) When the reference system's blood saturation stabilizes at a target plateau level, blood sampling may begin. Time to reach stability can vary considerably based on study conditions including but not limited to *pulse oximeter probe* site, participant perfusion, participant respiratory pattern, as well as the magnitude of the plateau change. Time to reach stability can take up to several minutes under some of these conditions. After an adjustment in FiO_2 to induce a change in plateau level, a period of at least 60 s should be allowed prior to sampling the first arterial blood draw of the new plateau to allow SaO_2 to reach stability at the *pulse oximeter probe* site, and to allow the SpO_2 of the reference *pulse oximeter equipment* (if used) and *pulse oximeter equipment* under test to reach stability.

c) Assess plateau stability in real time during study data collection, using multiple parameters including but not limited to SaO_2, SpO_2 of a reference *pulse oximeter equipment*, SpO_2 of the device under test, end tidal oxygen concentration, and FiO_2. Record the methods for assessing plateau stability during the study in the test report.

d) Assess plateau stability during data analysis. Assess the variability of consecutive sets of blood samples/data collection points for stability of SaO_2, and SpO_2 of the *pulse oximeter equipment* (if used) and the *pulse oximeter equipment* under test. Ensure that consecutive samples within a plateau are at least 20 s apart. Determine stability by an observer, a software-based algorithm or a combination of the two.

e) Record the methods for assessing plateau stability during data cleaning/processing in the test report, including % of session samples not included in analysis, exclusion criteria, as well as participant/session selection for inclusion in the dataset (e.g. were these consecutively enrolled participants or chosen from a pool of participants post hoc), and the time period/dates over which study data were collected.

NOTE 1 The stability acceptance criteria can include but are not limited to:
- variation in of the reference SpO_2 values during the plateau;
- slopes of the plateau;
- predictions from a model; and
- direct assessment of the participants' vital signs and/or comfort.

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f) The study protocol should describe the specific conditions of the test (e.g. optimal laboratory conditions, participant motion, low pulse amplitudes). Warming means can be utilized to improve circulation and pulse amplitude at a *pulse oximeter probe* site.

g) Arterial lines should be placed by a clinician using jurisdictional guidelines for arterial puncture. Arterial lines should be placed by a clinician using guidelines (e.g. CDC) for infection prevention. Aftercare should include 5 min to 15 min pressure, pressure bandage >1 h, instructions for limited activity and Principal Investigator's contact information.

NOTE 2 Further detail of the proper techniques and maintenance of the arterial line are beyond the scope of this document.

h) After a catheter is placed in the artery, the *pulse oximeter probes* to be evaluated are attached to the participant's fingers, forehead, nose, ears or other body surfaces as appropriate. *Pulse oximeter probes* can be covered with opaque material to prevent optical interference (light from one *pulse oximeter probe* or another source reaching the photodetector of an adjacent *pulse oximeter probe*).

i) The protocol should specify criteria and methods for determining stability of the SpO_2 at the *pulse oximeter probe* site.

EXAMPLE 1 A stable plateau on the *pulse oximeter equipment* under test.

EXAMPLE 2 A stable plateau on a reference *pulse oximeter equipment*.

EXAMPLE 3 A real-time measurement of expired respiratory gases.

j) Carbon dioxide may be added to the inspired gas mixture to maintain normal carbon dioxide levels and to prevent respiratory alkalosis secondary to hyperventilation caused by hypoxia.

k) Arterial pressure, ECG, heart rate, $EtCO_2$, respiratory rate and FiO_2 should be monitored.

l) F_iO_2 is reduced or increased to bring the participant near target levels. Desaturation to the lowest level (e.g. 70 % SaO_2) is conducted in a stepwise *process* targeting a number of saturation plateaus (periods in which the saturation is relatively stable). The number of saturation plateaus finally accepted as valid is represented by a value M.

m) When combined across participants, these M plateaus should result in a distribution of collected and pooled data pairs spanning the specified SaO_2 range. See also DD.2.3.3.

n) Within each saturation plateau level, draw N blood samples and pair with the corresponding SpO_2 values.

o) Time in low O_2 saturation decades (i.e. 60 % to 70 %, 70 % to 80 %, and 80 % to 90 %) should be limited to the minimum amount of time required to obtain stable test/reference data and as tolerated by the study participant.

EXAMPLE 4 A study design is shown in Table DD.1 and Figure DD.1. In this example, $M = 5$ and $N = 4$. The values in this example are not intended to be limiting in the number of plateaus or numbers of samples per plateau.

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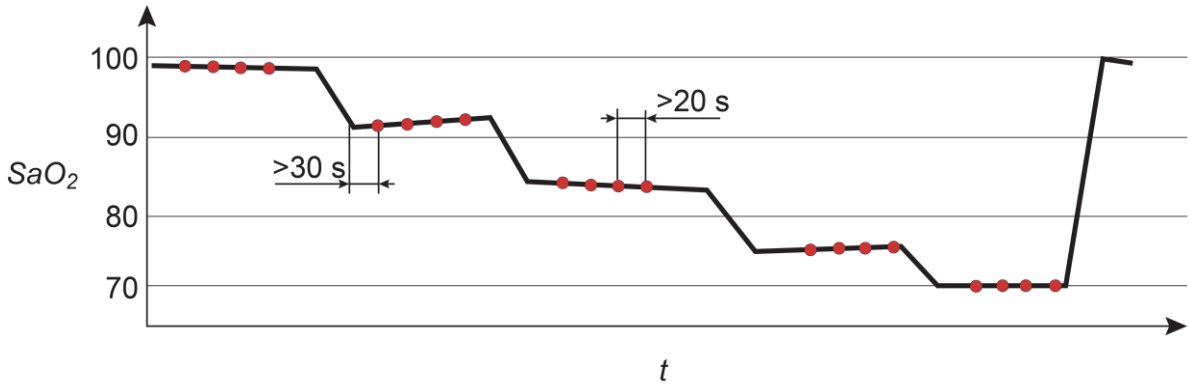
Table DD.1 — Example of target plateaus and ranges

SaO_2 plateau range %	Target number of samples
100 to 97	4
97 to 92	4
92 to 85	4
84 to 78	4
77 to 70	4
Total	20



Red dots indicate SaO_2 values at the time of the blood sampling in an idealized, single desaturation run. The values in this example plot are not intended to be limiting in the number of plateaus or numbers of samples per plateau.

Figure DD.1 — Example of desaturation-time profile

p) For each participant, $M \times N$ blood draws provide (SaO_2, SpO_2) data pairs for analysis (see DD.2.3.2). These data pairs are either acquired simultaneously or correlated in time to accommodate physiological and *pulse oximeter equipment* delays.

NOTE 2 The values of M and N can vary by participant, given the ability to reach and maintain the targeted plateau levels.

q) When the reference system's blood saturation stabilizes at an acceptable plateau level, blood sampling can begin. After a change in plateau level, readings should be allowed to stabilize for at least 30 s to allow SaO_2 to reach equilibrium at the *pulse oximeter probe* site.

r) Care should be taken for the sampling, handling, and analysis of blood to ensure the SaO_2 *accuracy* of the CO-oximetry measurement. *Procedures* for the sampling, handling and analysis of blood are found elsewhere[126][127].

s) The protocol should specify the time interval between successive samples within a plateau to ensure that samples are independent. In determining this interval, consideration should be given to allowing the blood circulation to flush and replace the haemoglobin at the *pulse oximeter probe* site and to the averaging time of the *pulse oximeter equipment*.

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DD.2.3.3 Data quality considerations

An invasive *controlled desaturation study* data analysis is performed as follows.

a) Paired SpO_2 and SaO_2 data points are pooled for all participants and the A_{rms} is calculated using the formula given in Formula (1) in 201.12.1.101.4.

b) Pooled data values are required to include SaO_2 values within 3 % of the endpoints of the SpO_2 *accuracy* range (e.g. 70 % to 100 %). SpO_2 *accuracy* calculations are required to include data pairs with at least one SaO_2 value in the 67 % to 73 % range and one in the 97 % to 99 % range. (see 201.12.1.101.3).

c) For *pulse oximeter monitors* that place an upper limit on displayed SpO_2 (e.g. 99 % or 100 %), a means that does not bias the A_{rms} result should be used.

EXAMPLE 1 Include only observations where SpO_2 readings are less than the upper display limit.

EXAMPLE 2 Statistically down-weight those values with $SpO_2 = 100$ % (e.g. treat observations of 100 % as censored, as is done in the analysis of survival data).

EXAMPLE 3 Configure the data-collection system to record values of $SpO_2 > 100$ %.

NOTE 1 A_{rms} describes the combined bias and *precision* of SpO_2 readings, and by limiting display values, the assumptions of a normal distribution are violated.

d) Points collected with SaO_2 values beyond the specified SpO_2 *accuracy* range (e.g. within 3 % of the endpoints) are excluded, unless the protocol specifically requires these samples to be included.

NOTE 2 A *CO-oximeter* has a specification and an inherent measurement uncertainty. SaO_2 values of 101 % and 102 % fall within the acceptable tolerance of the *CO-oximeter* and can be recognized as valid readings and not excluded from the analysis.

NOTE 3 Arbitrarily including or excluding such data points allows *manufacturers* to manipulate the SpO_2 *accuracy* specification to their advantage.

e) Data pairs can be rejected if, determined retrospectively, they were taken during conditions that were outside of the scope of the testing as specified in the protocol.

EXAMPLE 4 An unstable SpO_2 plateau.

EXAMPLE 5 If the study record indicated that there were difficulties with the blood draw (e.g. excessive bubbles).

EXAMPLE 6 The *CO-oximeter* experienced error conditions.

f) The total number of acceptable data pairs acquired during the study needs to be sufficient to demonstrate statistically the specified SpO_2 *accuracy*. For example, about 20 blood samples are acquired from each of at least 24 participants, resulting in at least 480 data pairs. Specific numbers of samples and participants can vary, if properly justified using statistical methods.

g) The distribution of SaO_2 values in the pooled data set should be made with comparable density over the full claimed range. For example, approximately 1/3 of the data should fall within each of the following ranges: 70 % to 80 %, 80 % to 90 %, and 90 % to 100 % SaO_2. See Figure DD.1.

DD.3 Considerations for invasive testing on *patients*

NOTE 1 There is guidance or rationale for this Clause contained in AA.2.21.

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The *SpO_2 accuracy* of *pulse oximeter equipment* is measured by comparing SpO_2 readings of the *pulse oximeter equipment* to simultaneously time matched arterial blood specimens (i.e. SaO_2 values) preferably from an indwelling catheter.

NOTE 2 Using single needle punctures as a source of arterial blood is likely to result in unstable SpO_2 values and can result in sampling inaccuracy due to *patient* discomfort and synchronization of blood sample with SpO_2 values.

In a clinical environment, the primary responsibility is *patient* care. SpO_2 measurements from *patients* in that environment when compared to measurements from a *CO-oximeter* in that environment can be degraded because data collection cannot always be well controlled. Both measurements are better controlled under laboratory conditions.

In a clinical environment, measurements from *pulse oximeter equipment* and *CO-oximeters* are often subject to non-optimal conditions, so the SpO_2 and SaO_2 values can be difficult to match reliably due to circulatory instabilities or dynamics.

The *patient's* clinical condition should be considered when placing any *pulse oximeter probe* in relation to the arterial sampling site. Whenever possible, the *pulse oximeter probe* should be observing blood that is part of the same circulatory stream as the artery from which blood is taken.

Generating the number of data pairs necessary to demonstrate statistically the specified *SpO_2 accuracy* over the specified range can require a large number of *patients*. Blood samples may be taken needed clinical care or solely for the purposes of the study, as specified in an approved study protocol.

The total number of acceptable data pairs acquired during the study needs to be sufficient to demonstrate statistically the specified *SpO_2 accuracy*. The distribution of reference values in the pooled data set needs to be made with comparable density over the full claimed range. Specific numbers of samples and participants as well as the analysis technique need to be justified using statistical methods. See CC.5.3.

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Annex EE
(informative)

Simulators, calibrators, and *functional testers* for *pulse oximeter equipment*

EE.1 General

The committees recognize that *functional testers* have become commonly available and are incorrectly perceived by some *responsible organizations* as being calibrators. This annex addresses appropriate uses of each type of tester.

A variety of equipment can be used to test *pulse oximeter equipment*. Some of this equipment is provided by the *manufacturers* of *pulse oximeter equipment*, some by independent tester *manufacturers*, and some by research laboratories. The committees conclude that it is helpful to suggest standard terms that can be used in describing this equipment, in the interest of improving *responsible organizations'* understanding of the capabilities of particular testers. The need for this discussion is made greater by two somewhat unusual characteristics of *pulse oximeter equipment*.

— Unlike many other types of *ME equipment*, *pulse oximeter equipment* is not designed to be calibrated after it leaves the factory.

— There is today no accepted method of verifying the correct calibration of a *pulse oximeter probe* and *pulse oximeter monitor* combination other than testing on human beings.

All available tools for testing *pulse oximeter equipment*, at the time of writing, are properly called *functional testers*. 201.7.9.3.1.101 requires the *technical description* of *pulse oximeter equipment* to state that *functional testers* cannot in general be used to measure the *SpO_2 accuracy of pulse oximeter probes* and *pulse oximeter monitors*. This annex is intended to clarify the reasons for this requirement as well as semantic issues. Terms such as simulator, calibrator and tester have several common meanings, which can contribute to misunderstanding of the actual capability of a particular item. The committees have recommended specific uses of the terms "calibrator" and "functional tester," when these terms are applied to pulse oximetry. This annex explains the difference between *functional testers* and other types of testing equipment and suggests the correct way to use of *functional testers*. It also explains why it is inappropriate to use measurements made with *functional testers* to support *SpO_2 accuracy* claims for *pulse oximeter probes* or *pulse oximeter monitors*. A *functional tester* can be used to demonstrate that *SpO_2* and pulse rate values calculated by a module that conforms to this document are not corrupted during communication to host equipment.

EE.2 What is a simulator?

In conventional usage, a simulator is test equipment that stands in for the human *patient*. For example, simulators for invasive and non-invasive blood pressure and for electrocardiograph signals are well accepted substitutes for a *patient.* The measurement *accuracy* for testing *ME equipment* using a simulator can be expected to be comparable to that seen monitoring *patients.* A finite possibility of some additional inaccuracy exists due to errors in the simulator.

There is, at the time of writing, no simulator for pulse oximetry that reproduces the optical properties of a broad range of *patients* well enough to warrant its use in determining the *SpO_2 accuracy* of any *pulse oximeter monitor* and *pulse oximeter probe* combination. Various simulators are available that are suitable for specific engineering purposes related to development and testing of *pulse oximeter equipment*.

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EE.3 What is a calibrator?

A calibrator, as the term is conventionally used, is test equipment that can be used to adjust the *accuracy* of *ME equipment*. A calibrator is typically a very accurate simulator that is used to adjust the *ME equipment's* calibration. Although *pulse oximeter equipment* has no means to adjust its calibration, the committees conclude that the least of semantic evils are to recommend a special use of the term "calibrator." A pulse oximetry calibrator (POC) is a high-fidelity simulator, capable of producing signals or optical responses indistinguishable from those that come from a human *patient* or test participant. With currently available *pulse oximeter equipment,* it is generally not possible to improve the *accuracy* of calibration of a specific combination of *pulse oximeter monitor*/*pulse oximeter probe* using a POC. The POC is used to measure the error with which the *pulse oximeter monitor* measures oxygen saturation on one or more simulated *patients*. If the error is found to be unacceptable, the typical cure is to replace the defective components or redesign the *ME equipment*. Another difference between the POC and other sorts of calibrators is the difficulty of reducing the error contribution of the POC to the level expected of calibrators. A calibrator should be at least four to ten times more accurate than the equipment being calibrated. Given the typical *pulse oximeter equipment* SpO_2 *accuracy* of ±2 % SpO_2, a POC should ideally characterize the *accuracy* of any *pulse oximeter monitor*/*pulse oximeter probe* combination with an error of less than 0,5 % SpO_2.

EE.4 How is *pulse oximeter equipment* calibrated presently?

Pulse oximeter equipment is unlike other *ME equipment*, in that as of this writing, no simulators have been proven adequate for use as *pulse oximeter equipment* calibrators. The interaction of light and human tissue upon which pulse oximetry depends is complex. At least one effort attempted to produce a properly *validated* POC[74][75] that modelled at least some of the optical intricacies, but no such effort has been successful.

Thus, the primary method of determining the SpO_2 *accuracy* of *pulse oximeter equipment* is to compare its readings with the functional oxygenation readings of a *CO-oximeter* (which determines by optical measurements in vitro the concentration of several forms of haemoglobin in arterial blood). See also DD.2.

Pulse oximeter equipment, as manufactured to date, is never calibrated in the same sense that an invasive blood pressure transducer can be calibrated. Various manual or automatic adjustments are possible for the *pulse oximeter equipment* (e.g. to set gains or cancel amplifier offsets) but these are all adjusted against ordinary electronic reference standards (e.g. an offset adjustment to bring the reading of a voltmeter to zero). The basic relationship between optical signals derived from the *patient* and the displayed value of SpO_2 is determined by the *manufacturer* for a particular combination of *pulse oximeter monitor* and *pulse oximeter probe.* This document refers to this as *modulation ratio* or *R* that is stored in firmware and never adjusted[126]. Most *pulse oximeter monitors* contain a set of *modulation ratios* where each *modulation ratio* is for a specific range of optical properties (i.e. range of wavelengths). Most contemporary *pulse oximeter equipment* uses the concept of a *modulation ratio* or the *ratio of ratios*, which can be approximated as indicated in Formula (EE.1).

$$R = \frac{\log_{10}(\max._{\mathrm{red}} / \min._{\mathrm{red}})}{\log_{10}(\max._{\mathrm{IR}} / \min._{\mathrm{IR}})} \approx \frac{(AC_{\mathrm{red}} / DC_{\mathrm{red}})}{(AC_{\mathrm{IR}} / DC_{\mathrm{IR}})} \approx \frac{\%mod_{\mathrm{red}}}{\%mod_{\mathrm{IR}}} \qquad \text{(EE.1)}$$

where

AC_{red} is the magnitude of the time-modulating component in the red wavelength signal;

AC_{IR} is the magnitude of the time-modulating component in the infrared wavelength signal;

DC_{red} is the magnitude of the non-modulating component in the red wavelength signal; and

DC_{IR} is the magnitude of the non-modulating component in the infrared wavelength signal.

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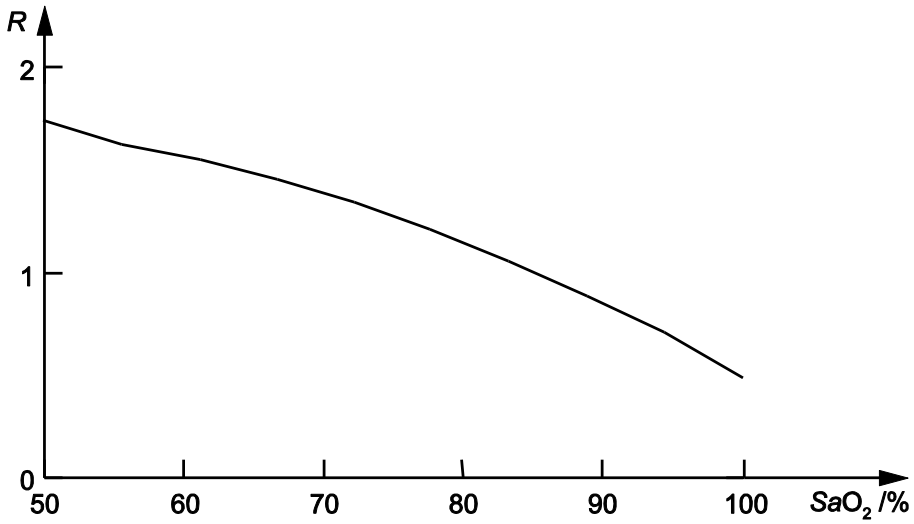
NOTE This approximate formula is cited only to provide a concrete example to support the following calibration curve discussion; accurate *pulse oximeter monitors* are designed around a variety of mathematical approaches, each of which requires some sort of empirical calibration curve.

An empirically determined calibration curve, such as that illustrated in Figure EE.1, allows the *pulse oximeter equipment* to derive the displayed SpO_2 from the observed *R*. The *procedure* to determine the calibration curve is called a *controlled desaturation study*. It typically involves providing healthy adult volunteer test participants with breathing mixtures having reduced oxygen content. Arterial blood is sampled and then measured with a *CO-oximeter*. The *CO-oximeter* readings are plotted against the *R* values observed when the blood was sampled. The *process* should be carefully conducted, to avoid a wide variety of possible errors. *Controlled desaturation studies* can also be conducted by comparing the readings of the *pulse oximeter equipment* under test to *transfer standard* that have previously been calibrated against *CO-oximeters*. This approach avoids the need to draw arterial blood but still always requires the use of human test participants.



Red/IR *modulation ratio*, *R*, as a function of arterial oxygen saturation

Figure EE.1 — Sample calibration curve for *pulse oximeter equipment*

This curve shows the observed value of *R* for various values of SaO_2 determined with a *CO-oximeter*. *Pulse oximeter equipment* software uses this curve to determine the relationship between an observed value of *R* and a displayed SpO_2 value for *pulse oximeter probe*s with specific range of optical properties (i.e. range of wavelengths).

EE.5 What is a *functional tester*?

Every *pulse oximeter equipment* tester intended to generate simulated signals on the market at the time of writing is a *functional tester*. Two principal characteristics of *functional testers* are as follows.

— An appropriate *functional tester* allows the *responsible organization* to determine whether the *pulse oximeter equipment* is performing as the *manufacturer* designed it to perform, without in any way determining whether the design was correct.

— A *functional tester* has limited ability to determine whether any *pulse oximeter probe* is performing as the *manufacturer* designed it to perform (more is indicated about the limitations below) and can never determine whether the design was correct.

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A *functional tester* presents the *pulse oximeter equipment* with a signal having a predictable value of *R*, so that the *responsible organization* can observe the resulting displayed value of SpO_2 and evaluate it in comparison to expectations for that particular *pulse oximeter monitor* model. If the tester *manufacturer* knows the calibration curve that has been designed into a particular *pulse oximeter monitor*, the *manufacturer* can accurately produce the *R* value which ought to lead to a particular value of SpO_2 (e.g. 85 %). Then the *pulse oximeter equipment* can be evaluated for its ability to reproduce the calibration curve that was designed into it. Any error exceeding the combined error specifications of the *pulse oximeter equipment* and the tester suggests that either the *pulse oximeter equipment* or the tester requires repair or replacement.

An accurate reading of SpO_2 on a *functional tester* never implies that the *pulse oximeter equipment* is accurate on human beings. All that is being evaluated by the tester is the *pulse oximeter monitor's* ability to reproduce the calibration curve that the *manufacturer* designed into it; this calibration curve is possibly not accurate. The following detailed observations are chosen to emphasize this point.

— Some *functional testers* are designed to connect electrically to the input of the *pulse oximeter monitor* in place of the *pulse oximeter probe*. This type of tester cannot evaluate the optical properties of the *pulse oximeter probe*, which are known to be very important for calibration. This family of testers has limited abilities to test the optical properties of *pulse oximeter equipment*.

— Some *functional testers* are electronic modulators having an optical interface to the *pulse oximeter equipment*. The *pulse oximeter probe* is applied to an optomechanical “finger” of some sort, and modulated optical signals are delivered to the detector of the *pulse oximeter probe*. While such testers can appear to evaluate the *pulse oximeter probe*, they usually only test the most basic properties of the *pulse oximeter probe*; that its light sources and detector are active and that no disabling shorts or open circuits exist. The same determination can be made by applying the *pulse oximeter probe* to the *operator's* index finger and observing that the *pulse oximeter equipment* displays some value of SpO_2. This type of *functional tester* simply uses the *pulse oximeter probe* as a tool to deliver a desired test signal to the electronics of the *pulse oximeter equipment*.

— Several brands of *functional tester* have an optomechanical “finger” containing a detector, which picks up light from the *pulse oximeter probe's* light emitter, and a light-emitting diode (LED) which delivers modulated light to the *pulse oximeter probe's* detector (see Figure EE.2). This is one example of the optically interfaced tester described above. If the *pulse oximeter probe's* red LED were of the wrong wavelength for the calibration curve in use, this definitely causes the *pulse oximeter equipment* to be inaccurate in actual use on *patients*. The *functional tester* is unable to detect this error, as does the *pulse oximeter equipment* under test, so that inaccurate *pulse oximeter equipment* can appear to be accurate. Some *manufacturers* produce *pulse oximeter probes* with a variety of different wavelengths. The *pulse oximeter monitor* uses something like a resistor within the *pulse oximeter probe* to select the correct calibration curve from a the set of available curves. The instruction for selection of the correct curve is given to the *pulse oximeter monitor* by means of a coding device, such as a resistor, which is carried by the *pulse oximeter probe*. An important quality control requirement in new or *remanufactured pulse oximeter probes* is close matching of emitter wavelength (and wavelength distribution) to the calibration code in the *pulse oximeter probe*. Currently available *functional testers* cannot verify the correctness of the value of the centre wavelength.

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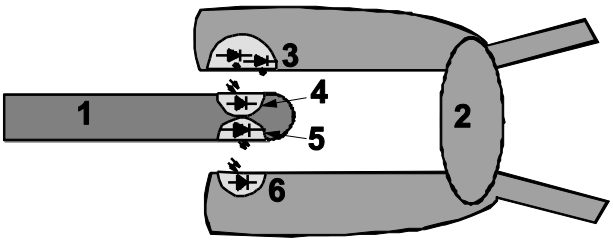

Key

1 test finger
2 *pulse oximeter probe*
3 light source
4 photodiode
5 LED
6 photodiode

Figure EE.2 — Interface of a *functional tester* that uses a photodiode and LED to interact with a *pulse oximeter probe*

— Some *functional testers* provide comprehensive tests for all possible shorts and opens in the *pulse oximeter probe*. While this is a valuable test, electrical integrity is a necessary, but not sufficient, condition for *accuracy*.

— In a *pulse oximeter probe*, the colour of the plastic cushion or bandage that touches the *patient's* skin has an important effect on the calibration of the *pulse oximeter equipment*. If a bandage were badly stained, this can affect the *SpO_2 accuracy* of the *pulse oximeter probe* in actual use. The types of *functional tester* described above are insensitive to the presence of the stain. This "bandage colour" issue actually identifies a larger sphere of concern. The *SpO_2 accuracy* of *pulse oximeter equipment* is affected strongly by the interacting optical properties of both the *patient's* tissue and every part of the surrounding optical environment. *Functional testers* are insensitive to such effects. A true pulse oximetry calibrator, when it appears, needs to reproduce faithfully this complex interaction. An implication is that the *accompanying documentation* of any POC that eventually comes to market should include a discussion as to which physical and physiological aspects of *pulse oximeter equipment* performance are replicated, and which are not.

One class of *functional testers* has inherent sensitivity to the wavelength distributions of the *pulse oximeter equipment's* emitter. Such testers work by modulating optically the light emitted by the *pulse oximeter probe's* emitter and conducting the modulated light to the *pulse oximeter probe's* detector. One such tester family works by modulating the amount of a dye solution that is forced between the *pulse oximeter probe's* emitter and detector (see Figure EE.3). Another such family uses a liquid crystal device to modulate the light in route from emitter to detector (see Figure EE.4). Such testers can be designed to cause wavelength-dependent modulation approximating the dependence of haemoglobin's optical absorption on wavelength. In principle they can also be designed to approximate the important effects of tissue scattering on *pulse oximeter monitor* calibration (although, the committees cannot find published evidence that this has been done). For such testers to approach the status of true POCs, they need to reproduce the optical interactions of human tissue with the coloured materials that surround the emitter and detector. At the present state of the art, the committees conclude that this class of *functional tester* should not be assumed to come any closer to being true *SpO_2 accuracy* testers than the other classes of *functional testers*. The *functional tester* can be assessed on the usual basis of comparing other analytic test equipment (e.g. cost) convenience, durability, and versatility.

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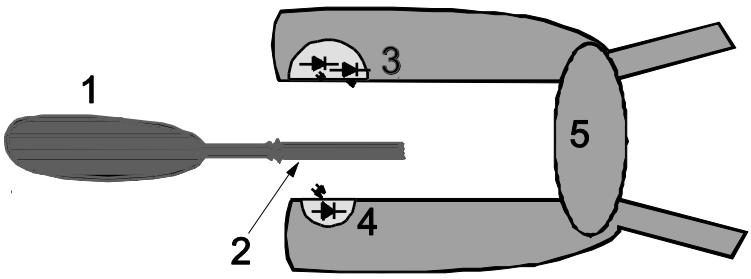
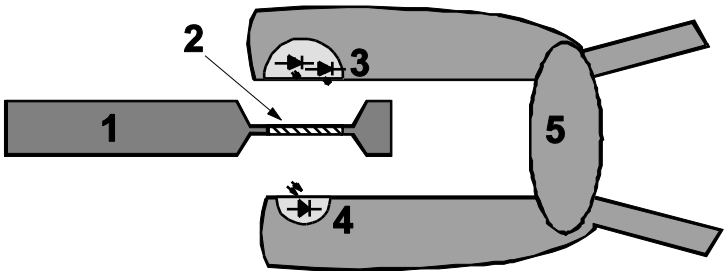


Key
1 bladder with dye mixture
2 variable-gap optical cell
3 light source
4 photodiode
5 *pulse oximeter probe*

NOTE By squeezing the bladder, the amount of dye forced between the plates of the optical cell varies.

Figure EE.3 — Interface of a *functional tester* that uses a dye mixture



Key
1 test finger
2 liquid crystal modulator
3 light source
4 photodiode
5 *pulse oximeter probe*

Figure EE.4 — Interface of a *functional tester* that uses a liquid crystal modulator

Some *manufacturers* of new or *remanufactured pulse oximeter probes* have claimed that their *pulse oximeter probes* are routinely shown to be accurate by testing with *functional testers*. Such evidence has so far been insufficient to reflect the true performance of *pulse oximeter equipment*, given the limitations of *functional testers*.

EE.6 Beyond *functional testers*

How does one know when a true *pulse oximeter equipment* calibrator has been developed? Such equipment can be recognized by the way it is used and by the nature of the published experimental results that *validate* its capabilities. Its properties would be as follows.

— The *pulse oximeter probe* of *pulse oximeter equipment* is applied to a part of the POC that approximates the dynamic optical behaviour of a body part for which the *pulse oximeter probe* is designed. The optical

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simulation should include the interaction that occurs between the *pulse oximeter probe's* materials and human tissue, in which light repeatedly leaves and re-enters the tissue, reflecting from the materials of the *pulse oximeter probe*.

— The POC can be set to simulate the optical behaviour of the simulated body part for a *patient* having selected oxygen saturation, SaO_2, causing the *pulse oximeter equipment* to display an SpO_2 reading.

— *Validation* experiments exists that has established that the reading induced in the *pulse oximeter equipment* by the POC matches within the stated simulation SpO_2 *accuracy* the reading that the same *pulse oximeter monitor* and *pulse oximeter probe* give on a *patient*. The basic *validation* experiment that should be done many times and under many conditions, is as follows.

  — Apply a *pulse oximeter probe* to a human being whose SaO_2 is determined by measuring arterial blood samples in a multi-wavelength oximeter (e.g. *CO-oximeter*).

  — Observe the SpO_2 value displayed by the *pulse oximeter equipment*. It does not matter whether this number is accurate.

  — Apply the same *pulse oximeter monitor* and *pulse oximeter probe* to the POC, with the POC set to simulate the same SaO_2 that was seen on the human participant.

  — Observe the SpO_2 value displayed by the *pulse oximeter equipment*.

  — Calculate the error of the POC as the difference between the two SpO_2 values. The *validation* of POC performance should include testing over the following ranges of conditions.

  — Use many different brands of *pulse oximeter equipment*, having emitters of the widest available variety of wavelength distributions.

  — Use many different *pulse oximeter probes* designed for use with the chosen body part, including the widest available variety of shapes and material colours. Testing should preferably include use of "challenge" *pulse oximeter probes* that are known to be very inaccurate in use on *patients*. The POC is required to cause the *pulse oximeter monitor* to exhibit exactly the same inaccuracy that the real *patient* does.

  — Testing should preferably include use of a particular class of challenge *pulse oximeter probes* that are known to produce very different SpO_2 readings when tested on different human volunteers having the same value of SaO_2. As an example of such a *pulse oximeter probe*, Figure EE.5 shows the reflectance spectrum of a particular blue bandage material which has been shown to give extremely variable performance from one *patient* to another. This is not a material that would be used in any commercial *pulse oximeter probe*; it was specifically selected to demonstrate the variable saturation dependent bias that would result when compared to a standard *pulse oximeter probe*. Figure EE.6 displays this extremely variable bias, compared to a standard *pulse oximeter probe*. This *pulse oximeter probe* cannot be accurate on all *patients*, no matter what calibration curve was used in the *pulse oximeter monitor*. If a POC has not been *validated* in testing with such *pulse oximeter probes* (which implies that a particular *pulse oximeter probe*/*pulse oximeter equipment* combination can appear accurate when tested on the POC but be inaccurate on many *patients*), this limitation should be disclosed clearly in the *accompanying documentation* of the POC, and the users of the POC should be advised of the importance of testing *pulse oximeter equipment* on a variety of human volunteers. In order to test meaningfully the behaviour of these variable-calibration challenge *pulse oximeter probes*, the POC would presumably have to be adjustable to simulate one of several different human participants.

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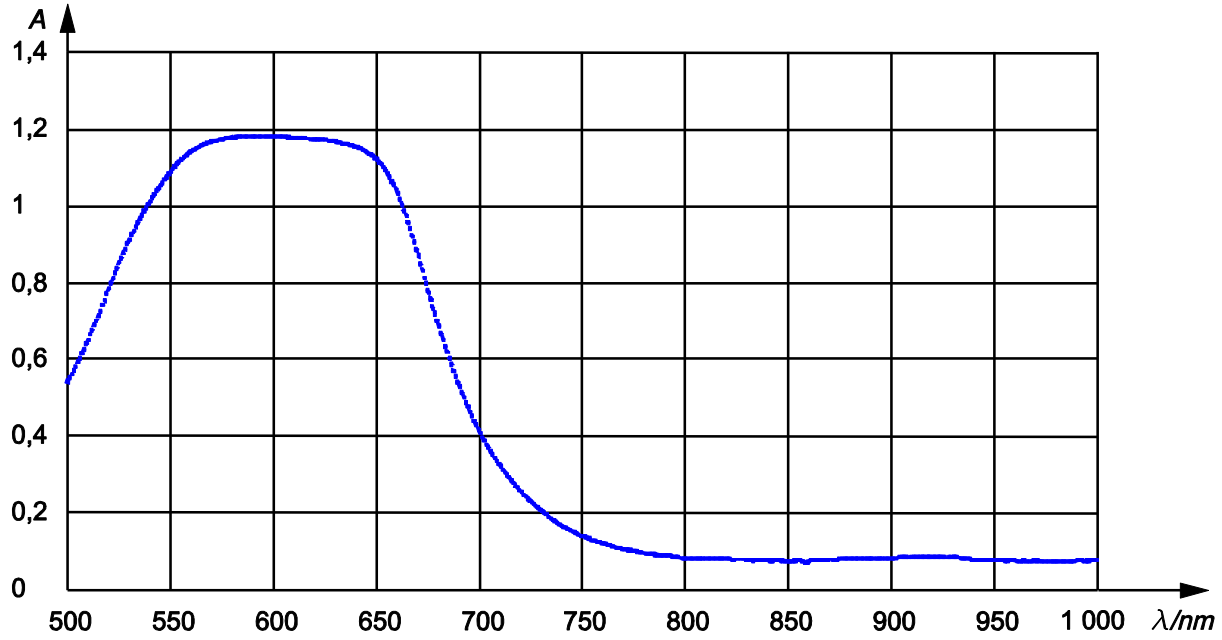
Test many *patients* or volunteer test participants at each SaO_2 level where the POC is specified for use. Special emphasis should be given to testing at the lowest values of SaO_2 at which the POC is specified for use, because *pulse oximeter equipment* errors tend to be larger at low saturation.

The *accuracy* specification of the POC does not include any component for inaccuracy of the *pulse oximeter equipment* under test (compare this to the typical *functional tester* specification, such as ±1 % SpO_2 ±(stated SpO_2 *accuracy)*). It is the purpose of the POC to determine the SpO_2 *accuracy* of the *pulse oximeter equipment*, without direct human testing. In this sense, the POC is *transfer standard*, with *controlled desaturation study* testing on humans retaining the role of "gold standard." See also JJ.



A absorbance, measured in reflection

λ optical wavelength

Figure EE.5 — Absorbency of blue bandage material (measured in reflection) used in a special test *pulse oximeter probe* with great *patient*-to-*patient* variability

Figure EE.6 a) shows a test of a blue test *pulse oximeter probe* (left index finger) as a function of reference *pulse oximeter probe* (left little finger).

Figure EE.6 b), shows a test of a *pulse oximeter probe* (left middle finger) as a function of the reference *pulse oximeter probe* (left little finger). A separate regression line is shown for each of the five test participants.

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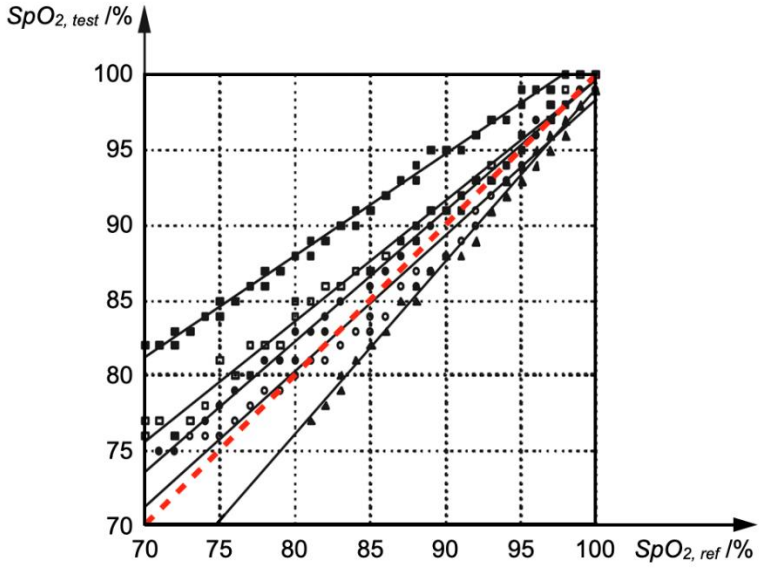
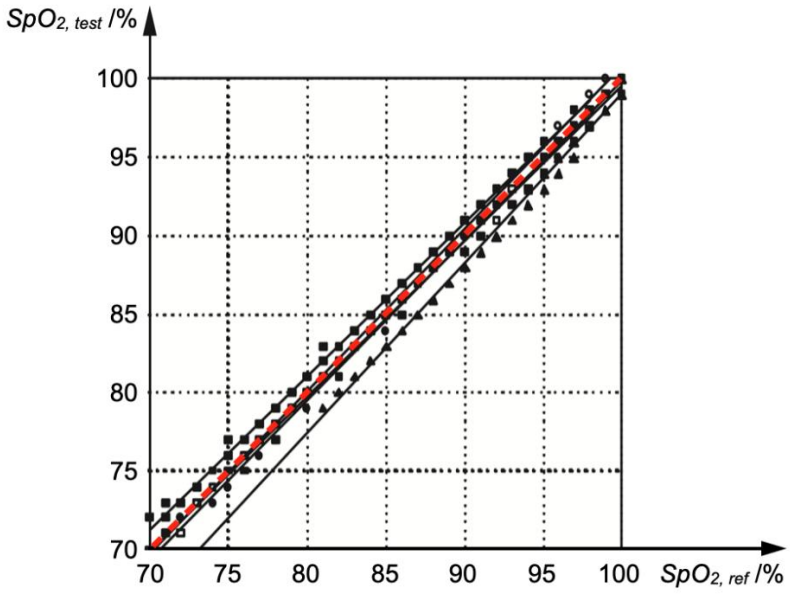


a) Comparison of a blue test *pulse oximeter probe* to a standard production *pulse oximeter probe*



b) Comparison of one standard production *pulse oximeter probe* to another

Key

$SpO_{2,\,\mathrm{test}}$ is the test SpO_2
$SpO_{2,\,\mathrm{ref}}$ is the reference SpO_2

SpO_2 readings observed in five participants are indicated by the solid and hollow squares, solid and hollow circles, and solid triangle points along the respective solid lines. The red coloured dashed line is the line of identity. See also CC.2.3.3.

Figure EE.6 —High-variability production *pulse oximeter probe* in a *controlled desaturation study* on five test participants

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Annex FF
(informative)

Concepts of *ME equipment* response time

FF.1 General

There can be trade-offs between accurately tracking the magnitude of changes in saturation and minimizing the effects of noise. In general, faster response times can cause *pulse oximeter monitors* to be more vulnerable to noise but can allow them to follow the actual saturation more closely. The response of some *pulse oximeter monitors* can be optimized for particular clinical situations. There are two important concepts in describing *pulse oximeter monitor* response. One is the *fidelity* in tracking saturation changes. The other is the *delay* from the time that an event occurs (the SaO_2 changes at the measurement site) until the display indicates the change or the generation of *alarm signals*. "Fidelity" and "delay" are influenced by *pulse oximeter equipment* design and *operator settings*. *Pulse oximeter equipment* design can include signal-processing and conditioning times and data transmission delays. Adjustable controls (e.g. averaging time and *alarm signal generation delay)* can be set.

FF.2 Fidelity

Fidelity can be described graphically by showing the range of responses of the *pulse oximeter equipment* to a change in saturation. Figure FF.1 illustrates a simulated response of *pulse oximeter equipment* to a change in saturation. Figure FF.2 illustrates the simulated effect of different averaging times on the response of the *pulse oximeter equipment*.

The variables δ and Δ in Figure FF.1 do not refer to any particular requirement in this document. They are illustrated here as possible points of interest, in that these are the likely areas of SpO_2 *accuracy* that can be affected by different averaging or filtering techniques response curves. The span depicted by the variable δ represents a time lag before changes in saturation are reflected in the processed SpO_2 value. This lag can be caused by, for example, the time required for data acquisition, signal conditioning and algorithm processing. The deviation denoted by Δ illustrates a lack of fidelity in reproducing the degree of change in a transient desaturation. Δ is generally affected by, for example, signal averaging or the *data update period.*

The importance of the errors, δ and Δ, introduced by the processing of the SpO_2 parameter as well as the detection of *alarm conditions* and the subsequent generation of *alarm signals* that the purchasers of *pulse oximeter equipment* need to consider for the applications in their clinical practice (see 201.7.9.2.1.101) c), are well illustrated in Reference [94].

FF.3 Effects of delays

Delays can be described graphically, for example, by showing the response of the *pulse oximeter equipment* using Figure FF.3. The time from t_1 to t_2 is the *alarm condition delay* and the time from t_2 to t_3 is the *alarm signal generation delay*.

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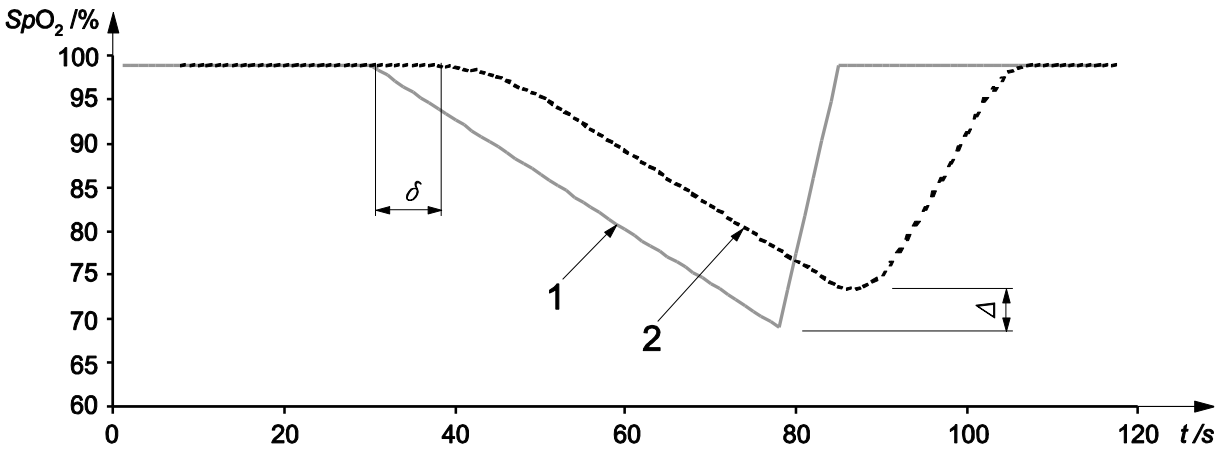
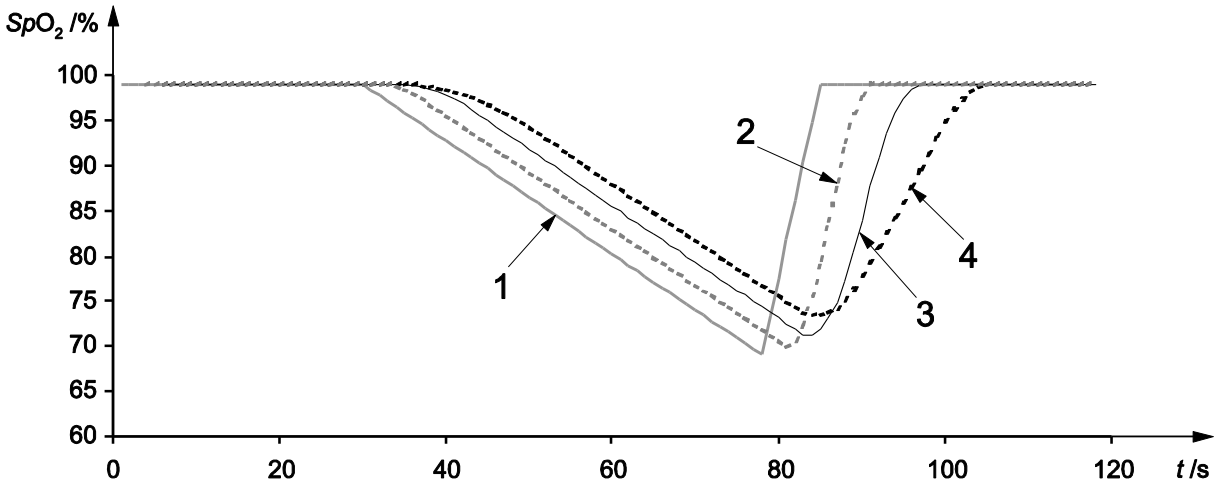


Key

1	SaO_2	Δ	is saturation deviation
2	displayed SpO_2	δ	is time delay
		SpO_2	is saturation
		t	is time

Figure FF.1 — Illustration of fidelity of *pulse oximeter equipment* performance in tracking saturation changes



Key

1 SaO_2
2 displayed SpO_2 faster averaging
3 displayed SpO_2 normal averaging
4 displayed SpO_2, slower averaging
SpO_2 is saturation
t is time

Figure FF.2 — Illustration of effect of different averaging times on fidelity

A possible *procedure* to measure the sum of the *alarm condition delay* and alarm *signal generation delay* of *pulse oximeter equipment* is described below.

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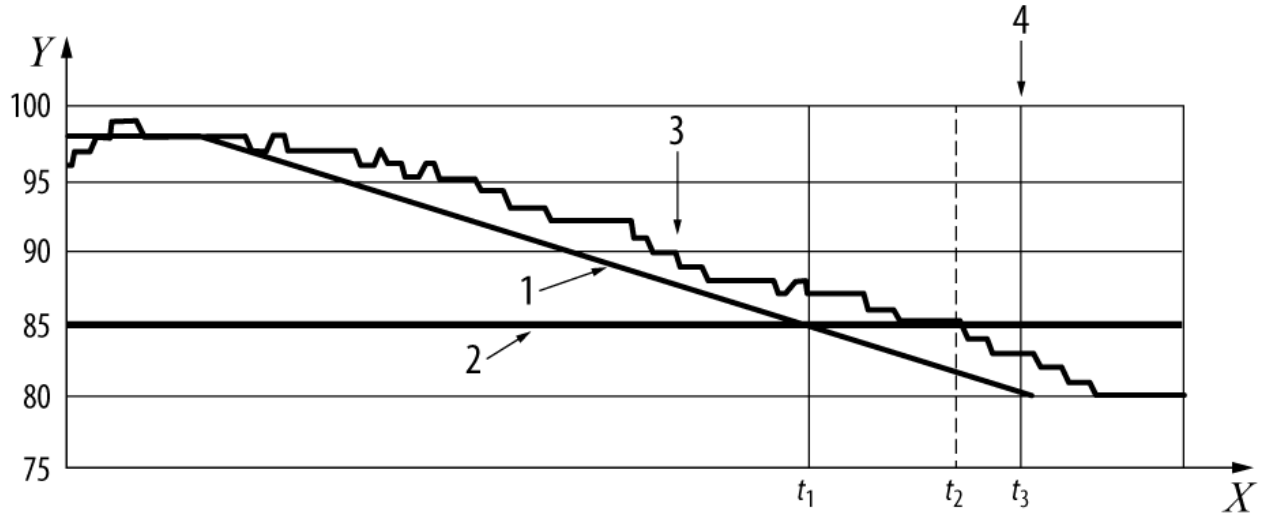
— A *functional tester* is set to start at a saturation level of e.g. 98 %.

— This level should be simulated for a time that is sufficient to allow stabilization of the *pulse oximeter equipment* under test.

— The *functional tester* then changes the saturation level in a linear ramp function with a prespecified slope (or any other prespecified function) down to a given end value (e.g. 5 % below the *alarm limit*).

— The sum of the *alarm condition delay* and *alarm signal generation delay* is defined as the time from having the simulated saturation passing the *alarm limit* threshold (e.g. 85 % or the default low saturation *alarm limit*) to the time the *alarm system* generates the appropriate *alarm signal*.

Figure FF.3 illustrates the components of *alarm signal generation delay*.



Key

1 *SaO_2*
2 *alarm limit*
3 displayed *SpO_2*
4 *alarm signal* generation
X is time
Y is *SpO_2*
t_1, t_2, t_3 see FF.3 description

Figure FF.3 — Graphic representation of components of *alarm system* delay

The delay due to the *pulse oximeter equipment* processing and averaging is $t_2 - t_1$, the *alarm condition delay*. The interval $t_3 - t_2$, the *alarm signal generation delay*, is attributed to the *alarm system* strategy and the communication time to the *alarm signal* generation equipment or *distributed alarm system* (e.g. *patient* monitor or central station). Thus, the overall *alarm system* delay time is $t_3 - t_1$.

Figure FF.4 represents a faster desaturation slope and a more realistic, noisier saturation signal. Curves 3 and 4 underestimate the depth of the fall in saturation. Curve 2, faster averaging, can cross a low saturation *alarm limit* sooner than curve 3, normal averaging, or curve 4, slower averaging, which likely does not cause an *alarm condition* at all. The benefit of normal and slower averaging is to smooth out the otherwise noisy signal and reduce the number of *false positive alarm conditions*.

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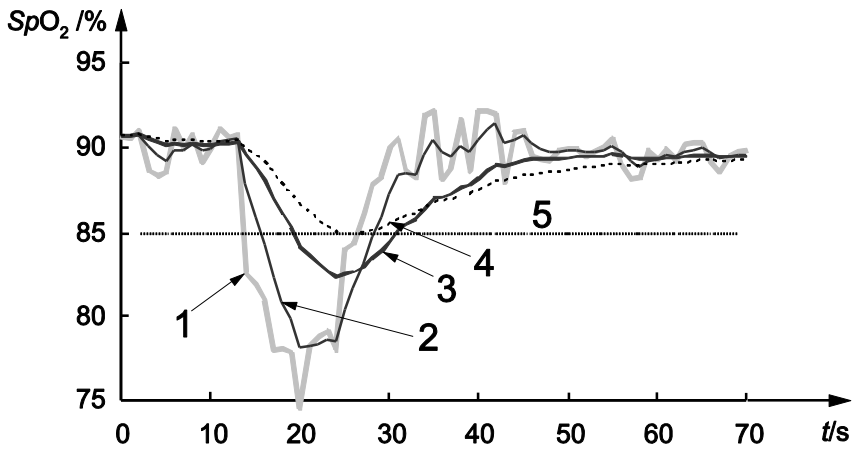

Key

1 unprocessed SpO_2
2 displayed SpO_2, faster averaging
3 displayed SpO_2, normal averaging
4 displayed SpO_2, slower averaging
5 *alarm limit*
t/s is time in seconds

Figure FF.4 — Illustration of the effects of different averaging times on a more rapid and noisier desaturation signal

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Annex GG
(normative)

Data interface requirements

GG.1 Background and purpose

Pulse oximeter equipment typically transmits data using proprietary interfaces and protocols. To help foster data availability and interoperability of *pulse oximeter equipment* in the medical device ecosystem, increased standardization of this interface is desirable, acknowledging that different applications need different levels of data. This document has sought to logically categorize the data that can be transmitted or received as parameters, identification data, settings data, configuration data, specification data, service monitoring data and *alarm system* data via a *functional connection*.

This Annex only applies to boards or systems with a *functional connection* that forms an electronic data interface that is intended to provide open access for other *ME equipment* or *ME Systems* to *pulse oximeter equipment* data.

The intention is that information that is provided to the user via the user interface (i.e. front panel) to ensure safe operation should also be available over the electronic data interface, especially if the interface is used for remote operation.

The following are the categories of data:

– **Parameters and units of measurement:** Parameters and units of measurement used within the *pulse oximeter equipment*

– **Equipment identification:** Information identifying the *pulse oximeter equipment.*

– **Equipment settings:** Settings relating to the control and operation of the *pulse oximeter equipment.*

– **Equipment configuration:** *equipment settings* that can be configured.

– **Equipment specifications:** Relevant specifications to be transmitted.

– **Service monitoring:** Indicators relating to preventative or corrective maintenance of the *pulse oximeter equipment* and its *accessories.*

– **External device information:** parameters and status information to be received by the *pulse oximeter equipment* from the system or other equipment.

All *pulse oximeter equipment* with a *functional connection* should provide the information to enable identification of the *pulse oximeter equipment* and the required parameters in Table GG.1.

The clinical context and rationales supporting the inclusion of many of the listed data items is provided in GG.3.

GG.2 Data definition

a) If *pulse oximeter equipment* falls under the requirement of 201.103.2 a), the *pulse oximeter equipment* parameter data shall include at a minimum the parameters listed as required in Table GG.1.

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b) In addition to these parameters, the pulse plethysmographic waveform and metadata should be transmitted. This metadata includes signal quality indicators, and *manufacturer*-specific indices.

Table GG.1 — Parameters and units of measurement

Parameter	Required/ Recommended/ Optional	Description	Type
SpO_2	Required	*SpO_2* value as calculated by the *pulse oximeter equipment* (see 201.3.261)	Value: (%)
SpO_2 algorithm type	Optional	Algorithm meta-data that can be helpful to determine whether a specific *pulse oximeter equipment* is fit for the intended use (e.g. a closed loop application[12])	Text string
Pulse rate	Required, if pulse rate is provided	Pulse rate as calculated by *pulse oximeter equipment*	Value: (beats/min)
Signal inadequacy	Required, if *pulse oximeter equipment* continues to report data while the *accuracy* can be compromised.	Indicator when the displayed *SpO_2* or pulse rate value is potentially incorrect (see 201.12.4.102)	As described by *manufacturer*
Signal inadequacy metadata	Optional	Information used by *manufacturer* to compute inadequacy indicator (e.g. wavelengths used in metric)	Text string (Red, IR, both)
Pulse waveform	Recommended, if provided.	Data samples representative of the displayed pulse plethysmographic waveform arising from the time-varying optical attenuation between the *pulse oximeter probe's* emitter and detector.	Waveform
Waveform metadata	Optional	Means by which the pulse waveform was prepared (*normalized*, filtered)	Text String
Raw plethysmographic waveform	Optional	Unprocessed plethysmographic waveform for each wavelength. This data need not be available to the *operator*	Waveform
Signal quality indicator	Recommended	A measure of the signal quality (e.g. *%mod*, noise indicator)	As described by *manufacturer*
The committees intend that many of the recommended parameters should become standard functionality on the data interface in the future, when shown by the *pulse oximeter equipment* on the display.			

c) If *pulse oximeter equipment* identification data is transmitted via a *functional connection*, the *pulse oximeter equipment* identification data shall include the parameters listed in Table GG.2.

1) This data may be provided by the combination of:

i) the *model or type reference*;

ii) serial number and the software version of the *pulse oximeter equipment;* or

iii) a unique device identifier (UDI).

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Table GG.2 — Equipment identification

Parameter	Required/ Recommended/ Optional	Description	Type
Manufacturer	See GG.2.c)1)	Identification of the *manufacturer* of the equipment	Text string
Model	See GG.2.c)1)	Identification of the product or model number of the equipment (monitor)	Text string
Serial number	See GG.2.c)1)	The identification number of the equipment	Text string
Unique device identifier	See GG.2.c)1)	unique device identifier for the equipment (e.g. as submitted to the GUDID)	Text string
Software version[a]	See GG.2.c)1)	Identification of the *pulse oximeter equipment* major software version as also accessible by the user on the *pulse oximeter equipment* display	Text string
Security certificate	Recommended, if the data interface is intended to be connected to a network that provides also external access. Optional, otherwise.	Means to allow a data consumer to verify the trustworthiness of the *pulse oximeter equipment* sending the data.	Text string

a More than one software version may be communicated from the equipment for different subcomponents.

NOTE The United States Food and Drug Administration UDI, consisting of a device identifier (DI) and a production identifier satisfies the requirements of Table GG.2.

d) If *pulse oximeter equipment* settings data is transmitted via a *functional connection*, the *pulse oximeter equipment* settings data shall include the parameters listed in Table GG.3.

1) Equipment settings may be made available through a *functional connection* continually or through a query-response.

2) If equipment settings are communicated by query-response, a change to equipment settings shall be immediately indicated via the *functional connection*.

Table GG.3 — *Operator*-adjustable equipment settings

Parameter	Required/ Recommended/ Optional	Description	Type	Adjustable via electronic interface?
Data update period	Optional	Interval in which the *pulse oximeter equipment* algorithm provides new valid data to the display or the *functional connection* (see 201.3.212)	Value: (s)	Typically not
Averaging time	Optional	*Operator* settable or preset averaging window used to calculate a parameter.	List of text strings (*manufacturer* specified)	Possible
Alarm signal inactivation state	Recommended	List of text strings (*alarm off*, *alarm pause*, *audio off*, *audio pause*, *acknowledged*)	List of text strings	Possible

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Parameter	Required/ Recommended/ Optional	Description	Type	Adjustable via electronic interface?
High *SpO2 alarm condition*	Recommended	Setting of the high *SpO_2 alarm limit*	Value: (% *SpO_2*)	Possible
Low *SpO2 alarm condition*	Recommended	Setting of the low *SpO_2 alarm limit*	Value: (% *SpO_2*)	Possible
High pulse rate *alarm condition*	Optional	Setting of the high pulse rate *alarm limit*	Value: (beats/min)	Possible
Low pulse rate *alarm condition*	Optional	Setting of the low pulse rate *alarm limit*	Value: (beats/min)	Possible
Active *technical alarm conditions*	Recommended	Currently active *alarm conditions* Examples of *alarm conditions* include the *pulse oximeter probe* is not connected to the *pulse oximeter equipment*, the *pulse oximeter probe* is not detected, detection of an artefact, no signal detected, low quality of signal, display error, *pulse oximeter equipment* inoperative, low battery, defective *pulse oximeter probe* or *probe cable extender*, unrecognized *pulse oximeter probe*, incompatible *probe cable extender*, line frequency interference, ambient light interference, low perfusion, inadequate display	List of text strings (*manufacturer* specified)	Not adjustable

e) If *pulse oximeter equipment* configuration data can be queried through a *functional connection*, the *pulse oximeter equipment* configuration data should include the parameters listed in Table GG.4.

Table GG.4 — Equipment configuration

Parameter	Required/ Recommended/ Optional	Description	Type
Pulse oximeter probe type	Optional. if available	Type of *pulse oximeter probe* for applicable *patient* cohort (e.g. neonate, paediatric, adult)	Text string
Pulse oximeter probe model number	Optional, if available	If applicable, specific *pulse oximeter probe* software and hardware version number connected to *pulse oximeter equipment*	Text string
Location intended for the *pulse oximeter probe*	Optional, if available	Intended location on the body of *pulse oximeter probe* type (e.g,. ear, nasal, finger)	Text string

f) If *pulse oximeter equipment* specifications data is transmitted via a *functional connection*, the *pulse oximeter equipment* specifications data should include the parameters listed in Table GG.5.

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Table GG.5 — Equipment specifications

Parameter	Required/ Recommended/ Optional	Description	Type
SpO_2 accuracy	Recommended, if known	*SpO_2 accuracy* applicable for the currently used *pulse oximeter probe*, expressed as A_{rms} as specified in 201.12.1.101.1 of this document	Value: (%)
Declared ranges of *SpO_2* and *SpO_2 accuracy*	Optional	*Declared range* of *SpO_2* and *SpO_2 accuracy* over the *declared range,* as specified in 201.12.1.101.1 of this document	Value pair [upper limit, lower limit] (% *SpO_2*) Value: (%)
SpO_2 accuracy under conditions of motion	Optional	*SpO_2 accuracy* under conditions of motion as specified in 201.12.1.102 of this document	Value: (%)
SpO_2 accuracy under conditions of low perfusion	Optional	*SpO_2 accuracy* under conditions of low perfusion as specified in 201.12.1.103 of this document	Value: (%)
Pulse rate *accuracy*	Recommended	Pulse rate *accuracy* as specified in 201.12.1.104 of this document	See respective section.
Declared ranges of pulse rate	Optional	*Declared ranges* of pulse rate as specified in 201.12.1.104 of this document	Value pair [upper limit, lower limit] (beats/min)

g) If *pulse oximeter equipment* service monitoring indicators data is transmitted via a *functional connection*, the *pulse oximeter equipment* service monitoring indicators data should include the parameters listed in Table GG.6.

Table GG.6 — Service monitoring indicators

Parameter	Required/ Recommended/ Optional	Description	Type
Remaining *pulse oximeter probe* life	Optional	Remaining *pulse oximeter probe* life, before *pulse oximeter monitor* inhibits operation	Value: (e.g. hours, days)
Remaining *pulse oximeter probe* end-of-life.	Optional	Remaining time before *manufacturer* determined end-of-life.	Value: (e.g. hours, days)
Next periodic maintenance date	Optional	Date service is required	ISO 8601 Date (YYYY-MM-DD)

NOTE If any component of the *pulse oximeter equipment* has a time-out condition, which can interrupt monitoring that needs to be conveyed to the *operator*. Remaining *pulse oximeter probe* life is intended to convey information as to when the *pulse oximeter probe* times out and that needs to be included in the data transfer (not intended to be implemented if *pulse oximeter monitor* does not have a time-out feature)

h) **External device information:** parameters and status information received by the *pulse oximeter equipment* from the system or other equipment.

i) *Pulse oximeter equipment* can also receive data via a *functional connection* and use this data to improve the performance or the usability of the *pulse oximeter equipment*. Examples are listed in Table GG.7

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Table GG.7 — External device information

Parameter	Required/ Recommended/ Optional	Description	Type
Blood pressure cuff status	Optional	Status of blood pressure cuff	Active (inflating or deflating),Not active
ECG data	Optional	HR or waveform can be helpful to improve pulse oximetry performance.	ECG samples HR numeric

If such received data can impact safety or performance, the *manufacturer* shall evaluate the associated *hazards* or *hazardous situations* and address those *hazards* in the *risk management process.*

Check conformity by inspection and by inspection of the *risk management file*.

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Annex HH
(informative)

Clinical context and rationales of data interface requirements

The SpO_2 measurement is a non-invasive surrogate for SaO_2, the arterial oxygen saturation. Oxygen saturation is used alone for *patient* assessment and can be used as an input to open-loop and closed-loop systems relating to oxygenation and ventilation. There is significant clinical value to provide data from *pulse oximeter equipment* for use in systems such as integrated clinical environment. AAMI 2700-1:2019, Annex B[19], provides examples of detailed clinical scenarios that illustrate adverse events that can have been averted with the use of such integrated *ME systems*. These scenarios include the use of safety interlocks (with and without *ME equipment* synchronization), *process* control, smart *alarm system*, clinical decision support, and physiological closed-loop control[12].

Device identification data can be helpful to facilitate the location of devices that need service activities to maintain safety or security. Example:

A serious vulnerability (9.3) (CVE-2015-7599) has been uncovered in the real time operating system (VxWorks) used by some standalone networked *pulse oximeter equipment*. Midland hospital IT department needs to determine if it has vulnerable *pulse oximeter equipment* and its location. Using a network tool they are able to query each networked device through their external data interfaces and to identify *pulse oximeter equipment's manufacturer*, model and their respective firmware versions and thus enabling at-risk *pulse oximeter equipment* to be updated or taken out of service.

24 CVE-2015-7599 190 DoS Exec Code Overflow 2017-02-07 2017-11-16 9.3e - Integer overflow in the _authenticate function in svc_auth.c in Wind River VxWorks 5.5 through 6.9.4.1, when the Remote Procedure Call (RPC) protocol is enabled, allows remote attackers to cause a denial of service (crash) or possibly execute arbitrary code via a username and password.

https://www.cvedetails.com/vulnerability-list/vendor_id-95/product_id-15063/Windriver-Vxworks.html

For future interoperability that allows that a *ME equipment* or *ME system* can dynamically determine whether connected *pulse oximeter equipment*, if fit for the intended combined use, the query of *pulse oximeter equipment* properties and *accuracy* levels can be important and therefore these concepts are captured above in Tables GG.4 and GG.5. Example:

Closed loop control of oxygen delivery to neonates - does the *pulse oximeter equipment* have sufficient *accuracy*?

Pulse oximeter equipment can monitor the *pulse oximeter probe* life and refuse operation once the lifetime is exceeded. The related information and pre-warning of expiration can be of importance also for connected *ME equipment* or *ME systems*. Therefore, these data per Table GG.6 is recommended to be available via the data interface (e.g. for remote monitoring or closed loop applications).

Monitoring of *pulse oximeter probe* to ensure pre-warning of expiration occurs and labelling encouraging availability of a replacement once *pulse oximeter probe* life exceeded to prevent gaps in monitoring.

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Annex II
(informative)

Using a *functional tester* to assess *pulse oximeter equipment* conditions of signal inadequacy over a range of transmitted light and optical modulation

II.1 Background and application

The method below describes in vitro testing of *pulse oximeter equipment* across a broad range of signal conditions to verify conformance with the signal inadequacy indicator requirements of 201.12.4.102.

This protocol and *functional testers* in general do not confirm or ensure *accuracy* of *pulse oximeter equipment* SpO_2 values to true SaO_2 values and should not be used for this purpose. As any test system, the tester itself introduces noise. The noise contribution from the tester varies for each *pulse oximeter equipment*. Simulators do not test the errors induced by all potential sources of signal inadequacy. One intent of using a *functional tester* is to establish a minimum performance expectation of the *pulse oximeter equipment* over a range of signal levels and pulse amplitudes to identify and correct problems during design.

The real-world performance of *pulse oximeter equipment* is always worse than that on a *functional tester* due to other confounding factors not simulated by the *functional tester*. This testing identifies some, but not all possible measurement errors.

There are different *pulse oximeter equipment* implementations available. A *functional tester* can have different limitations for different implementations. Not every *functional tester* is suited for each *pulse oximeter equipment* even if it offers the calibration curve for a specific *pulse oximeter equipment*. In particular, testing at low signal strength can reveal the limits of the *functional tester* rather than the performance limits of the *pulse oximeter equipment*. Any defect of a *functional tester,* such as noise or nonlinearity, can increase observed errors.

Ideally, for verifying the signal inadequacy indicator, one would test *pulse oximeter equipment* over every possible combination of light levels and %*mod*. In this application, the test *procedure* has been limited to be practical. Conformance to acceptance criteria can be by visual inspection or electronic data capture.

Appendix EE describes many types of available *functional testers*. The examples provided in this annex are based upon an electro-optical *functional tester*[3]. Other *functional testers* with electrical or optical interfaces can also be used to complete this testing.

II.2 Test method

II.2.1 Parameters

Percent modulation- typically a raw number, represents a means to quantify the intensity of pulse variability expressed in %, and abbreviated 'mod' relates to the *%mod* of the detected signal at the *pulse oximeter probe* site (generally is 100 times AC/DC and sometimes referred to as *percent modulation*, pulse amplitude, Perfusion Index, or PI). Typical normal *%mod* values range from 1 % to 8 %, but this assumes that the *pulse oximeter monitor* solely takes the IR into account for perfusion, which is not true for all *pulse oximeter monitors*.

The signal level is a measure of how much light is measured by the detector, expressed in the arbitrary units of digital counts, in general, a low level of counts represents low detected light levels and high counts

[3] The specific electro-*functional tester* used was a ProSim 8 Functional Tester with SPOT Light SpO_2 accessory that provides an optical interface (Fluke Biomedical, Everett, WA USA)

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represents high light levels. It is important to account for factors that influence the relationship between counts and detected light levels, such as LED drive current and amplification of the photocurrent signal.

II.2.2 Testing *procedure*

a) Start at a nominal signal level (DC) and *%mod* that supply signals to the *pulse oximeter monitor* such that it reads properly (for finger probes, this is commonly when ~20 % of the light is transmitted and the %*mod* is ~10 %). Set SpO_2 at 90 % and a pulse rate (PR) of 80 bpm. 90 % SpO_2 was chosen to make changes in either direction visible. The value from the *pulse oximeter monitor* is the starting SpO_2 reading.

b) Reduce %*mod* and signal level independently in a stepwise fashion until the *pulse oximeter monitor* no longer posts an SpO_2. The amount of reduction in each parameter can be chosen by balancing pragmatic aspects of testing time verse resolution, pausing at each test point for at least 1 min to allow the *pulse oximeter monitor* a chance to settle.

c) If readings do not settle, this is indicative of the *pulse oximeter monitor* nearing its measurement limit. Report the mean value.

d) If you obtain a valid reading, calculate the error and record count and *%mod* for each test point.

e) Determine the error by calculating the difference between the SpO_2 reading on the *pulse oximeter monitor* and the starting SpO_2 reading on the *pulse oximeter monitor* observed in 1) (i.e. a known good set point that has a high-count level and a high *%mod*). When no number is reported by the *pulse oximeter monitor*, the point noted as having dropped out (DO).

In this test, the red and IR light transmission values of the *functional tester* are equal. It is also recommended to run the same test with the red transmission at 1/6 that of the IR, when practical. This simulates the relationship in many darkly pigmented people.

Table II.1 contains the result of testing low-cost fingertip *pulse oximeter equipment*. Each data point contains the difference in the reported SpO_2 from the *pulse oximeter equipment* compared to the set-point of the *functional tester* (method 3a). Typically, *pulse oximeter equipment* is expected to be accurate at high settings when light transmission is around 20 % and %*mod* is around 10 %. Here errors above ±4% were chosen as the threshold for when the signal inadequacy indicator should become active. SpO_2 values in error by greater than 4% are shown in bold and an asterisk (*) is included if the signal inadequacy indicator was active.

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Table II.1 — Results following the outlined testing *procedure*

Count level	MOD							
	0,05	**0,1**	**0,2**	**0,5**	**1**	**2**	**5**	**10**
100 000	NT	NT	NT	NT	NT	NT	NT	NT
70 000	NT	DO	DO	DO	DO	DO	DO	DO
50 000	NT	DO	**-9***	0	0	0	0	0
35 000	NT	DO	-2	1	0	0	0	0
25 000	NT	DO	3	0	1	1	0	0
14 000	NT	DO	**5***	4	4	0	0	0
10 000	NT	DO	DO	DO	**5***	**5***	0	0
7 000	NT	DO	DO	**7**	4	2	0	0
5 000	NT	DO	DO	DO	4	-3	0	0
3 500	NT	DO	DO	DO	DO	DO	DO	DO
2 500	NT	NT	NT	NT	NT	NT	NT	NT

The numbers are the calculated error at the test point. The error in bold represents test points where the error exceeds the threshold. An asterisk (*) indicates the signal inadequacy indictor was active. Test points shown as DO indicate the *pulse oximeter equipment* did not post a number (drop out). Test points shown as NT indicates that test point was not evaluated.

II.2.3 Interpretation of sample data

The proper behaviour of a *pulse oximeter equipment* in this test is to provide an indication if insufficient %*mod* or low signal levels result in reading errors beyond an allowable threshold. Accordingly, the observations noted in Table II.1 can be simplified into four categories as summarized in Table II.2. Table II.3 presents the categorized results from this example, noting the associated observed behaviours.

Table II.2 — Categorization of the observations shown in Table II.1

Category	Description
A	Absolute error of displayed value is less than or equal to the allowed deviation threshold (≤ threshold)
B	Absolute error of displayed value is greater than the threshold (> threshold) AND the indicator is active, OR no value is displayed (readings dropped out, DO)
C	Absolute error of the displayed value is greater than the threshold (> threshold) AND the indicator is NOT active
D	Condition not tested (NT)

The signal inadequacy indicator should be active whenever the error exceeds the deviation threshold. In this ideal case, Table II.3 would not have any test points shown as “C”. Due to uncertainties of the *functional tester*, the maximum allowable test points shown as "C" are two. For the example provided, only one test point exceeded the deviation threshold without activating the signal inadequacy indicator (test point, 7 000 counts and 0,5 %*mod*). This *pulse oximeter equipment* would be considered in conformity with 201.12.4.102.

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Table II.3 — Categorization of the observations shown in Table II.2 using a deviation threshold of ±4 digits

Count level	MOD							
	0,05	0,1	0,2	0,5	1	2	5	10
100 000	D	D	D	D	D	D	D	D
70 000	D	B	B	B	B	B	B	B
50 000	D	B	B	A	A	A	A	A
35 000	D	B	A	A	A	A	A	A
25 000	D	B	A	A	A	A	A	A
14 000	D	B	B	A	A	A	A	A
10 000	D	B	B	B	B	B	A	A
7 000	D	B	B	**C**	A	A	A	A
5 000	D	B	B	B	A	A	A	A
3 500	D	B	B	B	B	B	B	B
2 500	D	D	D	D	D	D	D	D

While demonstrating a conforming implementation of the signal inadequacy indicator, the example *pulse oximeter equipment* exhibits reduced *accuracy* in the part of the table that represents a combination of low count level and %*mod*, while in cases with high count level and %*mod* the *accuracy* is within ±4 digits. This lower left region of Table II.3 relates generally to signal levels associated with dark skin and thick sites when combined with low %*mod*.

II.3 Testing for pulse rate

The same *procedure* outlined in II.2.2 can be used to test pulse rate and the same data interpretation method outlined in II.2.3 can be used to interpret the results. The deviation threshold should be adjusted to account for testing a different parameter.

II.4 Calibrating *pulse oximeter equipment* for transmitted light

Count level is a *functional tester* dependent measurement and used in this example. However, percent transmission, T, is a more useful parameter that is not *functional tester* dependent. The following outlines a method for calibrating an electro-optical functional tester such that measurements can be made using T instead of count level. Setup and use of various *functional tester* is related to each other if they are to compare data from different tests. The most difficult parameter to relate between *functional tester* is transmission, T. For this test, the electro-optical functional tester was checked using the plethysmogram data from a known *pulse oximeter monitor*. The sensor was placed on the electro-optical functional tester finger shell with nothing inside, creating a hole with T=100 while maintaining the optics spacing used by a normal SPOT. A reading of the count level (ADC counts) of the *pulse oximeter monitor* was recorded. The sensor was then placed on the actual electro-optical functional tester. The *functional tester* was set to mod=0 and stepped through T values until the count level of the *pulse oximeter monitor* was at 10 % of the recorded value. This setting was 80. It is important to ensure that the LED drive current and photocurrent amplification remain the same (or that their influence on the measured ADC counts is accounted for) to compare the ADC counts. For this example, to simulate a T=5 %, the electro-optical functional tester should be set at Transmission = 40, for 1 % at Transmission = 8, etc. The mod setting was also checked using the reference *pulse oximeter monitor* wave outputs. A similar *process* would be followed to use any *functional tester*.

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Annex JJ
(informative)

Using a *transfer standard* in *pulse oximeter equipment* development

JJ.1 Non-invasive testing

The *SpO_2* performance of *pulse oximeter equipment* can be assessed by comparing *SpO_2* readings to values obtained using a *transfer standard*, because its calibration is directly traceable to a *CO-oximeter*. Even though the *transfer standard* is traceable to *CO-oximeter*, it cannot be used in place of a *CO-oximeter* for demonstrating conformance to 201.12.1.101.3. *Transfer standard* only can be used to assess *pulse oximeter equipment* performance for developmental or calibration purposes.

Transfer standard can be used during the product development *process* to gather data on participants under controlled laboratory conditions for engineering purposes, assisting in the *pulse oximeter equipment* calibration *process* performance assessment where the final performance *verification* is not yet needed, or other reasons. *Transfer standard* can also provide value in clinical testing of *pulse oximeter equipment*.

In a clinical environment, the primary responsibility is *patient* care. *SpO_2* measurements from *patients* in that environment can be degraded because data collection cannot always be well controlled. *SpO_2* measurements are better controlled under laboratory conditions.

In a clinical environment, measurements from *pulse oximeter equipment* are often subject to non-optimal conditions and are difficult to match reliably, due to circulatory instabilities or dynamics.

The *patient's* clinical condition should be considered when placing *pulse oximeter probes*. Whenever possible, the *pulse oximeter probes* of the test and reference or *transfer standard* should be observing blood that is part of the same regional circulation.

Generating the number of data pairs sufficient to demonstrate statistically the specified *SpO_2 accuracy* over the specified range can require a large number of *patients* or observations.

The total number of acceptable data pairs acquired during the study needs to be sufficient to demonstrate statistically the specified *SpO_2 accuracy*. The distribution of reference values in the pooled data set needs to be made with comparable density over the full claimed range. Specific numbers of samples and participants as well as the analysis technique need to be justified using statistical methods.

The A_{rms} value obtained with non-invasive testing is expressed relative to the "gold-standard" (i.e. *CO-oximeter SaO_2* values) and includes the error of the reference or *transfer standard*.

JJ.2 Considerations for testing on volunteers

In non-invasive testing, the *SpO_2 accuracy* of *pulse oximeter equipment* is measured by comparing *SpO_2* readings of the *pulse oximeter equipment* to values obtained with *transfer standard*. This method utilizes healthy adult volunteers who consent to induced hypoxia as part of the experimental *procedure*.

Since the calibration of the transfer standard *pulse oximeter equipment* is directly traceable to a *CO-oximeter*, the transfer standard *pulse oximeter equipment* can be used as *transfer standard*. Even though the transfer standard *pulse oximeter equipment* is traceable to *CO-oximeter*, it cannot be used in place of a *CO-oximeter* for demonstrating conformance to 201.12.1.101.3. *Transfer standard* can be used for development purposes for *pulse oximeter equipment*.

The method for non-invasive laboratory testing on healthy adult volunteers follows the protocol described in DD.2 for invasive tests with the following exceptions.

— The reference values for *SpO_2* readings are obtained from *transfer standard* (i.e. replacing *SaO_2* values as measured with a *CO-oximeter*).

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— Blood sampling is not utilized.

  — The calibration of the *transfer standard* and the treatment of the data analysis are traceable to a *CO-oximeter*.

  — The total number of acceptable data pairs acquired during the study needs to be sufficient to demonstrate statistically the specified *SpO_2 accuracy*.

— For example, one possible profile for acquiring data follows the plateau scheme described in DD.2.3.3 e) through DD.2.3.3 h) (i.e. about 20 sampling periods) during plateaus, are used.

— Other profiles are possible (i.e. continuous data collection during gradual changes in saturation) independent of plateaus, relating sample pairs in time.

Specific numbers of samples and participants as well as the analysis technique need to be justified using statistical methods.

JJ.2 Requirements for *transfer standard*

The candidate *transfer standard* should meet the following conditions:

a) *pulse oximeter equipment* comprised of a *pulse oximeter monitor*, a *probe cable extender*, and a *pulse oximeter probe* (each component specially kept together as a single *transfer standard*);

   1) These components should be individually *marked* and state that when combined become a *transfer standard*.

b) be compared to *CO-oximeter SaO_2* data over the induced hypoxia range of 70 % to 100%, with the resultant $A_{\mathrm{rms}} \leq 2.0$ (see 201.12.1.101 for statistical details);

   1) Testing should include at least 48 adult participants with an even distribution of light, medium and dark pigmentation, conforming with 201.12.1.101.6 f).

c) development of a 'Transfer Equation,' which is stated on the *transfer standard* (see Formula JJ.1);

d) have an averaging time in determining *SpO_2* that is a minimum of 6 s and maximum of 10 s;

e) provide electronic data output at a minimum resolution of 1 Hz (i.e. once every second) of *SpO_2,* perfusion index, and pulse rate values; and

f) conform with the *differential bias* performance thresholds specified in 201.12.1.101.2.

The 'Transfer Equation' is calculated based on the *SpO_2* of the *transfer standard* under test being converted using Formula (JJ.1):

$$Y = mX + b \qquad \text{(JJ.1)}$$

where

X is the *SaO_2* from the *CO-oximeter*;

Y is the *SpO_2* value from the *transfer standard*;

m is the slope; and

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b is the Y intercept.

The 'Transfer Equation' is derived from a dataset [developed from a) through f)] of paired SpO_2 values from the *transfer standard* and SaO_2 values from a *CO-oximeter*. A linear regression of the SaO_2 values from the *CO-oximeter* against the SpO_2 values from the *transfer standard* is performed to obtain m (the slope) and b (the Y intercept) for Formula JJ.1 above. The derived slope and offset can then be used with future SpO_2 values from the *transfer standard* to obtain calculated SaO_2 values.

Formula (JJ.2) below shows how Formula JJ.1 above can be rearranged to show the *transfer standard* calculated SaO_2 value (x) in terms of the transfer equation slope (m), offset (b), and SpO_2 value displayed by the *transfer standard* (Y).

$$x = \frac{Y - b}{m} \tag{JJ.2}$$

where

x is the calculated SaO_2 from the *transfer standard*;

Y is the SpO_2 value from the *transfer standard*;

m is the slope from the 'Transfer Equation'; and

b is the Y-intercept of the 'Transfer Equation'.

Y represents the value displayed on the *transfer standard*, "*m*" and "*b*" are the slope and y-intercept of the 'Transfer Equation'. "*x*" represents the calculated *transfer standard* SaO_2 value (i.e. the assumed SaO_2 that the *CO-oximeter* would have determined). This calculated SaO_2 from the *transfer standard* can then be used as a reference for *accuracy* assessment of another *pulse oximeter equipment's* SpO_2 values.

Test data from a *pulse oximeter equipment* under test is compared to the *transfer standard* in the following manner. The *transfer standard* data are screened for unstable periods (e.g. data from rapid re-saturation periods are removed from analysis). The remaining data, collected in 1 s intervals, are analysed and graphed. The 1 Hz data is distilled to 17 to 30 points, similar to the stability period used when sampling arterial blood. The SpO_2 *accuracy* of the *pulse oximeter equipment* under test is derived from the Formula (JJ.3).

$$A_{\mathrm{rms}} = \sqrt{\frac{\sum_{i=1}^{n}(SpO_{2i} - SaO_{2\mathrm{ref},i})^2}{n}} \tag{JJ.3}$$

where

A_{rms} is the *accuracy* root mean square;

SpO_{2i} is the *pulse oximeter equipment* under test SpO_2 value during sample *i*;

$SaO_{2\mathrm{ref},i}$ is the *transfer standard* SaO_2 value calculated using the "Transfer Equation" sample *i*; and

n is the number of points.

Bias is the mean difference of the *pulse oximeter equipment* under test SpO_2 value from the *transfer standard* SaO_2 value, calculated using the "Transfer Equation" derived from Formula (JJ.4).

$$Bias = \frac{\sum_{i=1}^{n}(SpO_{2i} - SaO_{2\mathrm{ref},i})}{n} \tag{JJ.4}$$

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JJ.3 Rationale for the use and *accuracy* of *transfer standard*

An analysis comparing the use of a *CO-oximeter* as a reference to the use of a *transfer standard* was performed and included a total of 4 644 paired datasets, 1 014 from 43 black participants and 3 630 from white participants, during 195 *controlled desaturation studies*, over the range 70 % to 100 % SaO_2. The committees analysed data collected over the period 2017 to 2020 from controlled desaturation studies that was used for regulatory purposes. The studies were used for regulatory submissions and met the ISO 80601-2-61:2017 requirements and FDA guidance recommendations[28].

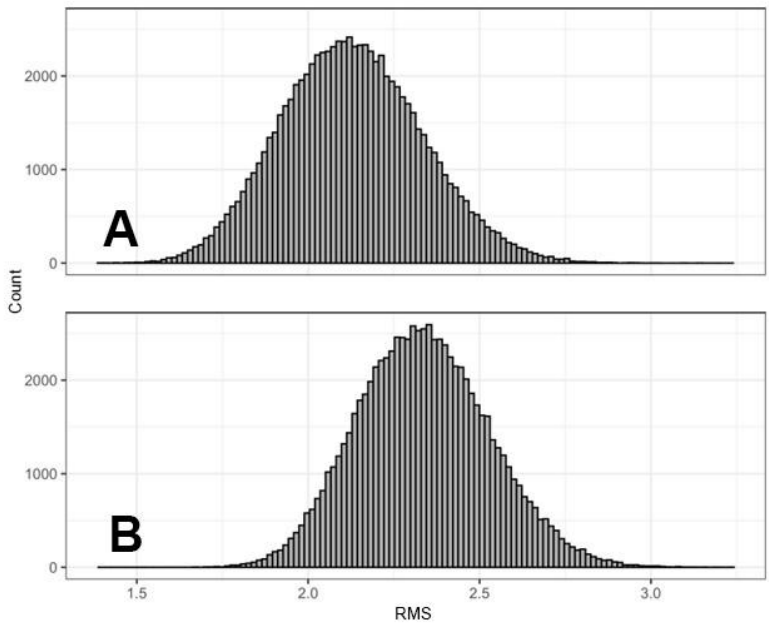
These analyses compared the A_{rms} of a *pulse oximeter sensor* when *transfer standard* is used as the reference to the A_{rms} of that *pulse oximeter sensor* when a *CO-oximeter* is used as the reference.

To gain confidence in that comparison, the population of 195 was bootstrapped[43] with samples of N=30 (10 dark + 20 light). Essentially this analysis investigated the error stack-up using the *transfer standard* as a reference instead of using *CO-oximeter* as the reference.

8 000 cluster bootstrap samples of size 30 were generated using the provided data, with participant's *pulse oximeter probe* as the clusters. This clustered bootstrap preserves any within-participant *pulse oximeter probe* correlation in the data. Bootstrapping was stratified by skin tone, with each sample of 30 consisting of 10 dark skinned participants and 20 light skinned participants. The number of paired *pulse oximeter equipment*-vs-reference measurements in the bootstrap samples was thus variable, with an average of 714 paired differences (the SD was 10,5, 95 % CI). For each sample, the root mean square (RMS) between *pulse oximeter equipment* and reference *CO-oximeter* and the RMS between *pulse oximeter equipment*-and-*transfer standard* as reference were calculated.

— The full data RMS values from these data are 2,08 for the *CO-oximeter* as reference and 2,36 for *transfer standard* as reference. With the full dataset, the *transfer standard* as a reference increases the RMS by about 0,28 over the reference standard.

— The difference is similar for smaller (n=30) samples, with the distribution shifted upwards by about 0,2 when the *transfer standard* is used as a reference. The RMS results are shown in Figure JJ.1 and Table JJ.1



Key

Count is the number of samples
RMS is root mean error per sample

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A SpO_2 as measured by a forehead *pulse oximeter equipment* vs reference SaO_2 as measured by *CO-oximeter*
B SpO_2 as measured by a forehead *pulse oximeter equipment* vs reference SaO_2 from a *transfer standard*

Based on 80 000 cluster bootstrap samples, with participants as clusters.

Figure JJ.1 — Distribution of root mean square error of SpO_2

Table JJ.1 — Quantiles of RMS values from bootstrap samples of size 30 (10 dark / 20 light skinned)

Forehead SpO_2 comparison	2,5 percentile	25 percentile	Median	75 percentile	97,5 percentile	Width of 95 % interval
CO-oximeter as reference	1,75	1,98	2,12	2,26	2,55	0,80
Transfer standard as reference	1,98	2,20	2,33	2,46	2,73	0,75

JJ.4 Non-invasive study use of a *transfer standard*

The *transfer standard* can be used to develop calibration curves of pulse oximeters and collect data where arterial blood is not readily available. This approach has been used effectively since 1990, during development of many *pulse oximeters*, which were then tested in comparison to *CO-oximeter* SaO_2 data as final *verification* for regulatory submissions.

Other benefits of the *transfer standard* are:

— An arterial blood study provides approximately 24 data points over the induced hypoxia range of 70 % to 100 % from each participant. However, the *transfer standard* supplies data at ~1 Hz, resulting in ~1,800 time matched data points during the same period of induced hypoxia range of 70 % to 100 % SpO_2.

— Usually less costly and more accessible.

— Promotes arterial vessel preservation by eliminating the need for invasive arterial sampling during calibration curve creation and other *pulse oximeter equipment* development.

— In an approved study protocol, collecting data is possible where sampling arterial blood is not practical or safe (e.g. neonates, children, or certain hospitalized adults). Thus, the data can be gathered on populations for intended clinical use of *pulse oximeter equipment*.

When used in combination with SaO_2 samples measured by a *CO-oximeter*, termed Hybrid approach, the *transfer standard* can provide a larger amount of data, non-invasively, so the effective sample size is increased without the *risk* and limitations of placing arterial catheters.

When a *transfer standard* is used, the transfer equation developed for that *transfer standard* is used as the comparator. The transfer equation removes bias error that would be introduced if the *pulse oximeter equipment* device-under-test (DUT) data were compared directly to the readings of the *transfer standard*.

A potential limitation of a *transfer standard* approach is the need to account for the unrecognized bias or error inherent in the *transfer standard*. Data obtained from a *transfer standard* study is not acceptable for final *verification* evidence in accordance with 201.12.1.101.3.

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Annex KK
(informative)

Reference to the IMDRF *essential principles* and labelling guidances

This document has been prepared to support the *essential principles* and labelling requirements of *pulse oximeter equipment*, and its *accessories* or parts as medical devices according to the International Medical Device Regulators Forum (IMDRF).

Conformance with this document provides one means of demonstrating conformance with the specific *essential principles* of IMDRF/GRRP WG/N47:2018[25] and labelling principles IMDRF/GRRP WG/N52:2019[26]. Other means are possible. Table KK.1 lists the clauses and subclauses of this document corresponding to the *essential principles* of IMDRF/GRRP WG/N47:2018, the replacement for Reference [7]. Table KK.2 lists the clauses and subclauses of this document corresponding to the labelling principles of IMDRF/GRRP WG/N52:2019.

NOTE 1 When an *essential principle* does not appear in Table KK.1, it means that it is not addressed by this document.

Table KK.1 — Correspondence between this document and the *essential principles*

***Essential principle* of IMDRF/GRRP WG/N47:2018[25]**	**Corresponding clause(s)/sub-clause(s) of this document**	**Qualifying remarks/Notes**
5.1.4	201.7.9.2.2.101, 201.11.8.101.1, 201.12.4.101, 201.13.101, 201.101.2 b)	
5.1.5 a)	201.12.1.101, 201.12.1.102, 201.12.1.103, 201.12.1.104, 206	Addresses accuracy of device output and requirements to identify and validate ergonomic aspects of the use of the device.
5.1.5 b)	206	
5.1.6	All	
5.1.7	201.4	
5.1.9	201.4, 201.10, 201.11, 201.13	
5.3.1 f)	201.12.1.101, 201.12.1.102, 201.12.1.103, 201.12.1.104	Covered with respect to accuracy of SpO_2 and pulse rate.
5.5.1	201.7.2.4.101, 201.7.9.2.14.101, 201.101, 201.103	Covered with respect to listed accessories, connecting accessories, and operator detachable components
5.5.2 a)	201.12.4, 201.12.101, 206	Only the part relating to design is addressed with respect accuracy of controls and instruments including indicating noncurrent data, inadequate signal, detecting probe faults and primary operating functions.
5.5.2 b)	206	

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Essential principle of IMDRF/GRRP WG/N47:2018[25]	Corresponding clause(s)/sub-clause(s) of this document	Qualifying remarks/Notes
5.5.2 c)	202	Only the part relating to design is addressed with respect to magnetic fields, external electrical and electromagnetic effects, and electromagnetic discharge.
5.5.2 h)	202	Only the part relating to design is addressed with respect to electromagnetic disturbances.
5.5.5	201.7.2.4.101, 201.7.9.2.14.101, 201.101, 201.103	Covered with respect to listed accessories, connecting accessories, and operator detachable components.
5.5.6	201.14.101	Covered with respect to design for protection against accidental adjustments and cyber security capabilities.
5.5.7	206	
5.6.4	201.101, 201.103	Covered for probe, extension cables and functional connections.
5.6.5	201.11.1.2.2	
5.7.1	201.13.101	Covered for probe and extension cables
5.7.2	201.11.8.101.1	Covered with respect to loss of a power source and warnings in the event of depletion of internal power source.
5.7.3	201.11.8.101.1	Covered with respect to loss of a power source and warnings in the event of depletion of internal power source.
5.7.5	202	
5.7.6	202	
5.8.4	201.14.101	
5.8.5	201.14.101	
5.9.1 a)	201.12.101	Covered for accuracy of SpO_2 and pulse rate. The part relating to stability is not addressed.
5.9.1 b)	201.7.4.3	Covered for of SpO_2 and pulse rate
5.9.1 c)	201.7.2.101, 201.12, 206	Covered for markngs; accuracy of controls and instruments, and primary operating functions.
5.10.1	201.7	

NOTE 2 When a labelling principle does not appear in Table KK.2, it means that it is not addressed by this document.

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Table KK.2 — Correspondence between this document and the labelling principles

Labelling principles of IMDRF/GRRP WG/N52:2019[26]	Corresponding clause(s)/sub-clause(s) of this document	Qualifying remarks/Notes
5.1.1	201.7.2.4.101, 201.7.2.101, 201.7.9.2.1.101, 201.7.9.2.2.101, 201.7.9.2.9.101, 7.2.9.2.14.101	Covered for markings on device and accessories, and information for IFU.
5.1.4	201.7.9.2.9.101, 201.7.2.101, 201.12.4.102	Covered for indicating keep dry, no alarms and signal inadequacy.
5.1.5	201.7.9.2.2.101	
5.2.1	201.7.2.101	
5.3.1	201.7.2.4.101, 201.7.2.101, 201.7.9.2.1.101, 201.7.9.2.2.101, 201.7.9.2.9.101	Covered for markings on device and accessories, and information for IFU.
5.3.11	201.7.9.2.9.101, 201.12.1.101, 201.12.1.102, 201.12.1.103, 201.12.1.104	
5.3.12	201.12.1.101, 201.12.1.102, 201.12.1.103, 201.12.1.104	
5.3.13	201.7.9.2.2.101	
5.3.17	201.7	
5.3.27	201.7.9.2.2.101, 201.7.9.2.14.101, 201.101	Includes instructions for restrictions of accessories, and effects of the accessories on the performance of the pulse oximeter equipment.

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